上海科技大学
ShanghaiTech University

脑科学与脑疾病
Brain Sciences and Brain Disorders

上节课的作业：同学提出问题

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问题组

10. 新生儿是怎么掌握语言的?
11. 在和父母/陌生人交流时新生儿大脑是如何活动的?
12. 人的意识是怎么产生的?
13. 睡眠是如何改变大脑的?
14. 怎么研究人的意识?
15. 如何研究人脑的可塑性?
16. 人类成年后大脑还有可塑性吗?
17. 如何用人类大脑控制机器?
18. 人吃饭/喝咖啡/运动前后大脑如何改变?
19. 如何基于人脑影像预测年龄、智商?

疾病组

20. 自闭症是怎么发现和治疗的?
21. 为什么有的人会得抑郁症?
22. 为什么会得阿尔茨海默症（老年痴呆症）?
23. 怎么治疗阿尔茨海默症?
24. 如何用fMRI帮助脑外科手术医生进行术前规划?
25. 如何用AI手段帮助精神科医生精准诊断?
26. 儿童多动症的大脑有什么异常吗?
27. 焦虑症和强迫症是病吗?
28. 帕金森病的最佳治疗方案是什么?
29. 如何早期筛查阿尔茨海默症?

提出问题请发给助教；如有其他问题，也请提出；分组相关的需求也请发给助教
我们将进行4-5人左右的分组；针对问题开展相关调研并视情况进行实验

目前分组情况

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课题序号	成员1	成员2	成员3	成员4	成员5	
29	赵汉卿	吴沂灵	李彦廷	高嘉志	李卓雅	陈一诺
28	范妤清	陈誉杰	计英昊			
19	王洁琳	冯靖尧				
18	陈姜联					
22	陈怡璇	慕欣	王宇瞳			

目前仅有15人提交，还有6人没提交？请尽量4-5人一组

提出问题清发给助教；如有其他问题，也请提出；分组相关的需求也请发给助教
我们将进行4-5人左右的分组；针对问题开展相关调研并视情况进行实验

这节课上我们将讨论...

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- 上节课：科学家们是如何研究大脑结构的？—— 脑图谱（离体脑ex vivo, 损伤模型lesion-based study, 根据结构的分区和根据功能的分区）
- 上节课：科学家们是如何研究大脑功能的？——fMRI和激活区检测理论（BOLD、GLM、对比contrast、Design matrix设计矩阵）
- 基本的统计思想（统计推断、假设检验、分布、自由度、p值）
- 研究人类大脑结构的sMRI及其基本分析技术
- 研究人类大脑功能的rs-fMRI及其基本分析技术
- 研究人类大脑白质结构的dMRI及其基本分析技术
- 研究人类大脑代谢的PET和大脑灌注的ASL及其基本分析技术

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脑科学与脑疾病

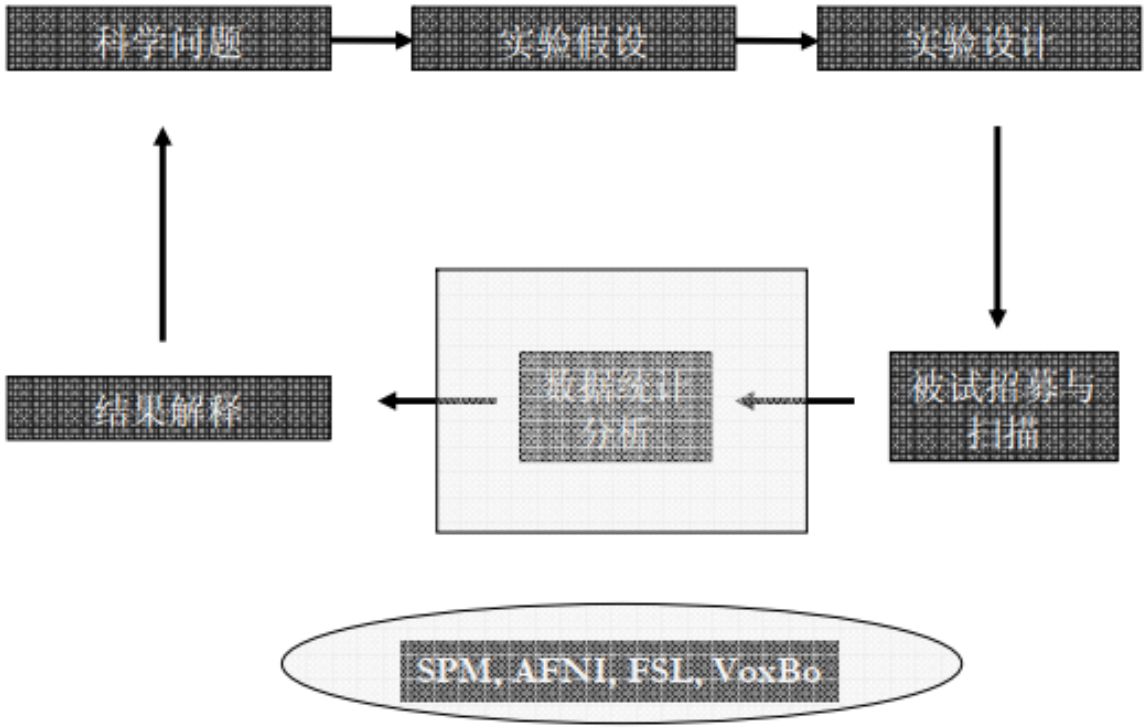
第四-六节课回顾与随堂测验：脑图谱、脑结构、fMRI、GLM

fMRI

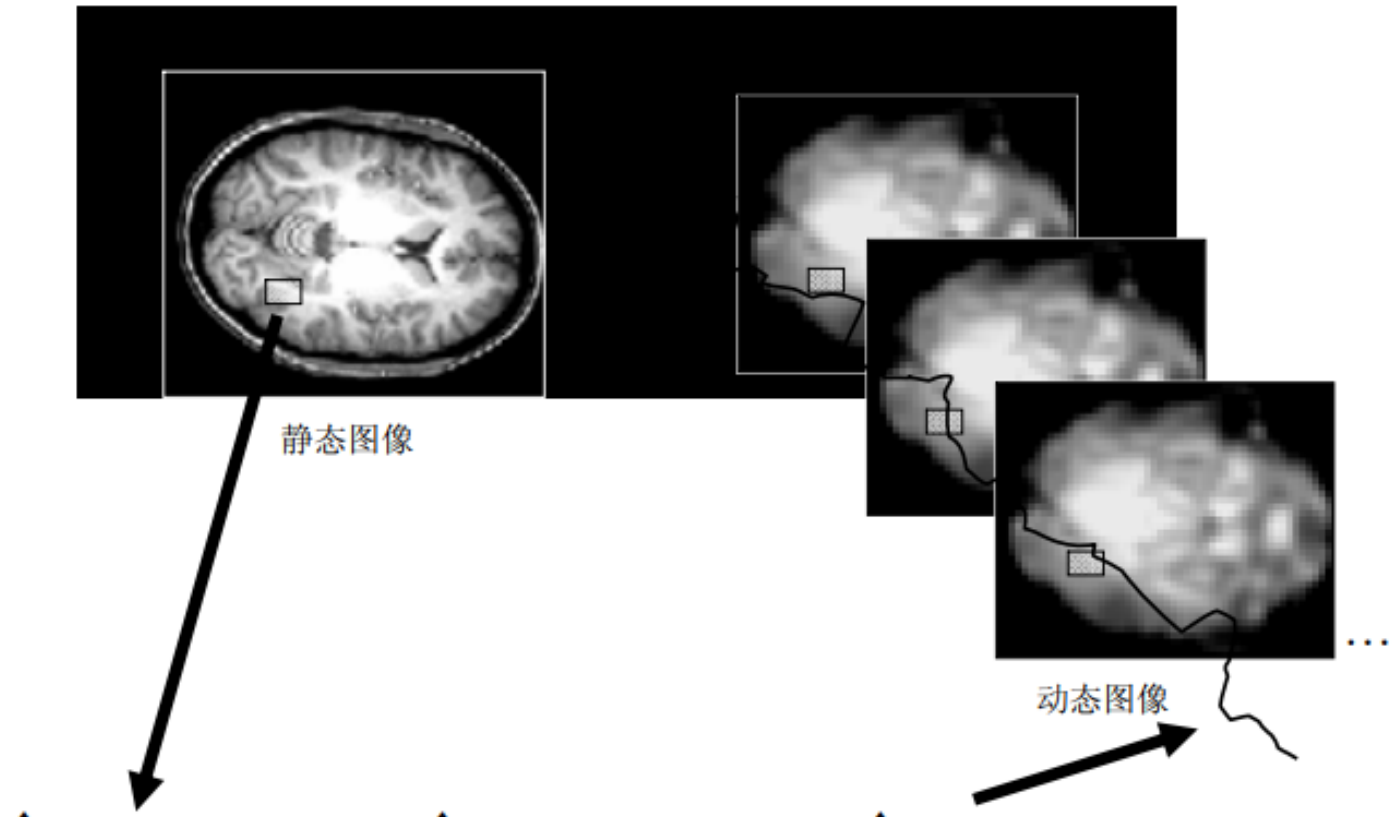
https://www.youtube.com/watch?v=drP8Nk67l1s

https://www.youtube.com/watch?v=GDx40b-FZiQ

fMRI研究框架



结构像与功能像



Safety first

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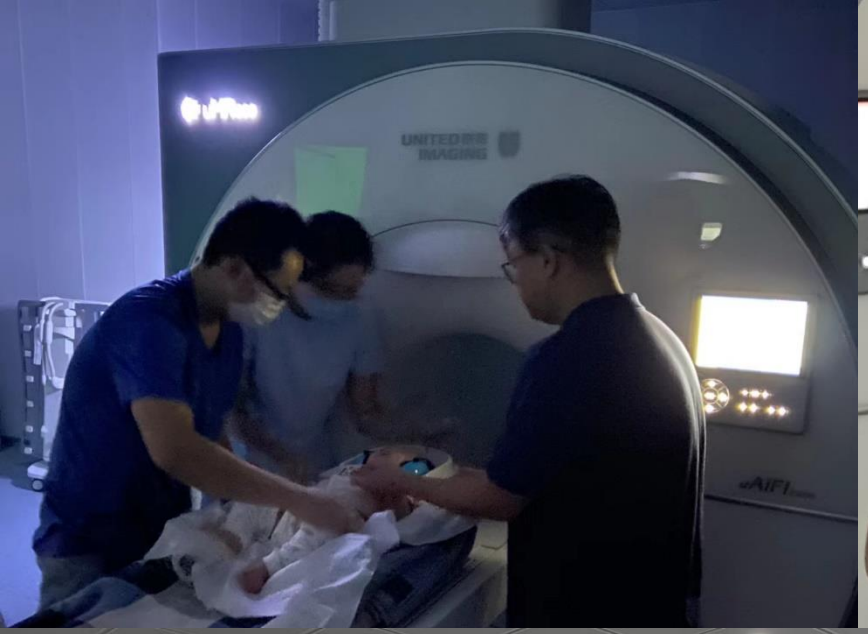
http://www.simplyphysics.com/mri_safety.html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脑科学与脑疾病
第六节课：基本的统计思想
（统计推断、假设检验、分布、自由度、p值）

https://www.youtube.com/watch?v=0oc49DyA3hU

Hypothesis Testing and The Null Hypothesis, Clearly Explained!!!

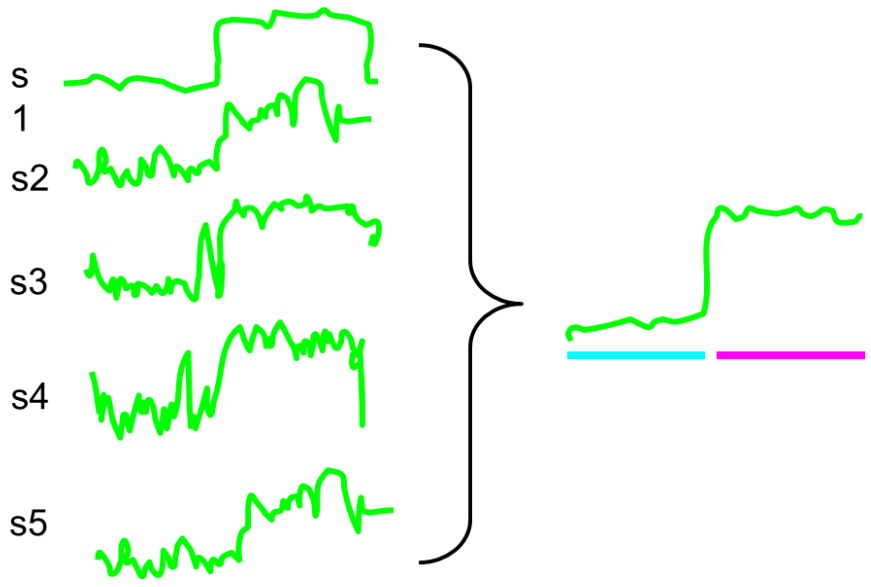
https://www.youtube.com/watch?v=vemZtEM63GY

p-values: What they are and how to interpret them

给大家的预热问题

- 我们为什么要做统计推断？直接计算差异大小就好了啊？
- 我们为何要做假设检验？明明没有假设还要假什么设？
- 为啥假设检验要和分布有关？特别是T分布又是什么？
- T检验里面的自由度表示什么？为什么叫“自由”度？
- 假设检验后得到的p值，为啥就和0.05扯上关系了呢？
- 什么是非参数统计和蒙特卡洛模拟？
- 什么是多重比较校正？多重比较以后为啥要校正？
- 怎样学好统计？

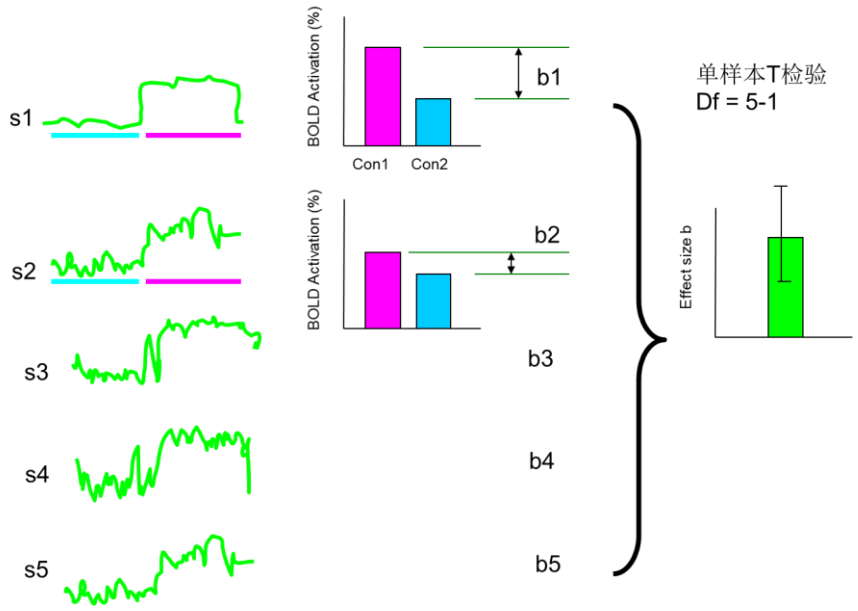
- 统计推断（Statistical inference）



Fixed effect analysis

- Assume that the experimental manipulation has same effect in each subject
- Uses data from ALL subjects to construct statistical test
  - Averaging/connecting across subjects before a t-test
- Sensitive to extreme results from individual subject
  - Strong effect in one subject can lead to significance even when others show weak or no effects
- Allows inference only to subject sample
  - you can say that effect was significant in your group of subjects but cannot generalize to other subjects that you didn't test

- 统计推断（Statistical inference）



Random effect analysis

- Assume that the experimental manipulation has different effect in each subject (but following normal distribution)
- Uses data from sampled subjects to construct statistical test
- Allow extreme results from individual subjects (remember: normal distribution)
- Allows inference to the entire population where the samples are from

我们为什么要做统计推断？直接计算差异大小就好了啊？

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- 假设我们看两组老人，一组文化程度高，一组低，想比较他们的智商。一组智商平均值95，一组智商平均值97，请问，两组是否显著差异？

- 假设我们有两组人，一组病人，一组正常人，看这两组是否有显著的脑激活水平差异，一组激活水平平均值为mean beta=0.8, 一组为mean beta=0.9. 请问，两组是否有显著激活水平差异？

- 假设我们看一种降压药物，用在10个被试上进行试验，吃之前血压mean为150，吃之后为140，请问，是否有显著下降？

假设检验原理：零假设

- 我们想检验病人组和正常人组是否有差异，但是无法得到病人组的总体分布，只有正常人组的总体分布。
- 假设病人是正常人，是从正常人总体里抽样出来的。
- 零假设：病人组均值和正常人组没有差别。

假设检验的根基：小概率事件不可能发生

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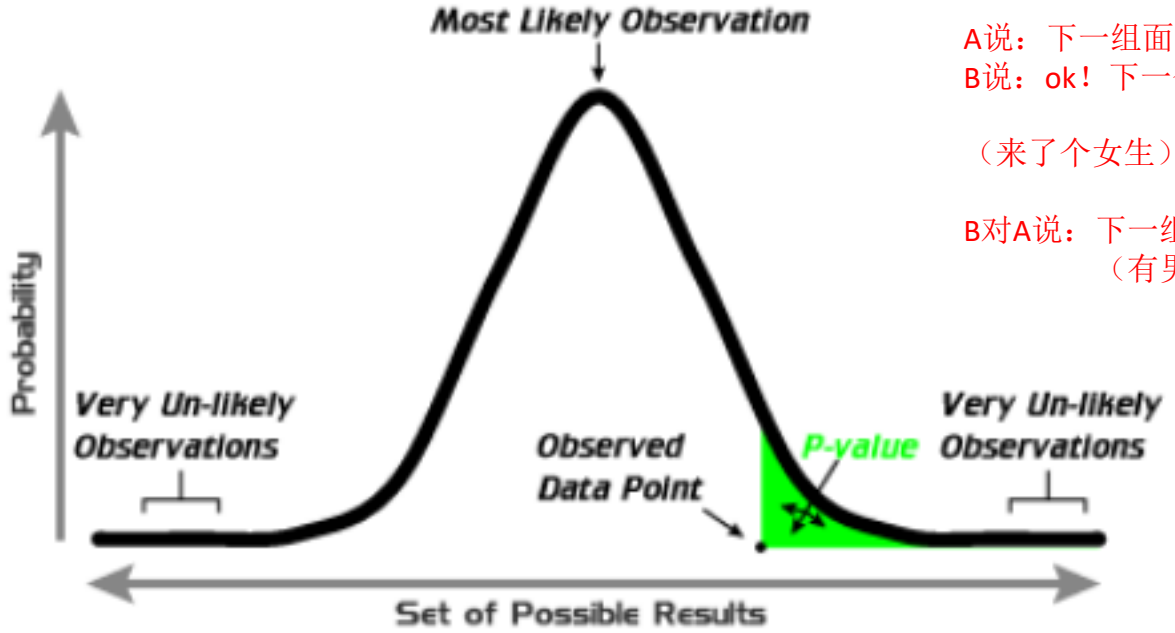
- 如果想检验P组和C组是否有差异，先假设两组无差异（H0）
- 构造统计量发现处于H0下参数分布的小概率处
- 根据“小概率事件不可能发生”的原则，认为样本不是来自于总体
- 拒绝H0，拒绝H0冒了多大风险？（p值）



A p-value (shaded green area) is the probability of an observed (or more extreme) result arising by chance

A说：下一组面试的人都是男生！
B说：ok！下一个请进来！

（来了个女生）

B对A说：下一组肯定不都是男生！
（有男有女）

一类错误和二类错误

降低了Type I error, 少报假警，
减少了p值，就提高了specificity = 1 - α

降低了Type II error, 报警都正确了，
就提高了sensitivity = 1 – β = power

- 一类错误：报假警，false positive, 本来没有激活但是你判断出来是激活的。p值就是一类错误概率(α)。
- 二类错误：不报警，false negative, 本来有激活可是你没有检测出来(β)。

两类错误是此消彼长的！
当然，理想情况是：既要降低Type I error, 又要降低Type II error.

		actual value: *p*	actual value: *n*	total
prediction outcome	*p'*	True Positive	False Positive	P'
	n'	False Negative	True Negative	N'
total		P	N	

单样本t检验

- 假设要求：样本来自正态总体且两两独立
- 比较样本均值是否显著不等于0

样本均值　　零假设总体均值（一般是0）

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}}$$

样本std　　样本数量

H0下，这个构造出的统计量服从df = n-1 的t分布

单样本t检验：对r的检验

相关系数 $r = \dfrac{\frac{1}{n-1}\sum (x_i - \bar{x})(y_i - \bar{y})}{s_x s_y}$

r to t转化公式 $t = r\sqrt{\dfrac{n-2}{1-r^2}}$

自由度 = n-2

双样本t检验

- 检验两组独立样本均值是否有差异。
- 两组必须独立。

组1样本均值

组2样本均值

一般为0

$$t = \frac{\overline{x}_1 - \overline{x}_2 - \mu_0}{\sqrt{2s_p^2/n}}$$

两组std相等的情况，s为两组共同的std，n为每组各自的样本数

$$t = \frac{\overline{x}_1 - \overline{x}_2 - \mu_0}{\sqrt{s_p^2/n_1 + s_p^2/n_2}} \qquad t = \frac{\overline{x}_1 - \overline{x}_2 - \mu_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$$

H0下，这个构造出的统计量服从df = 2n-2 的t分布

配对t检验

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- 单样本t检验的扩展，但是检测对象变为“两组配对样本观测值的差”
- 比较差值和0是否有显著差异。

配对样本观测值差异的平均数

$$t = \frac{\overline{d} - \mu_0}{s_d / \sqrt{n}}$$

配对样本观测值差异的标准差

H0下，这个构造出的统计量服从df = n-1 的t分布

F检验

- 假设各组数据均服从正态分布，相同标准差
- 可以用于三组或这多组之间的均值比较

对于one-way ANOVA, F检验可以写成：

$$F = \frac{\text{explained variance}}{\text{unexplained variance}}, \qquad F = \frac{\text{between-group variability}}{\text{within-group variability}}.$$

$$F = \frac{\sum_i n_i(\bar{Y}_{i\cdot} - \bar{Y})^2/(K-1)}{\sum_{ij}(Y_{ij} - \bar{Y}_{i\cdot})^2/(N-K)},$$

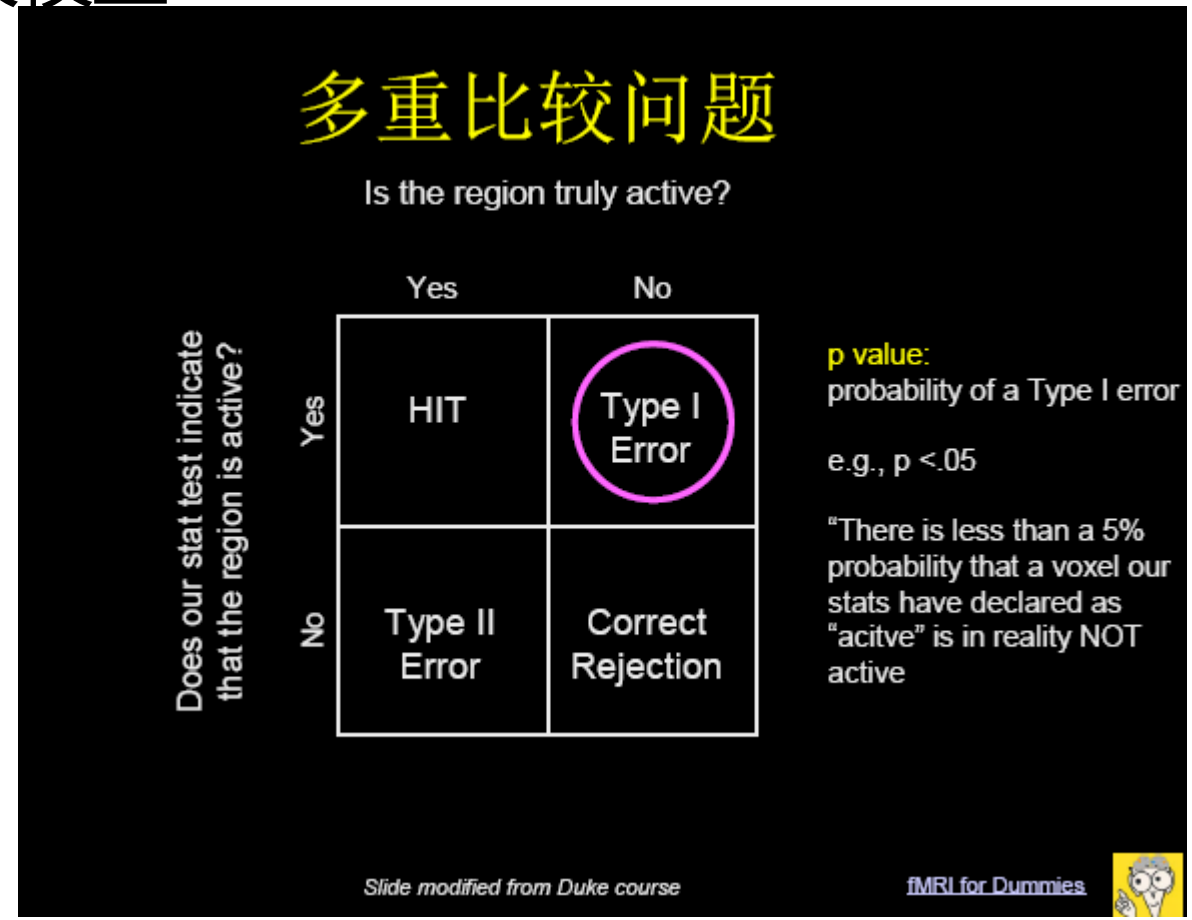
每组人数　组数　每组均值　总样本数

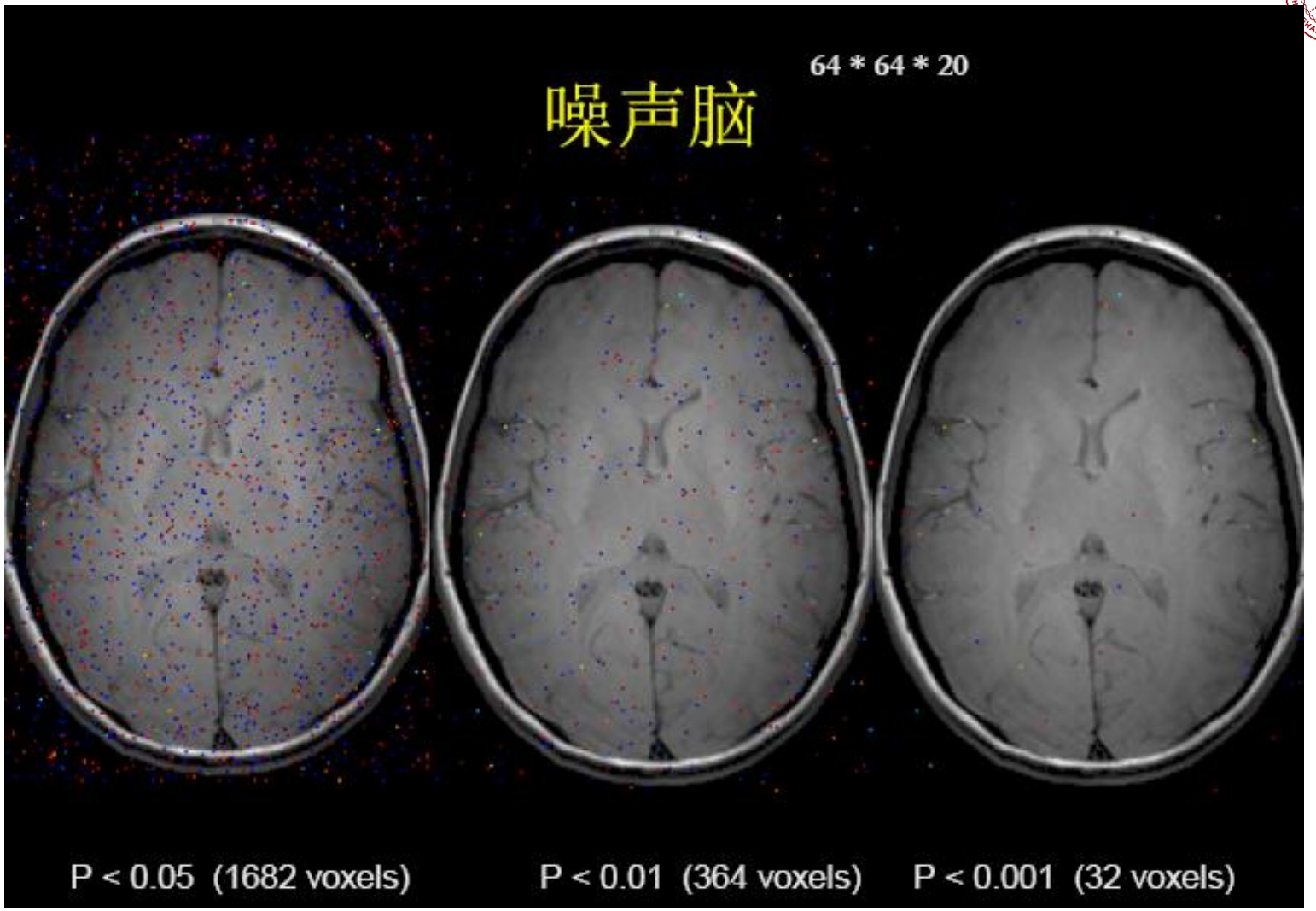
H0下，这个构造出的统计量服从df = K-1, N-K 的F分布

多重比较校正

https://egap.org/resource/10-things-to-know-about-multiple-comparisons/

- 何为多重比较校正




64 * 64 * 20
噪声脑
P < 0.05 (1682 voxels)
P < 0.01 (364 voxels)
P < 0.001 (32 voxels)

- 什么时候需要多重比较校正？
  - 逐体素(voxel-by-voxel)进行假设检验时；
  - 多个ROI逐一进行假设检验时；
  - 先做完ANOVA，然后做Post-hoc analysis时；
  - 有多个生理参数/量表得分需要分别做相关时；
  - ……

- 怎样避免？
  - 通过先验知识，降低voxel/ROI个数

多重比较校正（Bonferroni）

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一人犯罪，全家抄斩！
极其严格，近乎完美！

虽然SPM根据随机场理论做了相应的改进，
但是仍然没有任何理由的极其严格！。

one voxel Type I error p = ?

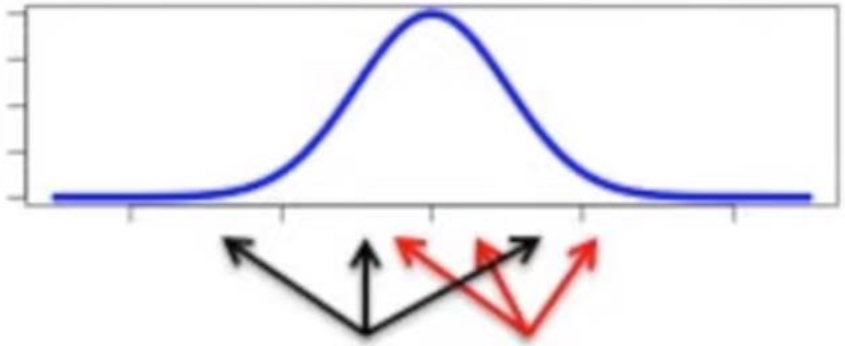
- number of voxels： N= 50,000
- overall correct detection = (1-p) (1-p)… (1-p) = $(1-p)^N$
- overall Type I error = **1 -** $(1-p)^N$ =~ Np
- Desired overall Type I error: Np = .05
- Required one voxel Type I error p = .05 / 50,000 = .000001

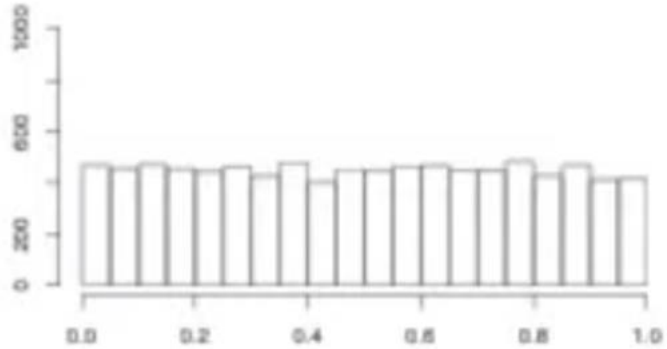
多重比较校正（FDR/AlphaSim）

- FDR（控制“激活”的体素中，只有少部分为假激活，目标合理！）。FDR的缺点：不同数据可能有不同的阈值。
- AlphaSim（基于蒙特卡洛模拟：模拟一个噪声脑，经过同样的处理步骤，得到所谓“激活区”的可能。模拟超过10000次。全噪声情况下，“激活”团块越大，越不可能实现）AlphaSim不适合的情况是：$p<0.001$时，仍然有大面积的激活（连成一个cluster了！）。

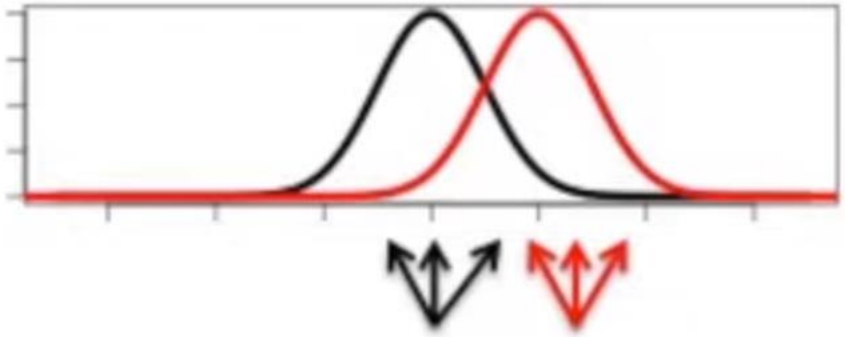
A better way to control the FWER is simulation

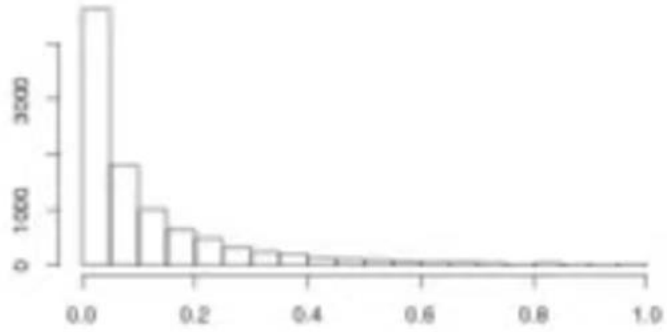
When samples come from the same distribution, the p-values are uniformly distributed…





When samples come from different distributions, the p-values are heavily skewed and closer to 0…





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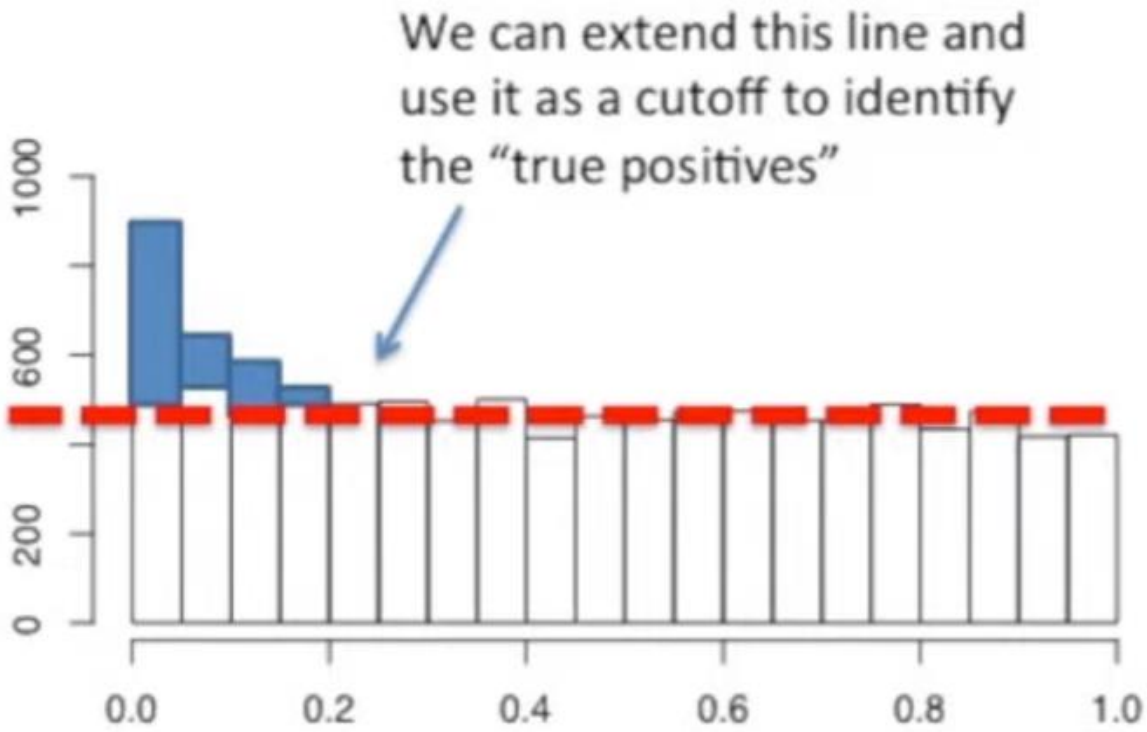
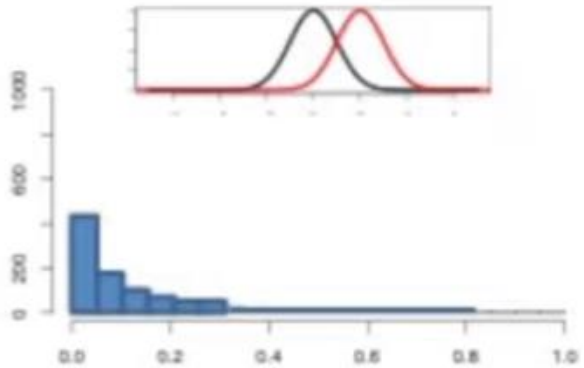

We can extend this line and use it as a cutoff to identify the “true positives”
1000
600
200
0
0.0
0.2
0.4
0.6
0.8
1.0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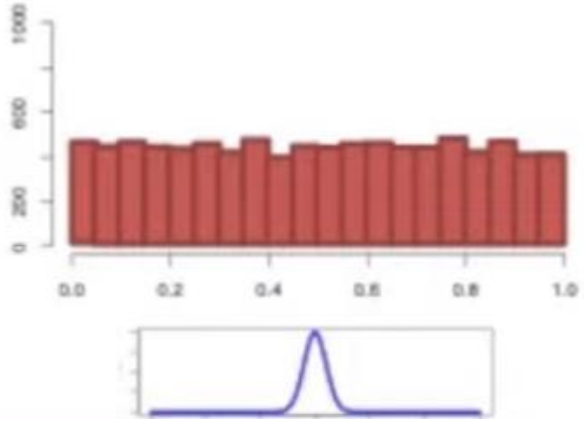
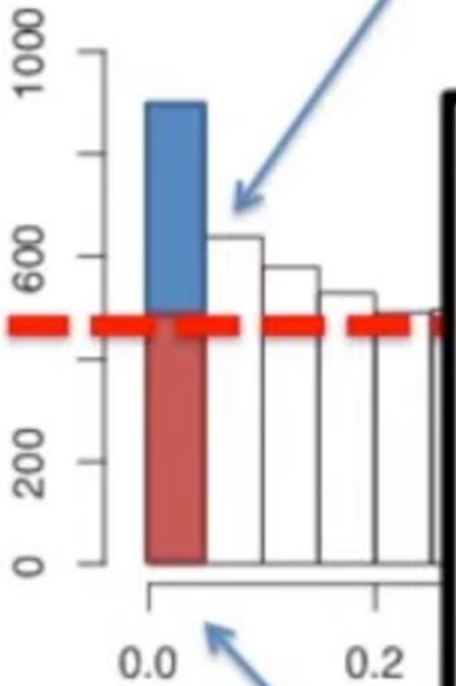

1000
600
200
0
0.0
0.2
0.4
0.6
0.8
1.0

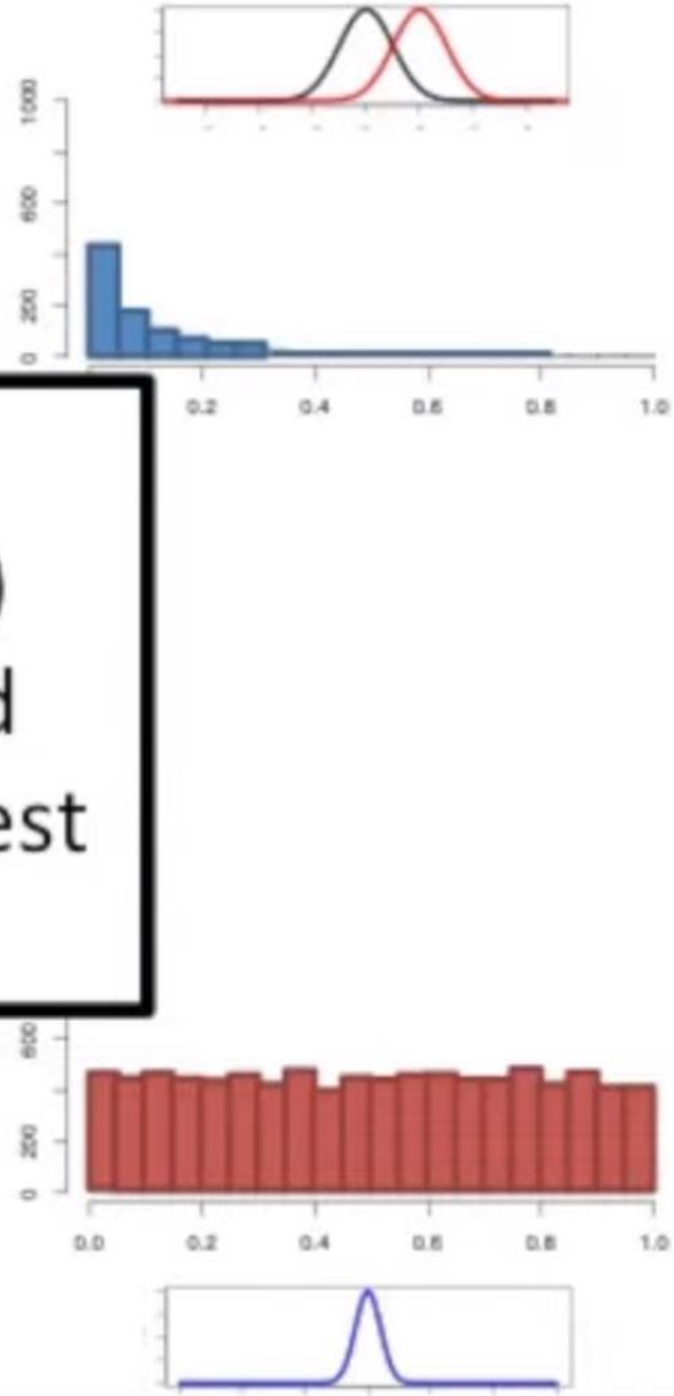
Press Esc to exit full screen

Roughly 450 p-values < 0.05 are above the dotted line.



One way to isolate the true positives (genes affected by the drug) from the false positives would be to only consider the smallest 450 p-values.

Roughly 450 p-values < 0.05 are below the dotted line.



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脑科学与脑疾病
第七节课：研究人类大脑结构的sMRI及其基本分析技术

https://www.youtube.com/watch?v=TT8FJUq-QKo

Voxel Based Morphometry (VBM vs. Freesurfer)

婴幼儿结构MRI数据分析技术研发

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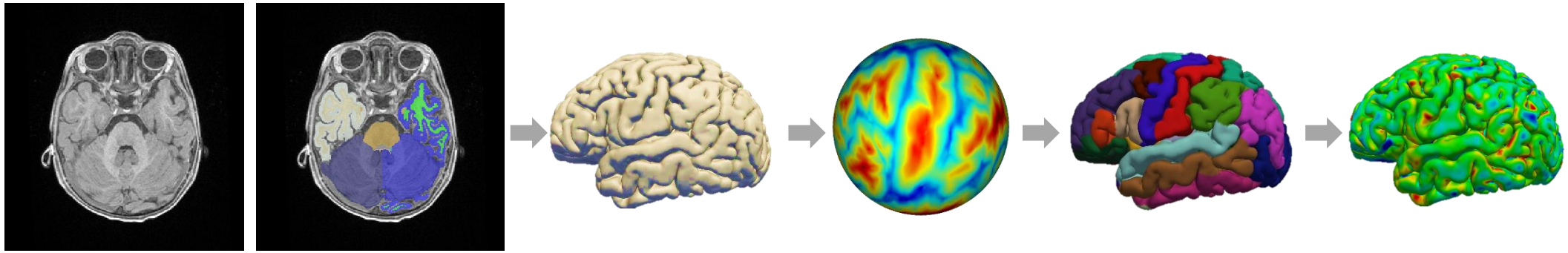
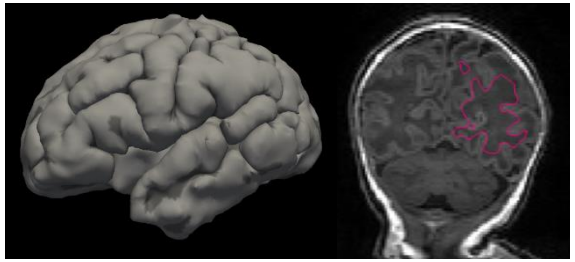
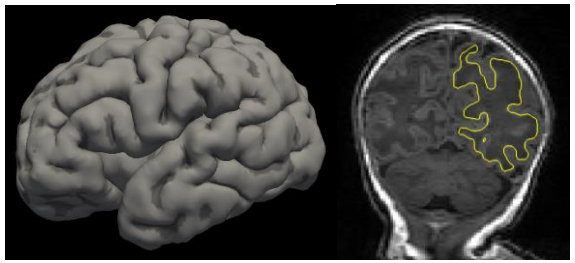
sMRI 处理流程

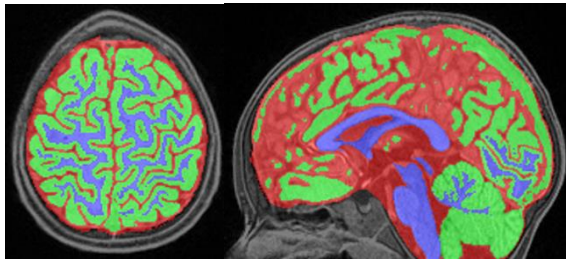




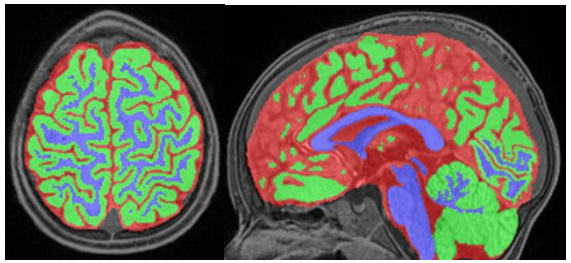
Infant FreeSurfer



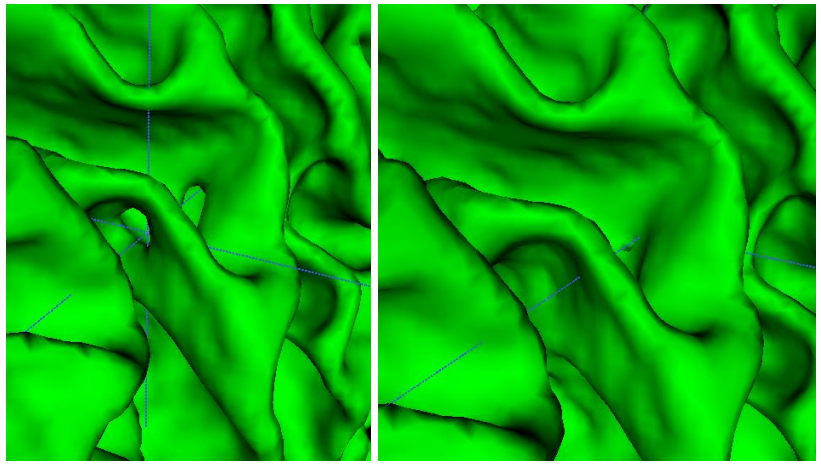
优于国际标准的处理流程



组织分割质量检验发现部分颅骨未剥离



优化分割算法提升组织分割精度

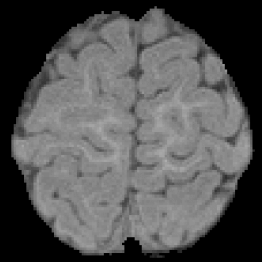


皮层质量检验及拓扑矫正

Infant Longitudinal Brain Structural MRI (T1-weighted)

3 month

6 month



9 month

12 month

24 month

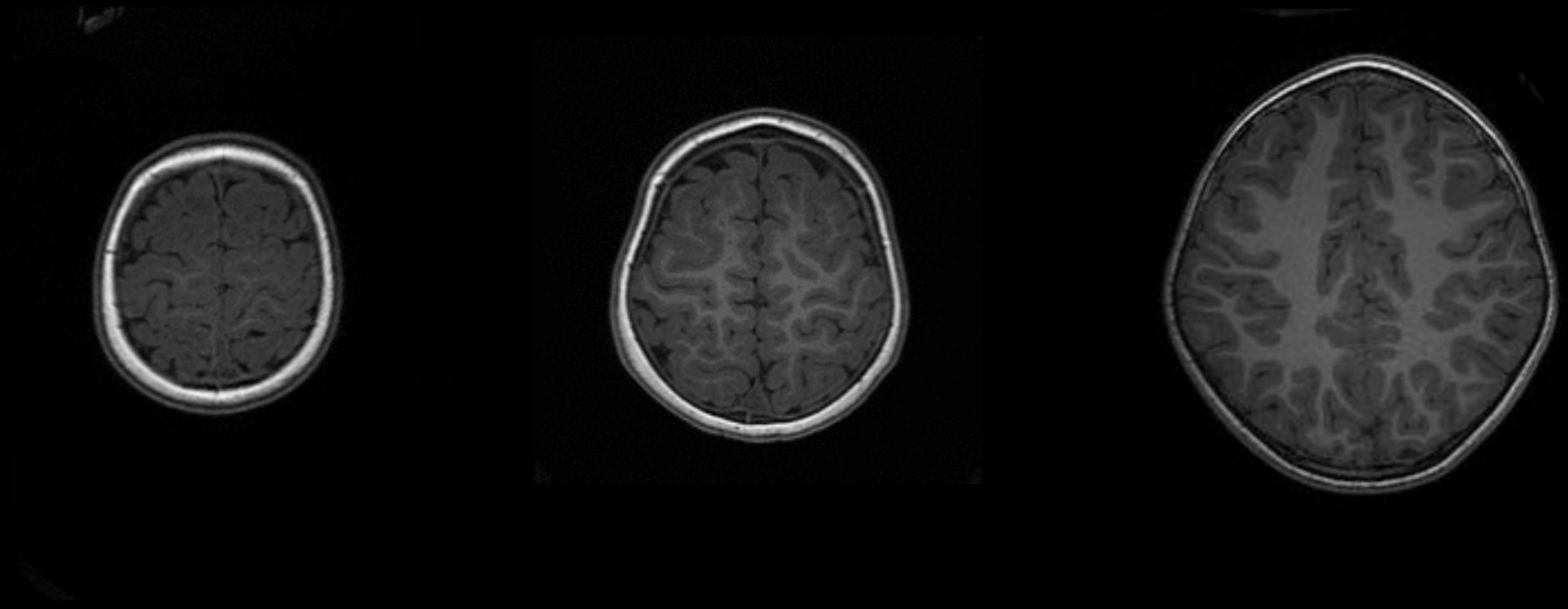
Research Lab IDEA

6个月

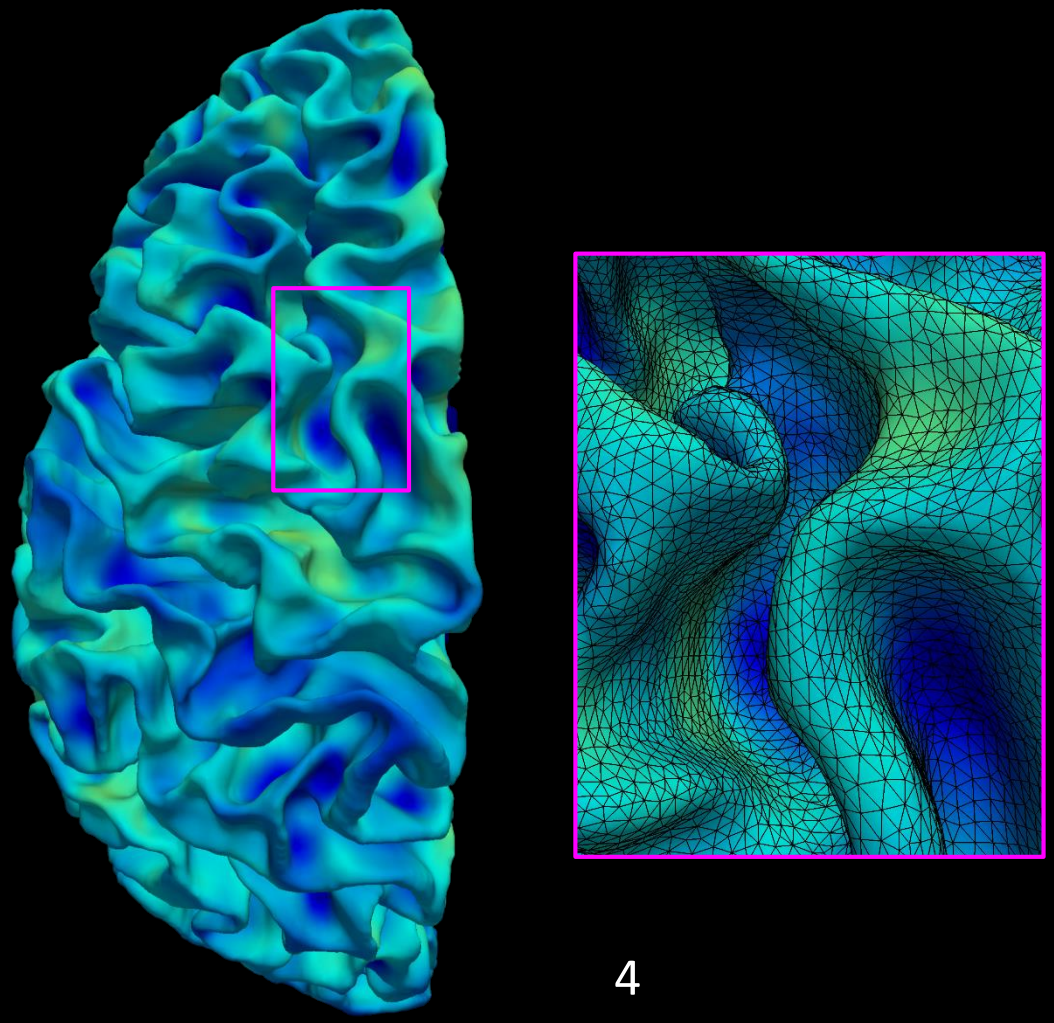
12个月

4岁

T1w

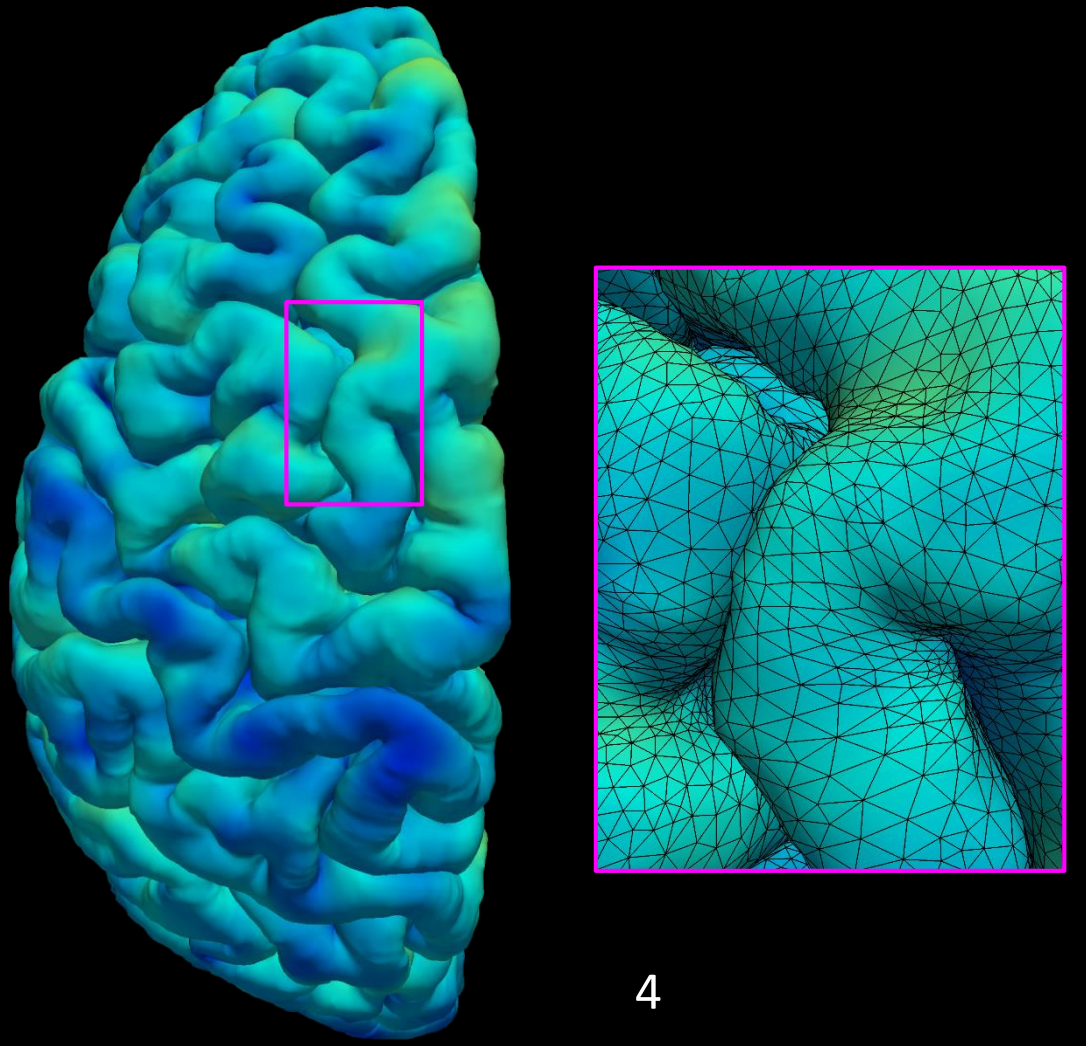


Cortical Surface Reconstruction



4

1

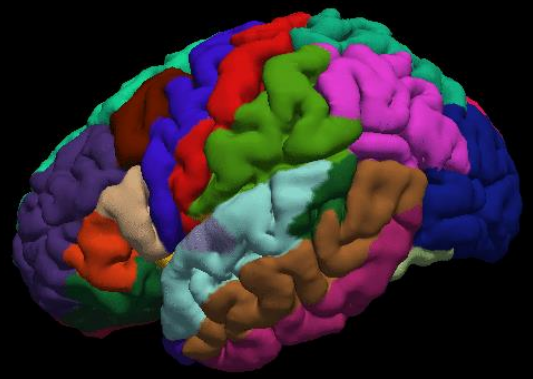


4

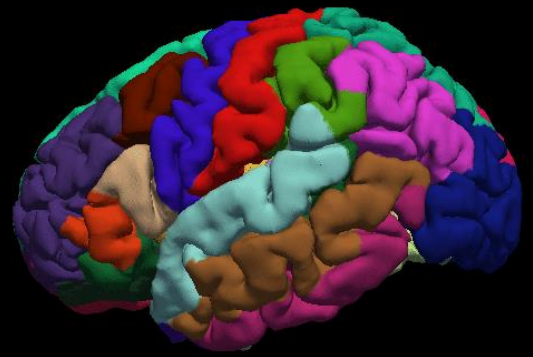
1

Li et al., Neuroimage 2014

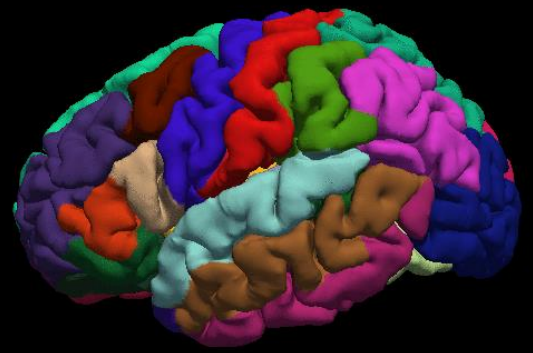
Infant Brain Parcellations (Brain Region Segmentation)



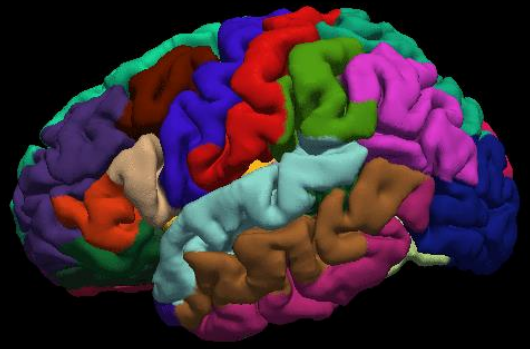
3 month



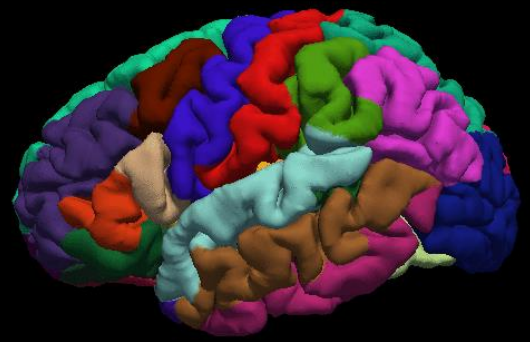
6 month



9 month



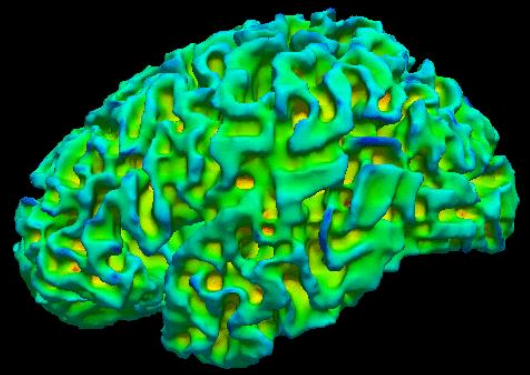
12 month



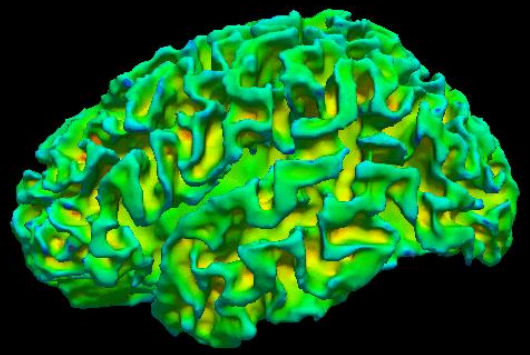
24 month

Research Lab IDEA

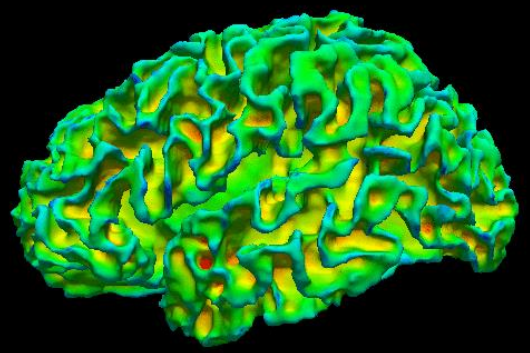
Infant Brain Geometric Measures (Curvature)



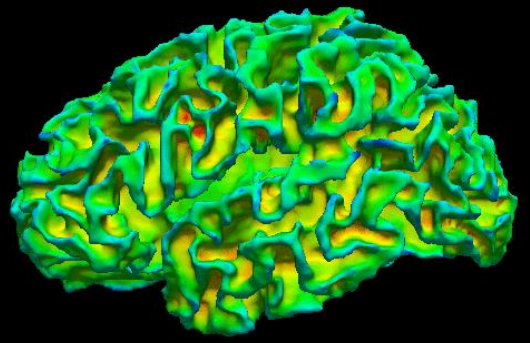
3 month



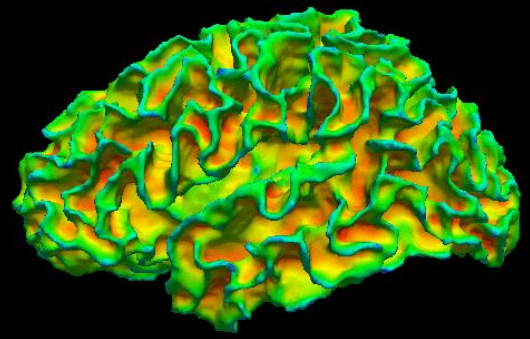
6 month



9 month

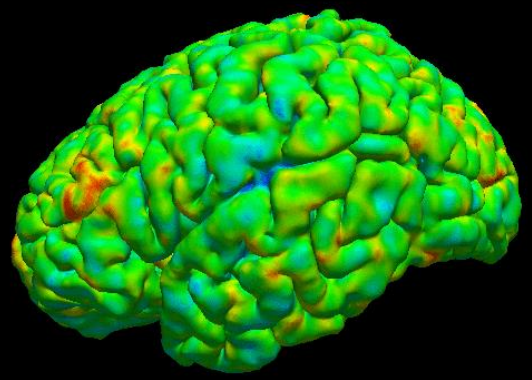


12 month

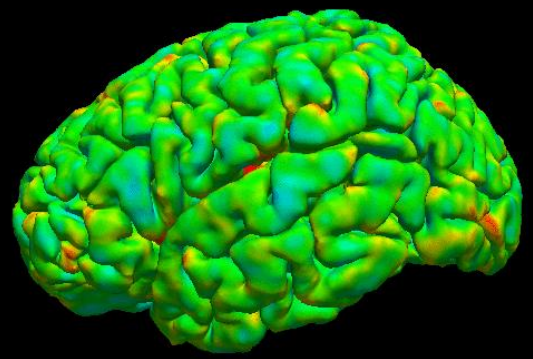


24 month

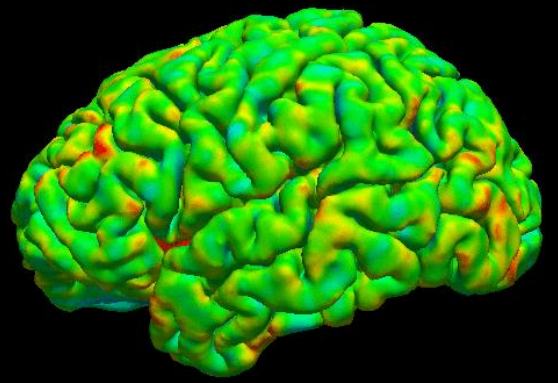
Infant Brain Geometric Measures (Cortical Thickness)



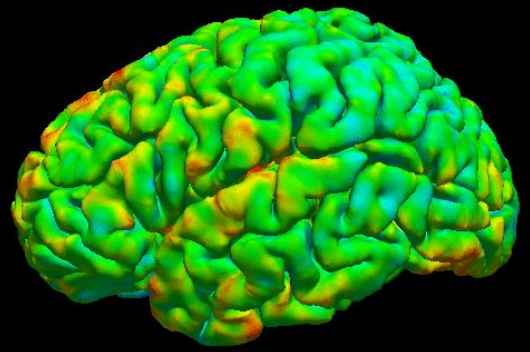
3 month



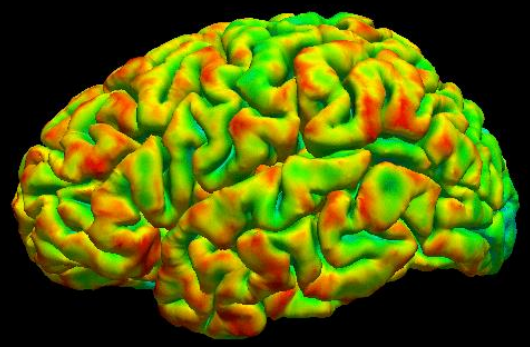
6 month



9 month

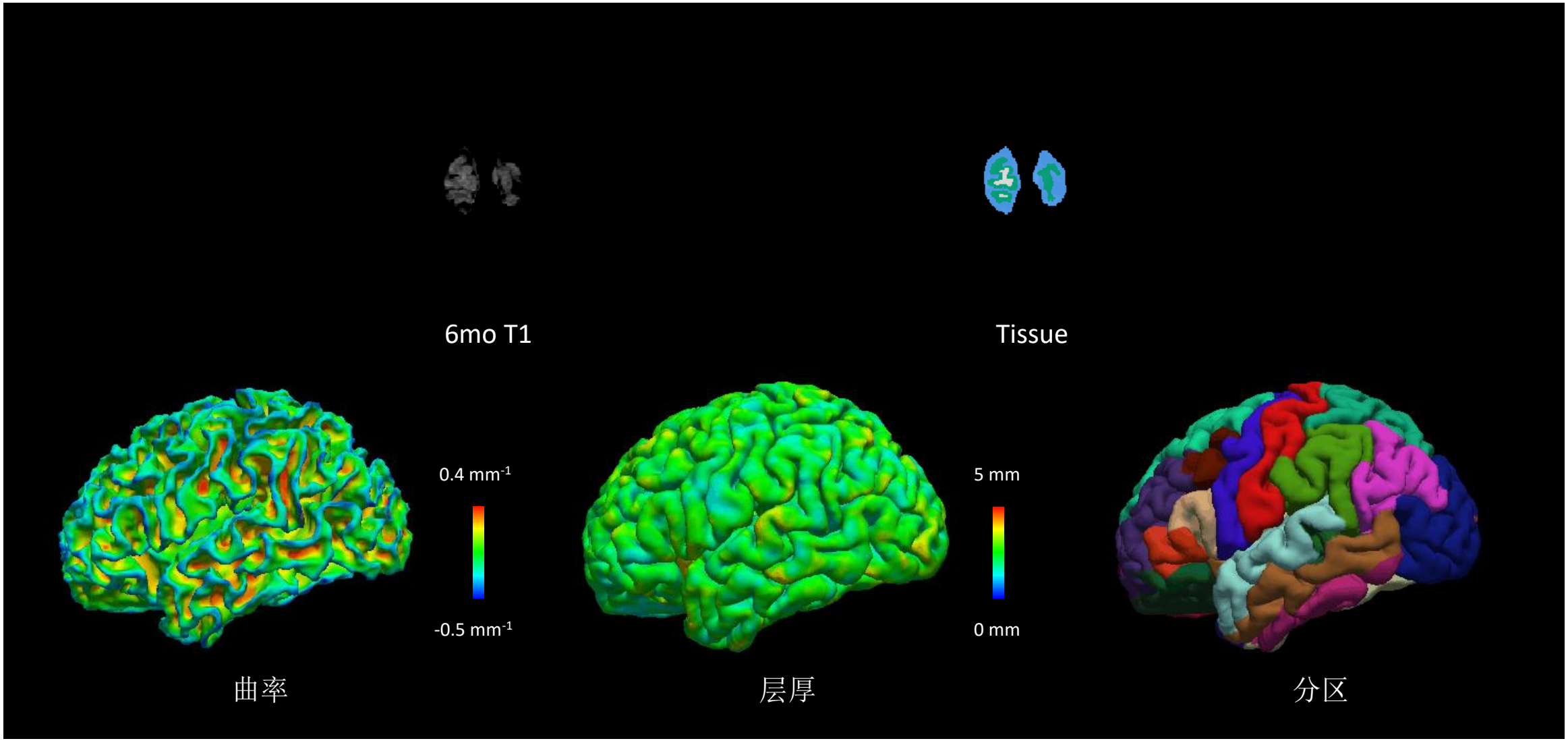


12 month

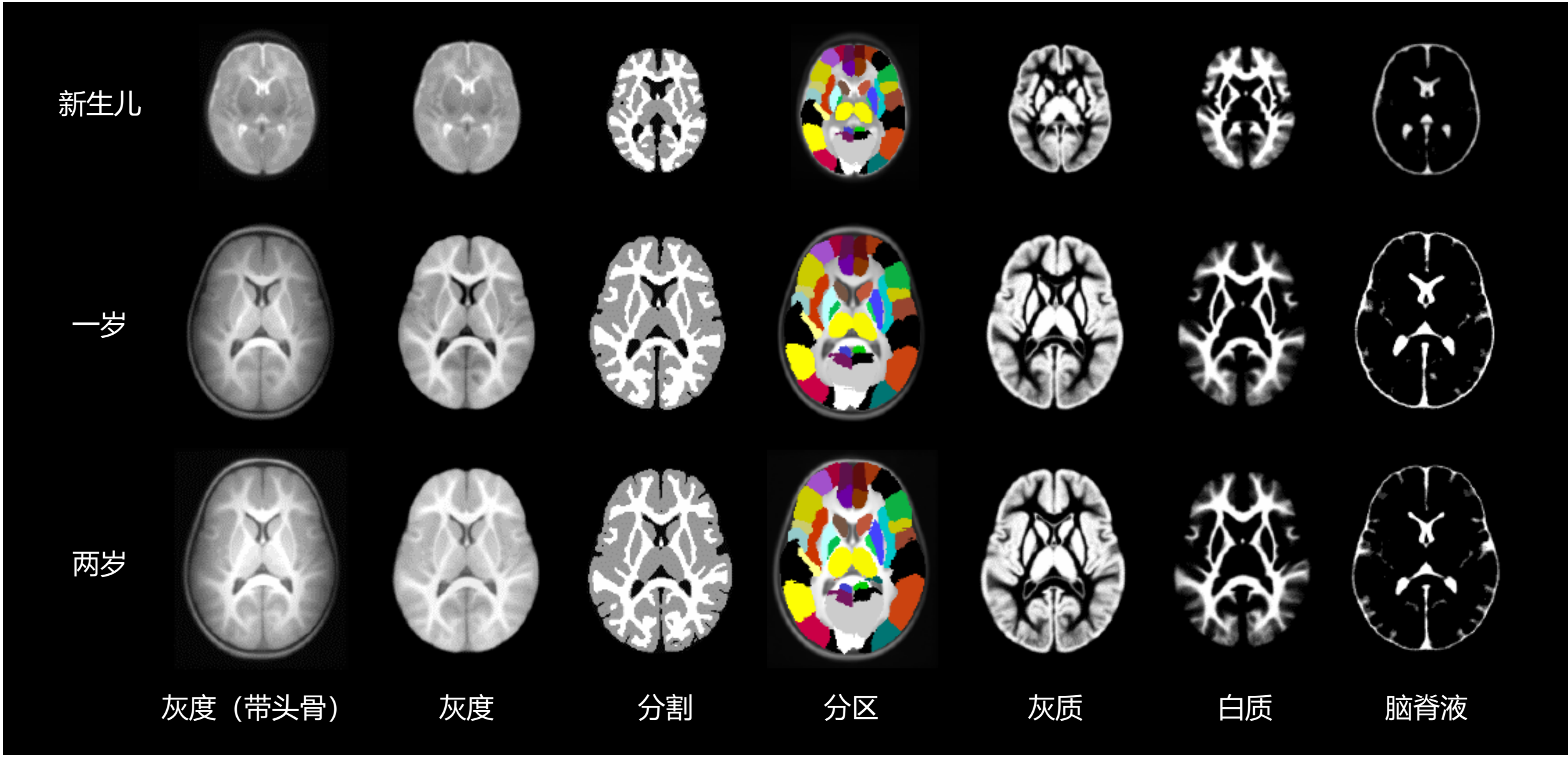


24 month

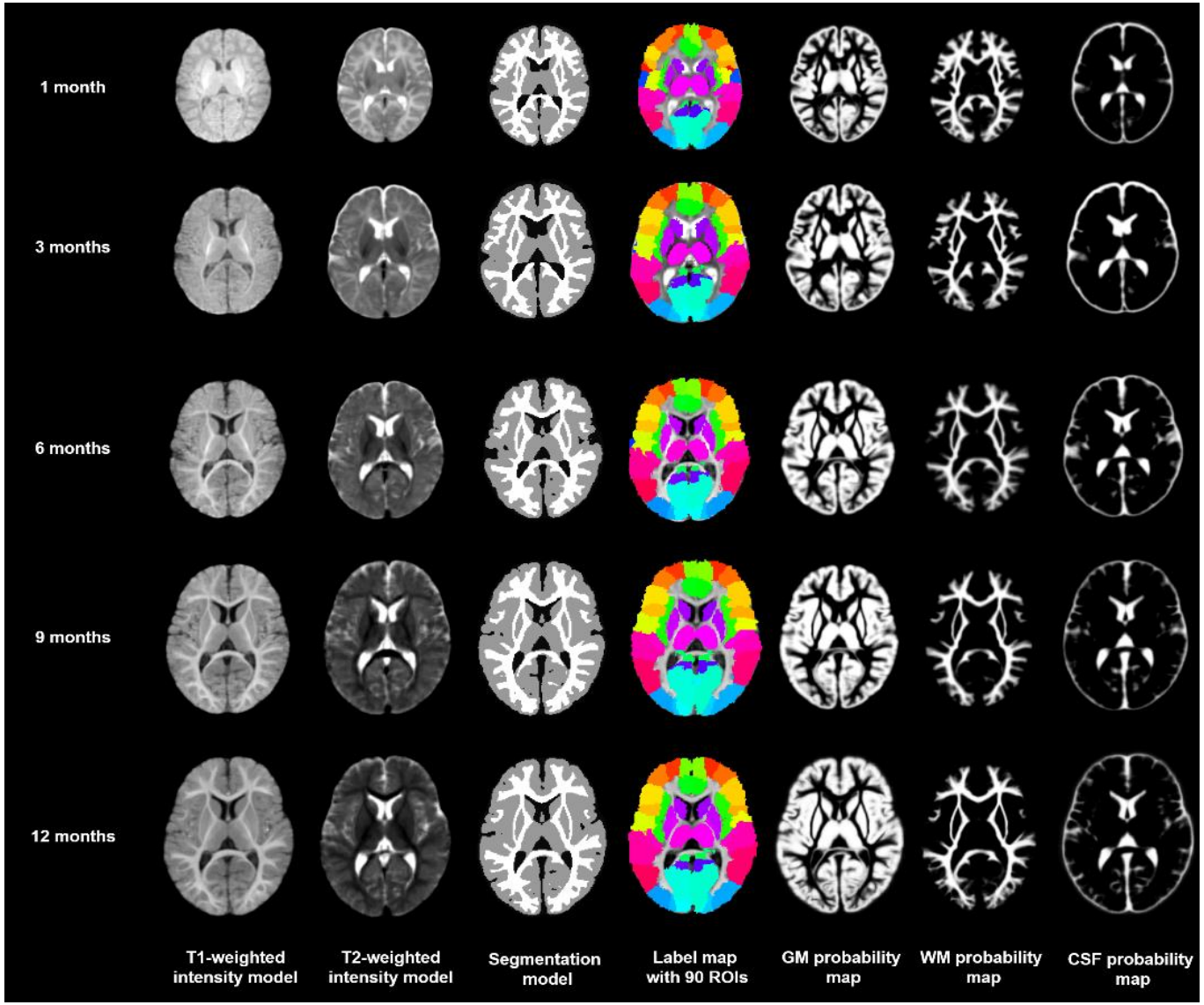
婴幼儿脑影像分析结果



婴幼儿脑结构图谱（0-1-2岁）



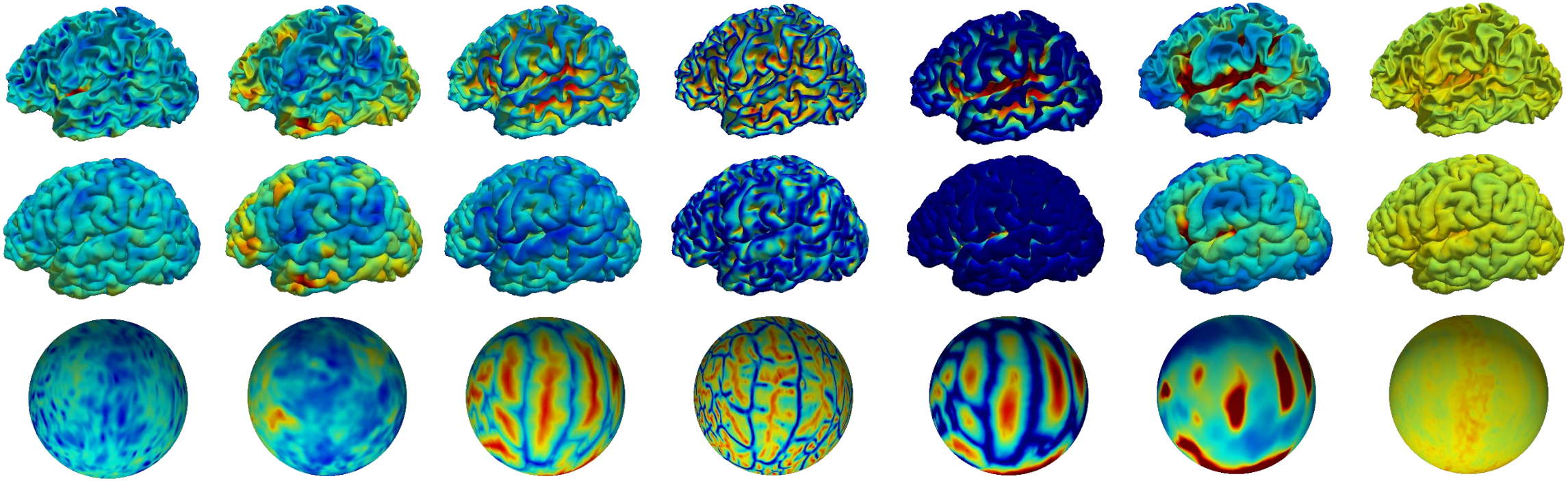
婴幼儿脑结构图谱（0-1岁）



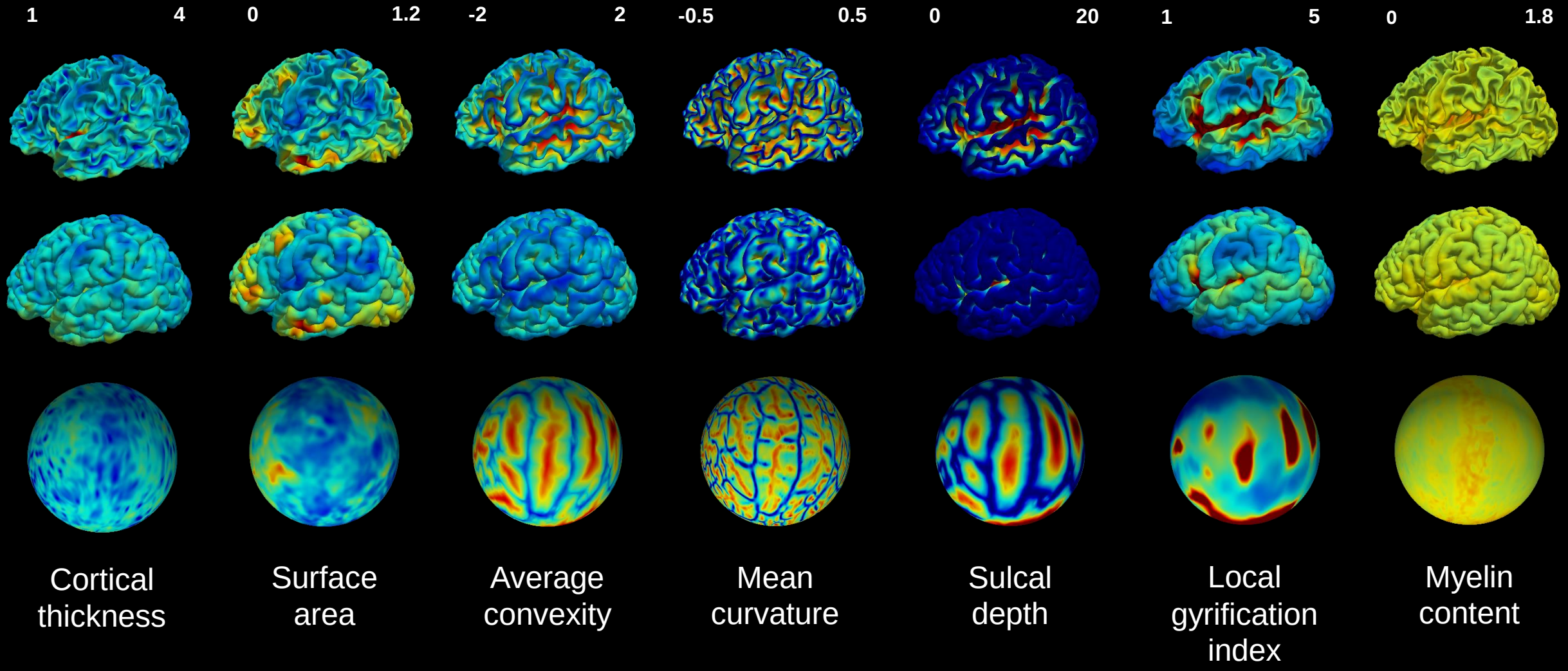
脑结构图谱

上海科技大学
ShanghaiTech University

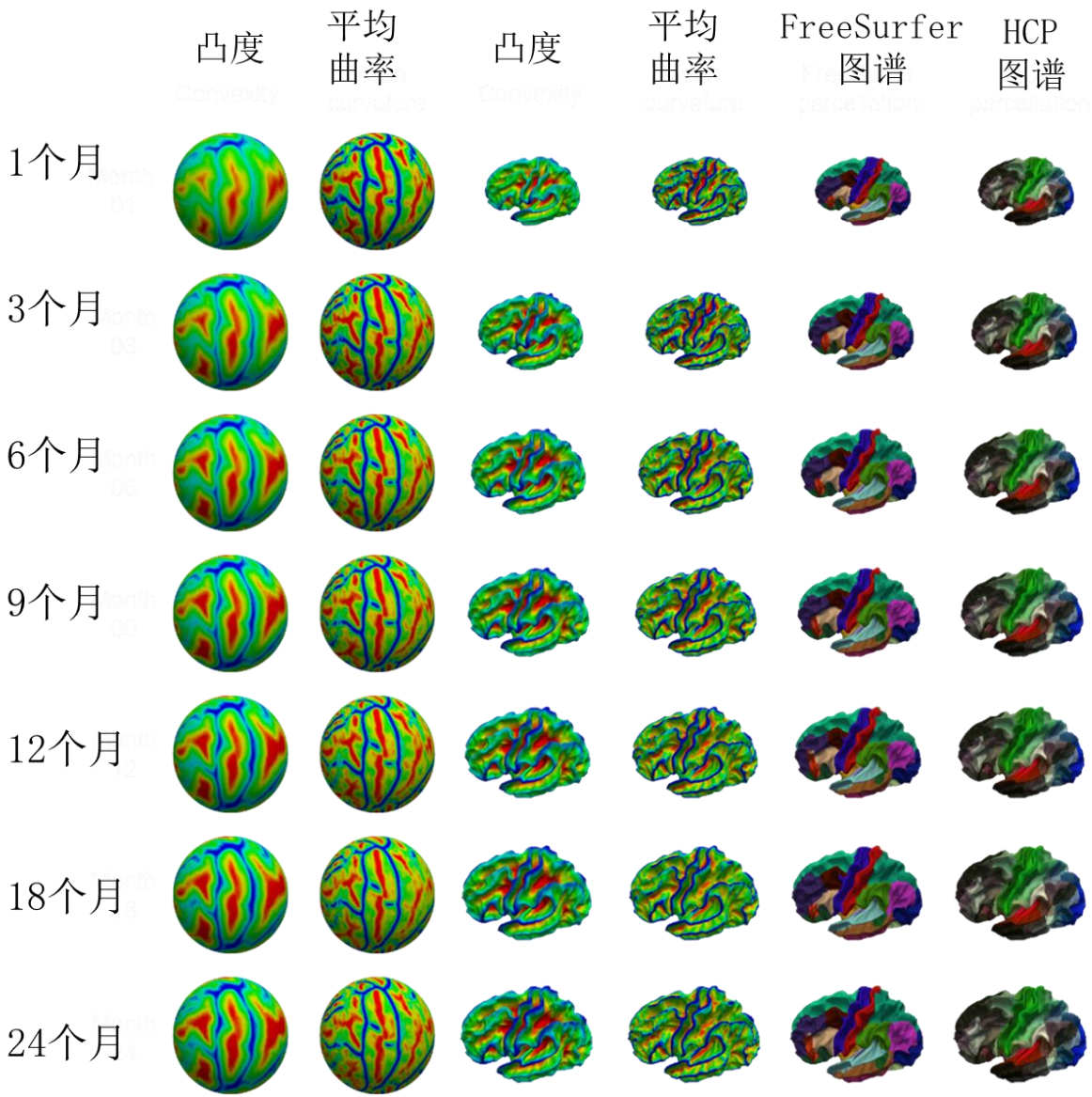
皮层厚度　皮层面积　平均凸度　平均曲率　脑沟深度　局部褶皱指数　髓鞘主体



Resampling and Computing Cortical Properties

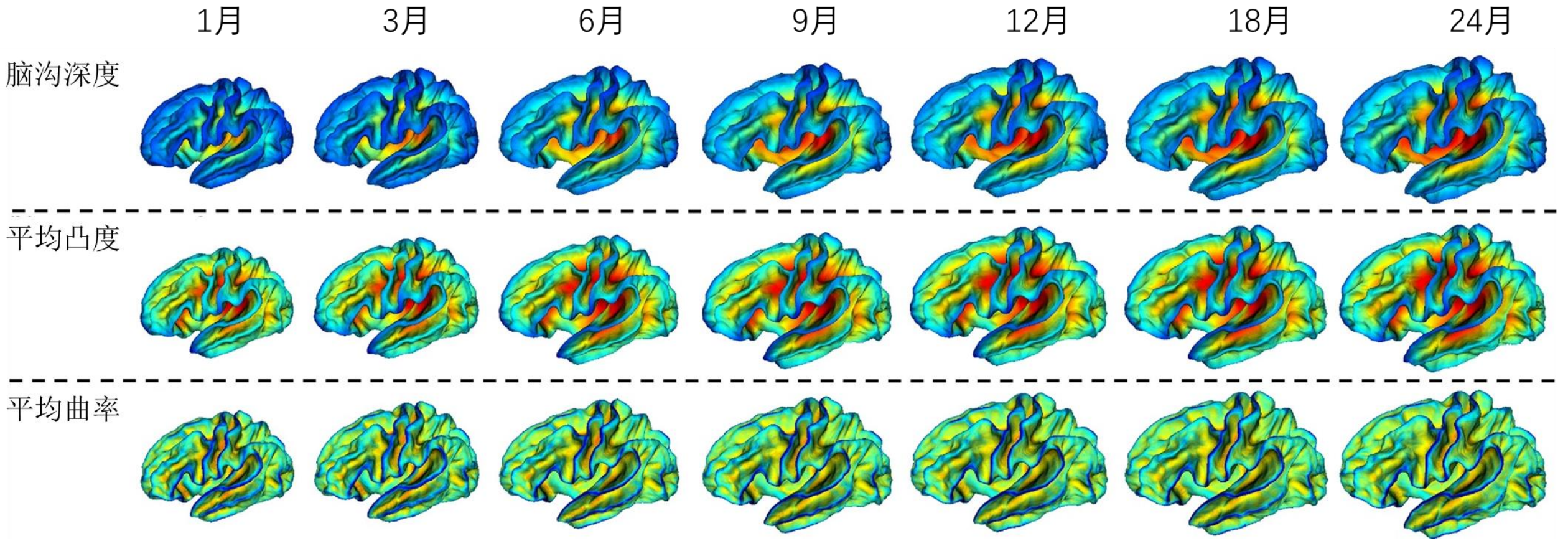


婴幼儿脑皮层图谱（0-2岁）



婴幼儿脑皮层图谱（0-2岁）

上海科技大学
ShanghaiTech University



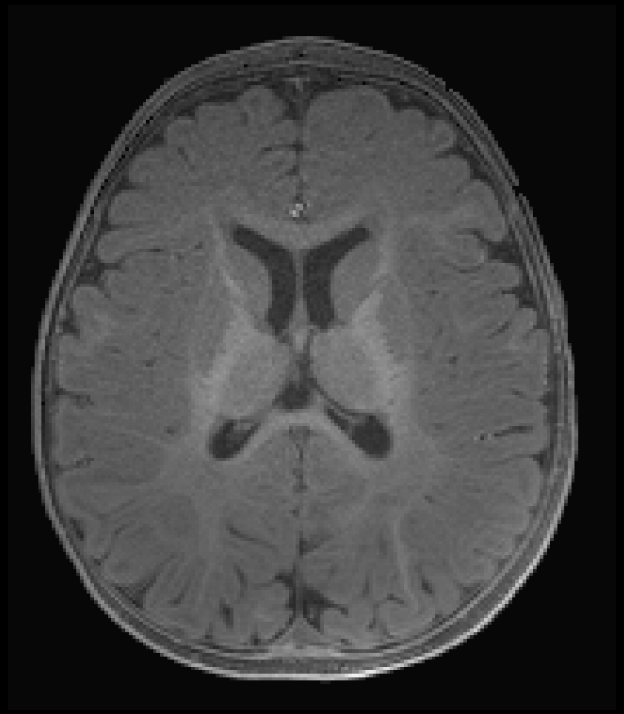
Demo

For a 6-month subject from BCP

Contact: li_wang@med.unc.edu

More details: http://www.unc.edu/~liwa

Original T1w MRI



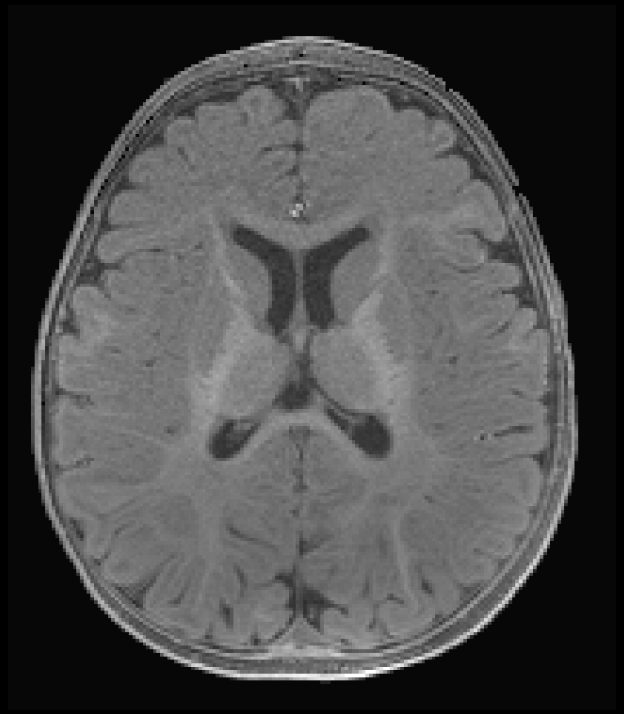
ID: **BCP_132476**; **186** days; **0.8*0.8*0.8** mm^3

Images were from BCP

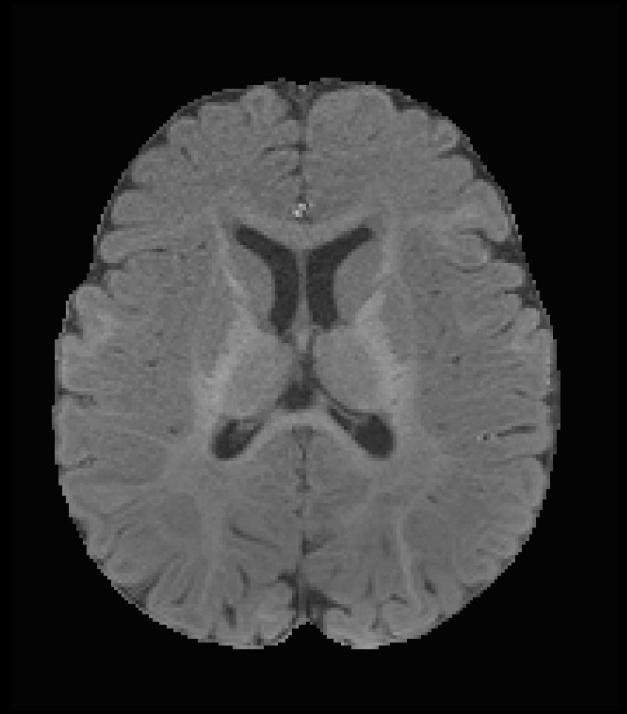
BRIC BIOMEDICAL RESEARCH IMAGING CENTER

Research Lab IDEA

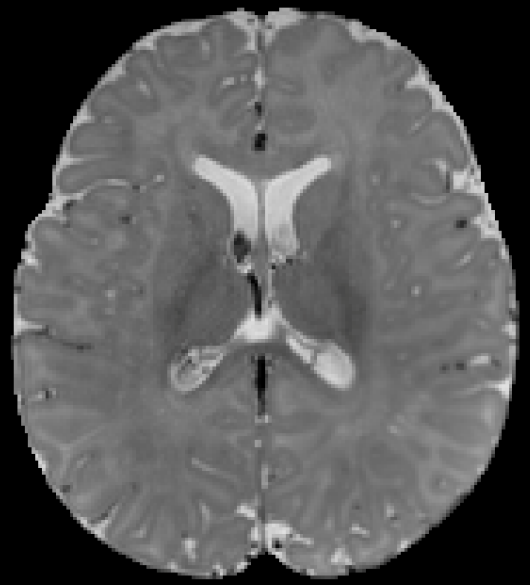
Preprocessing: Inhomogeneity corrected T1w MRI



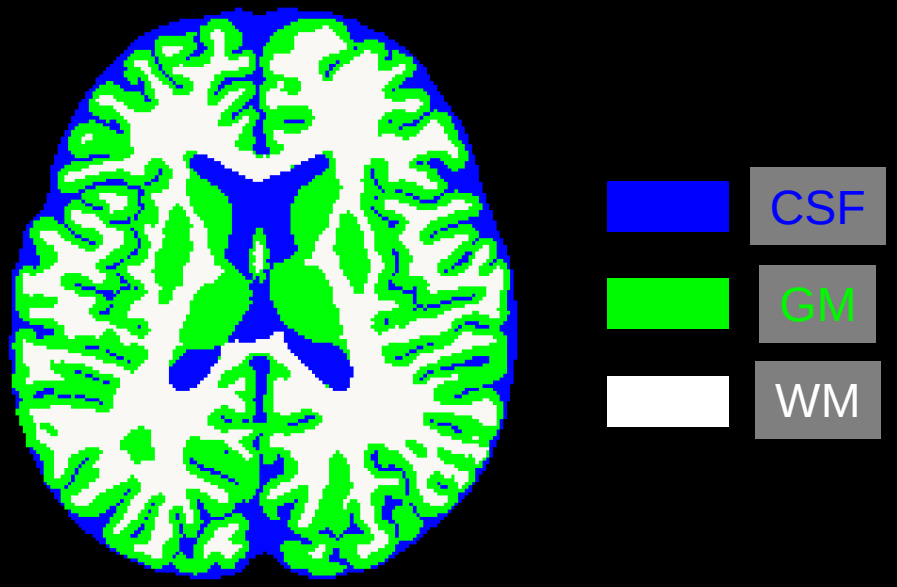
Brain Extraction: Extracted T1w MRI



Brain Extraction: Extracted T2w MRI

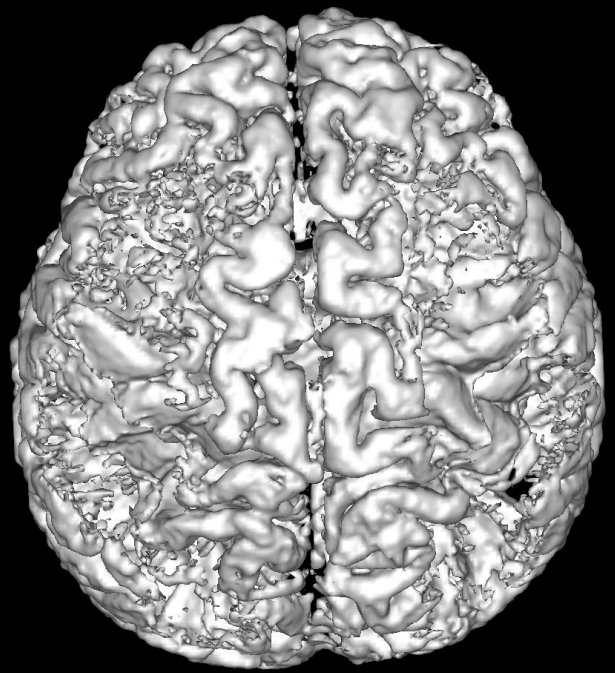


Tissue segmentation

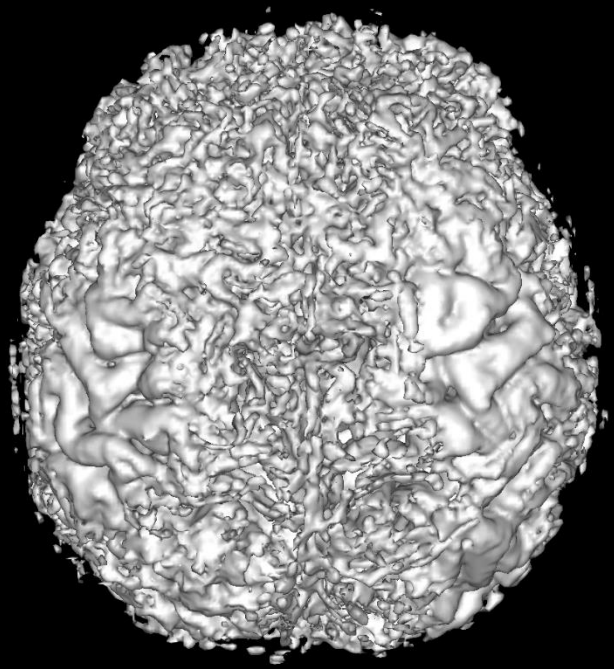


Comparisons

FSL



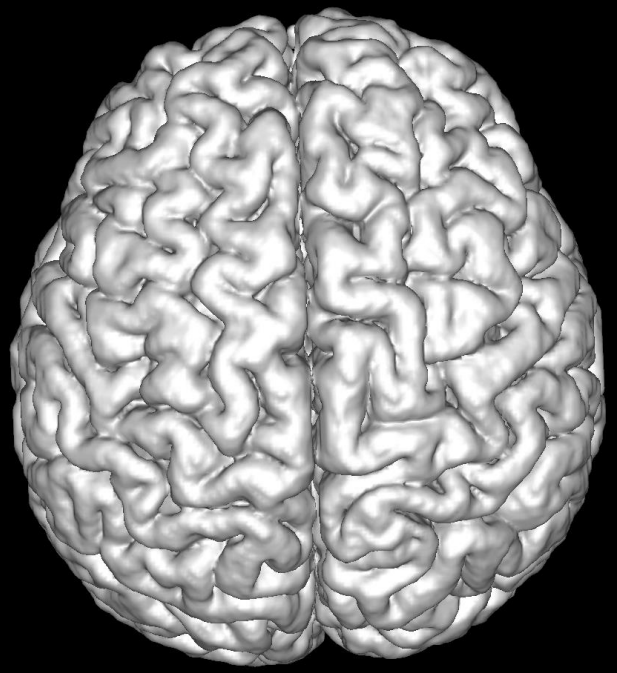
Outer cortical surface



Inner cortical surface

Comparisons

iBEAT

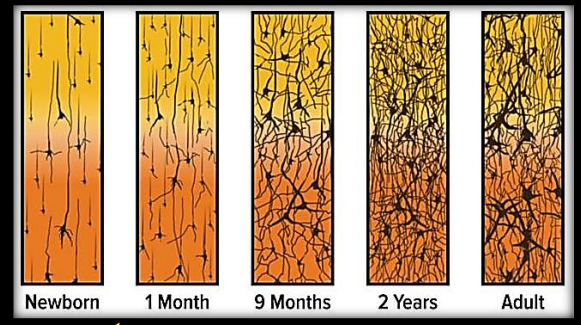


Outer cortical surface

Inner cortical surface

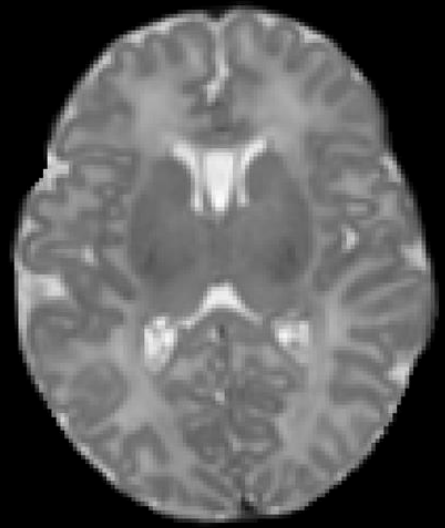
Background

- **Early Brain Development**
  - Dynamic, nonlinear, and region-specific
  - Neurodevelopmental disorders emerging
  - Poor imaging contrast and dynamic appearances



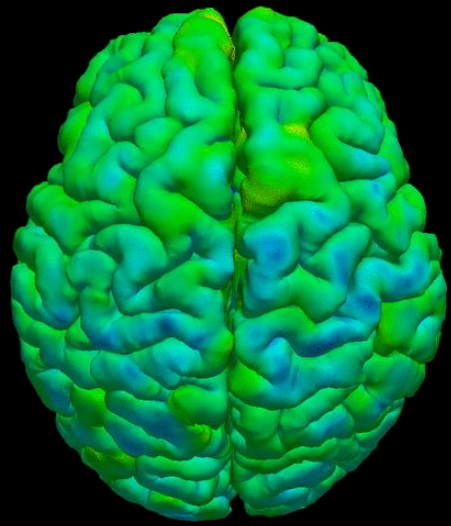
00 Months

T1 images



00 Months

T2 images



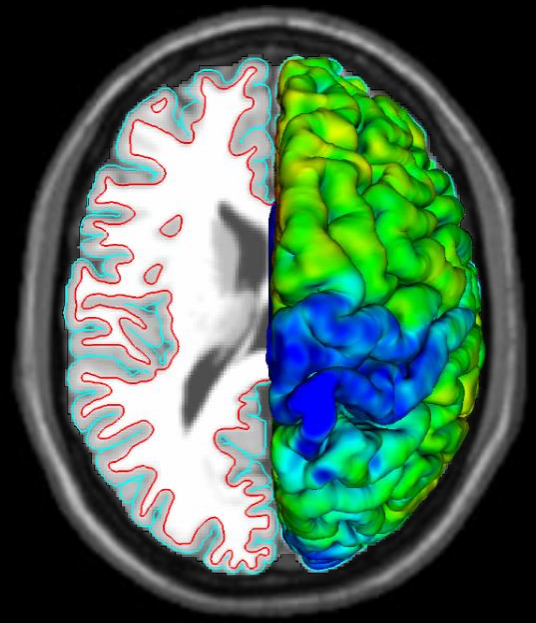
00 Months

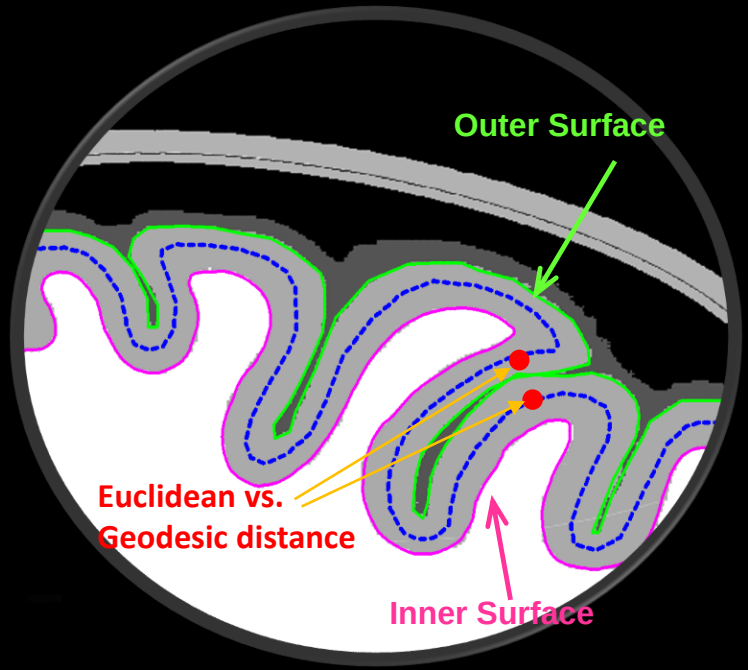
Cortical surfaces

Background

- Volume-based analysis
  - Treat brain as an image;
  - Best suitable for subcortex;

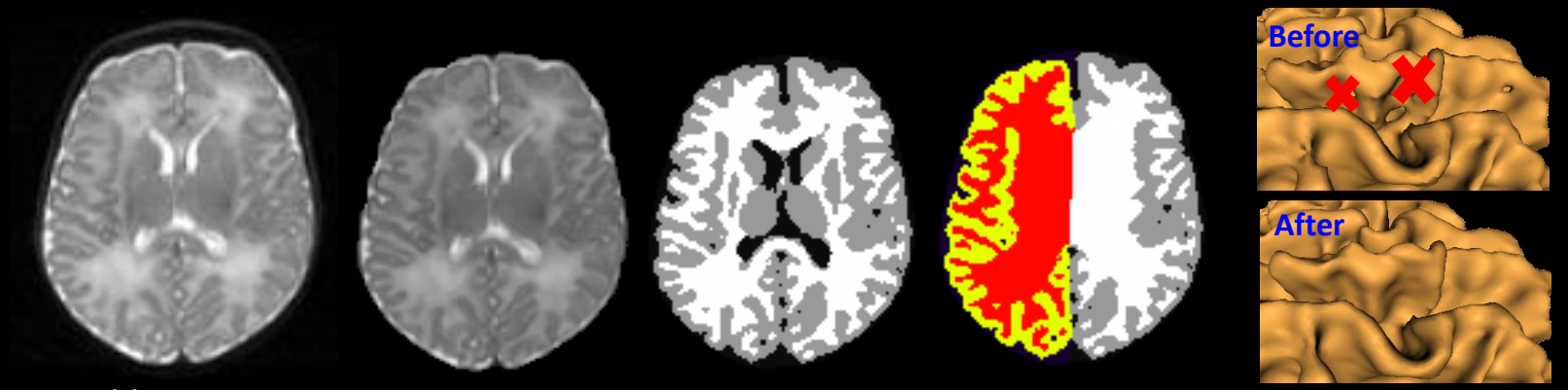
- Surface-based analysis
  - Respect cortex topology;
  - Facilitate spatial smoothing, alignment, mapping, multimodal integration and visualization;
  - Derive biologically-distinct cortical attributes:
    - cortical thickness, surface area, cortical folding, sulcal depth, myelination, etc;
  - Best suitable for cerebral cortex as shown in Human Connectome Project (HCP)





Our Infant-dedicated Pipeline

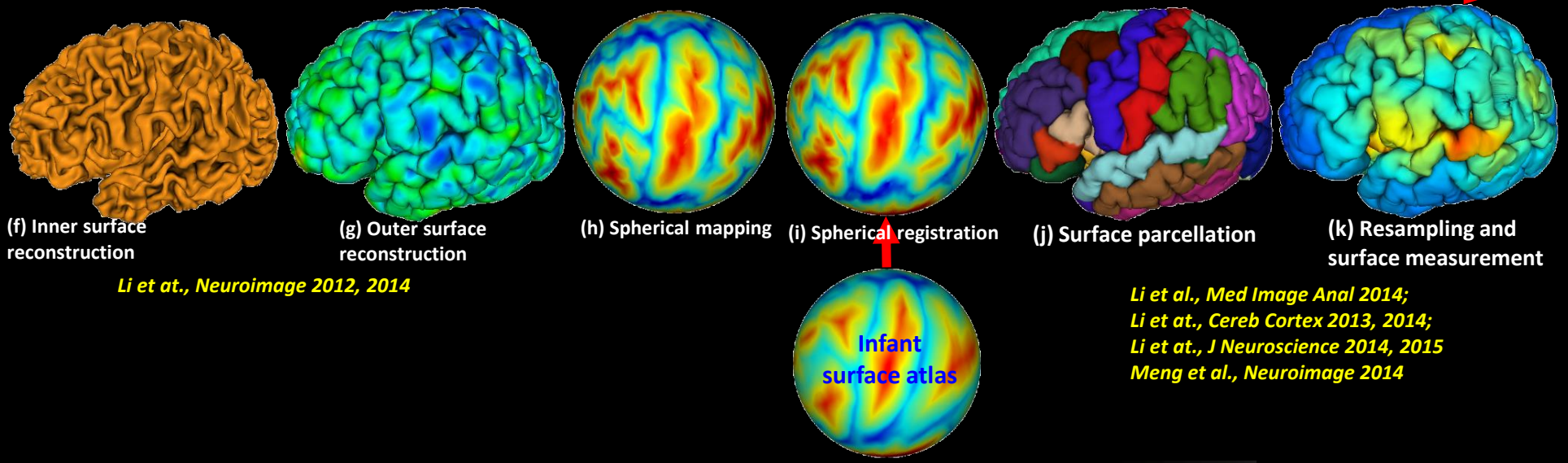
- Contact: gang_li@med.unc.edu
- More details: http://www.unc.edu/~gangl



(a) MR image | (b) Preprocessing | (c) Learning-based tissue segmentation | (d) Hemisphere separation | (e) Learning-based topology correction

Shi et al., Neuroimage 2012;
Wang et al., Neuroimage 2012, 2013, 2014, 2015

Li et at., Neuroimage 2014; Hao et al., MICCAI 2016



(f) Inner surface reconstruction | (g) Outer surface reconstruction | (h) Spherical mapping | (i) Spherical registration | (j) Surface parcellation | (k) Resampling and surface measurement

Li et at., Neuroimage 2012, 2014

Li et al., Med Image Anal 2014;
Li et at., Cereb Cortex 2013, 2014;
Li et at., J Neuroscience 2014, 2015
Meng et al., Neuroimage 2014

Li et at., Med Image Anal 2015

Validated on 2000+ infant MR images,
Highlighted in National Institute of Mental Health (NIMH)'s 2015-2020 Strategic Plan

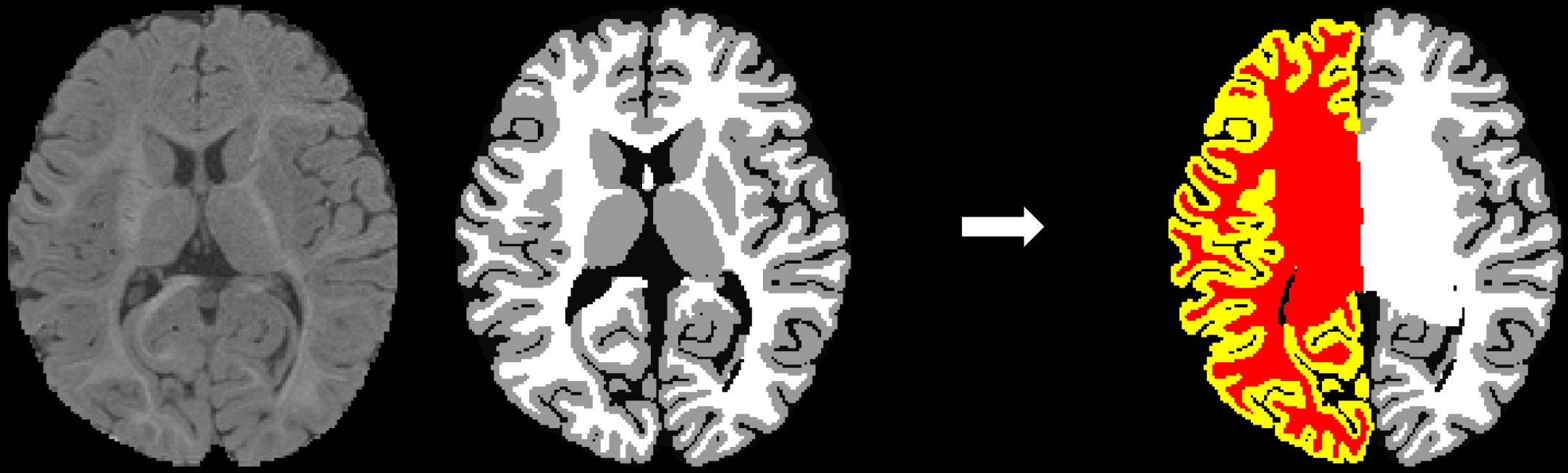
Step 1. Hemisphere Separation and Subcortex Filling

Command: HemisphereSeparation

Inputs: IntensityImage, TissueImage, Age

Outputs: LeftTissueImage, RightTissueImage

Rationale: Alignment with an age-specific volumetric atlas



Input

Output

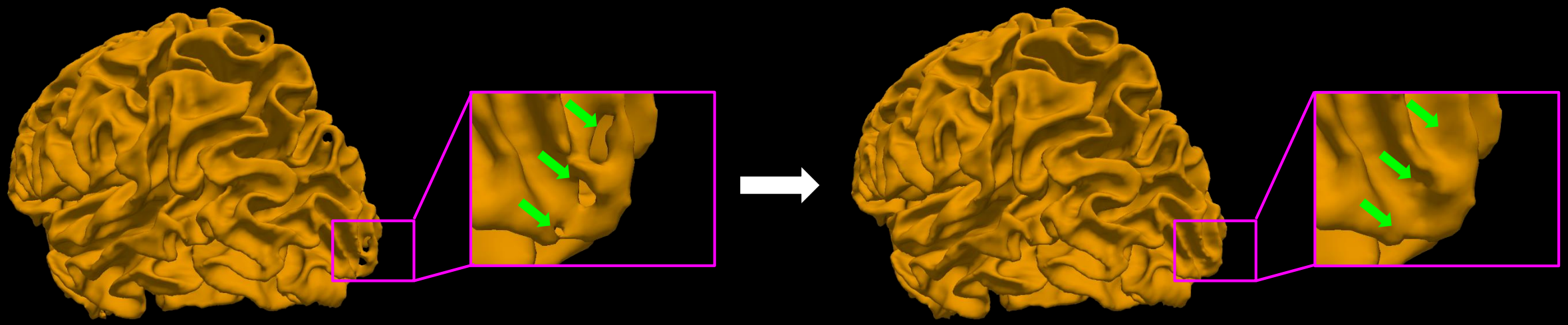
Step 2. Topological Correction

Command: TopologicalCorrection

Inputs: LeftTissueImage (or RightTissueImage), Age

Outputs: CorrectedTissueImage

Rationale: Topology preserving level set and sparse representation



Input

Output

Li et at., Neuroimage 2014; Hao et al., MICCAI 2016

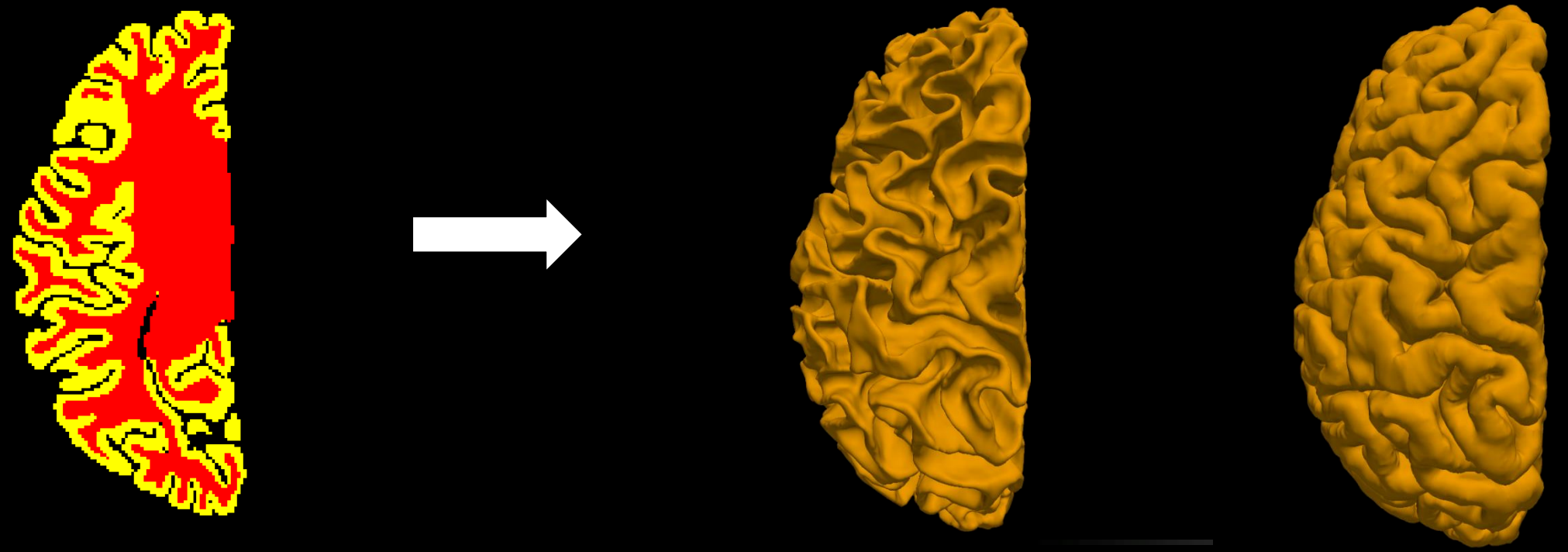
Step 3. Cortical Surface Reconstruction

Command: CorticalSurfaceReconstruction

Inputs: CorrectedTissueImage, Age

Outputs: InnerSurface, OuterSurface

Rationale: Marching cubes and topology preserving deformable surface

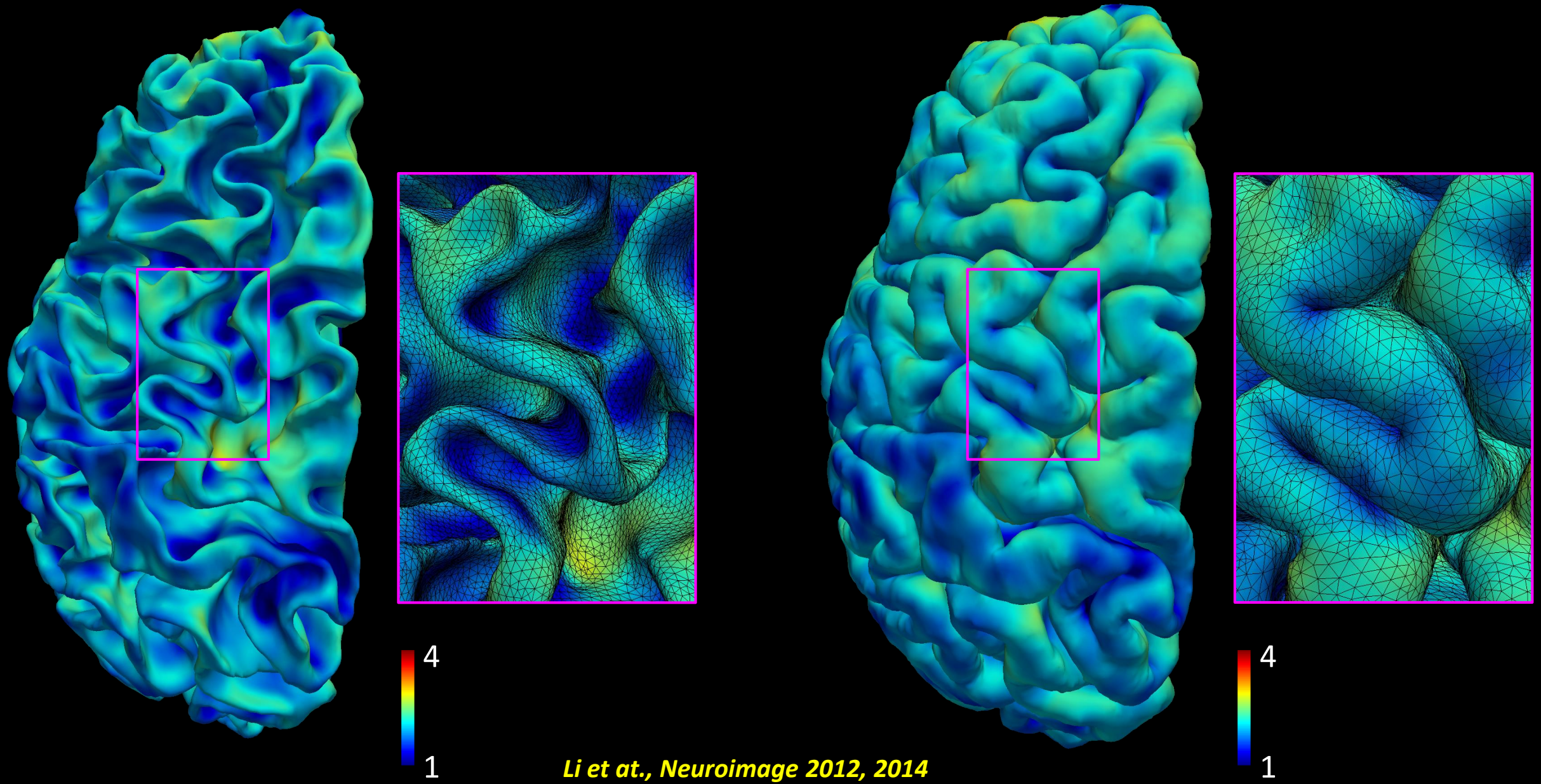


Input

Output

Li et at., Neuroimage 2012, 2014

Step 3. Cortical Surface Reconstruction



Li et at., Neuroimage 2012, 2014

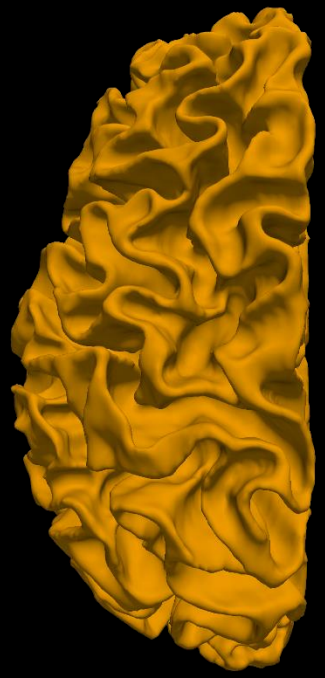
Step 4. Spherical Mapping

Command: SphericalMapping (or mris_sphere from FreeSurfer)

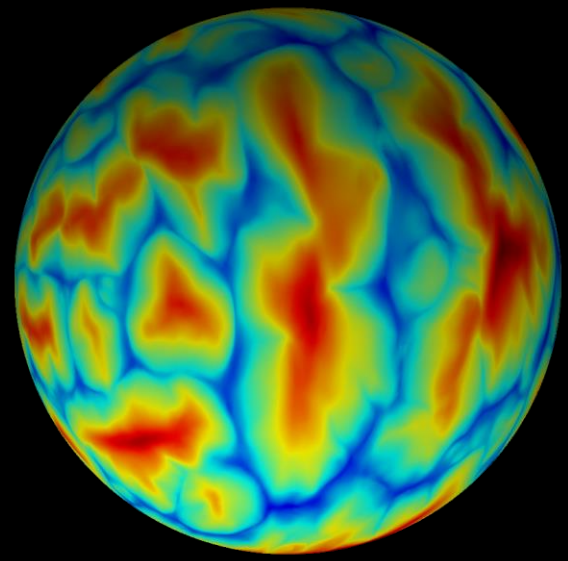
Inputs: InnerSurface

Outputs: SphericalSurface

Rationale: Inflation while preserving triangular angles and distances



Input



Output

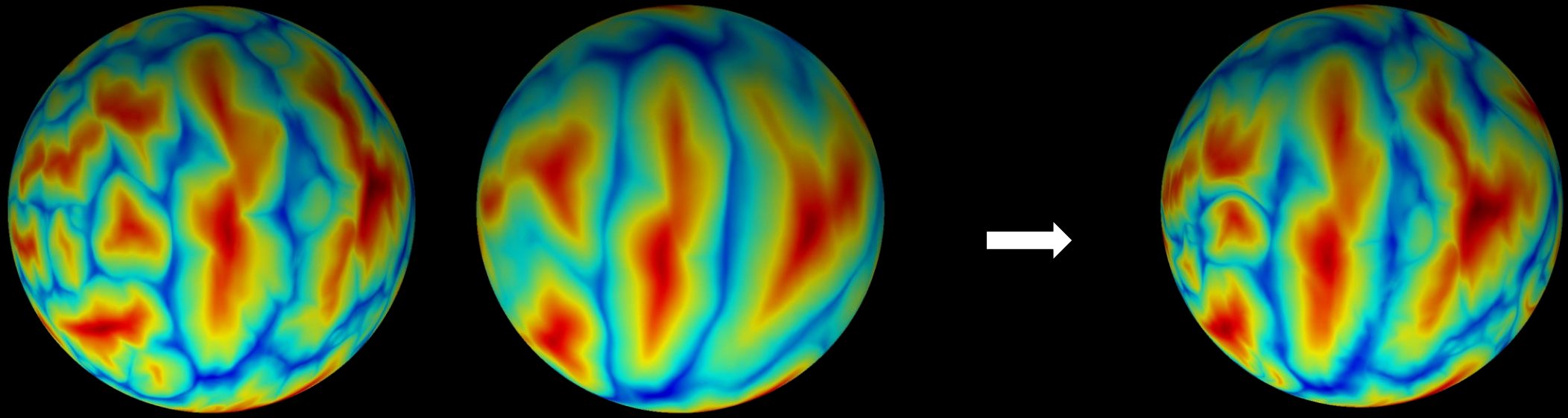
Step 5. Spherical Registration with Infant Surface Atlas

Command: mris_SD_pairwise_register (from Spherical Demons)

Inputs: SubjectSphere, AtlasSphere, Age

Outputs: AlignedSubjectSphere

Rationale: Moving subject vertices to find atlas vertices with similar properties



Subject spherical surface

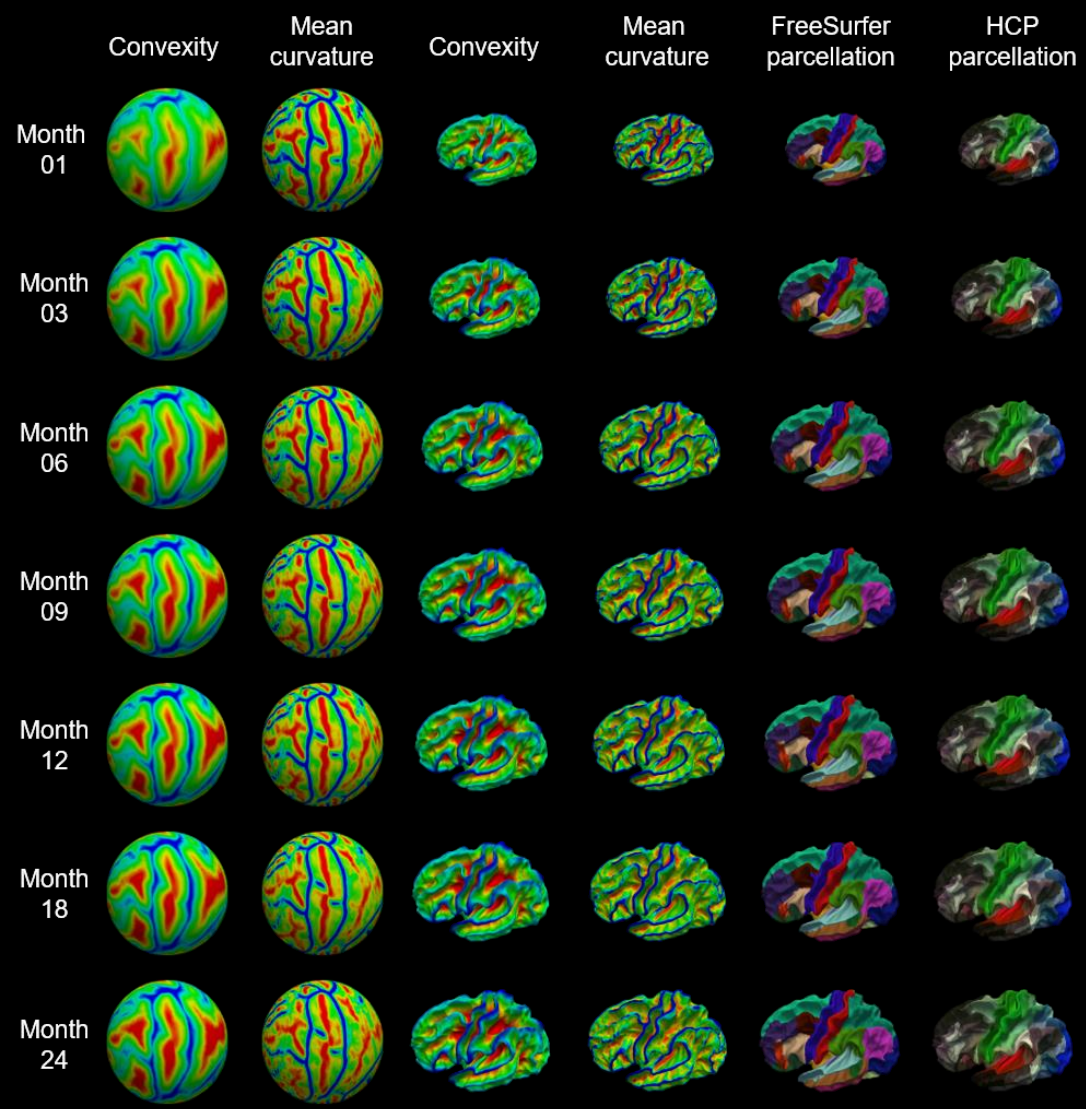
Atlas spherical surface

Aligned subject spherical surface

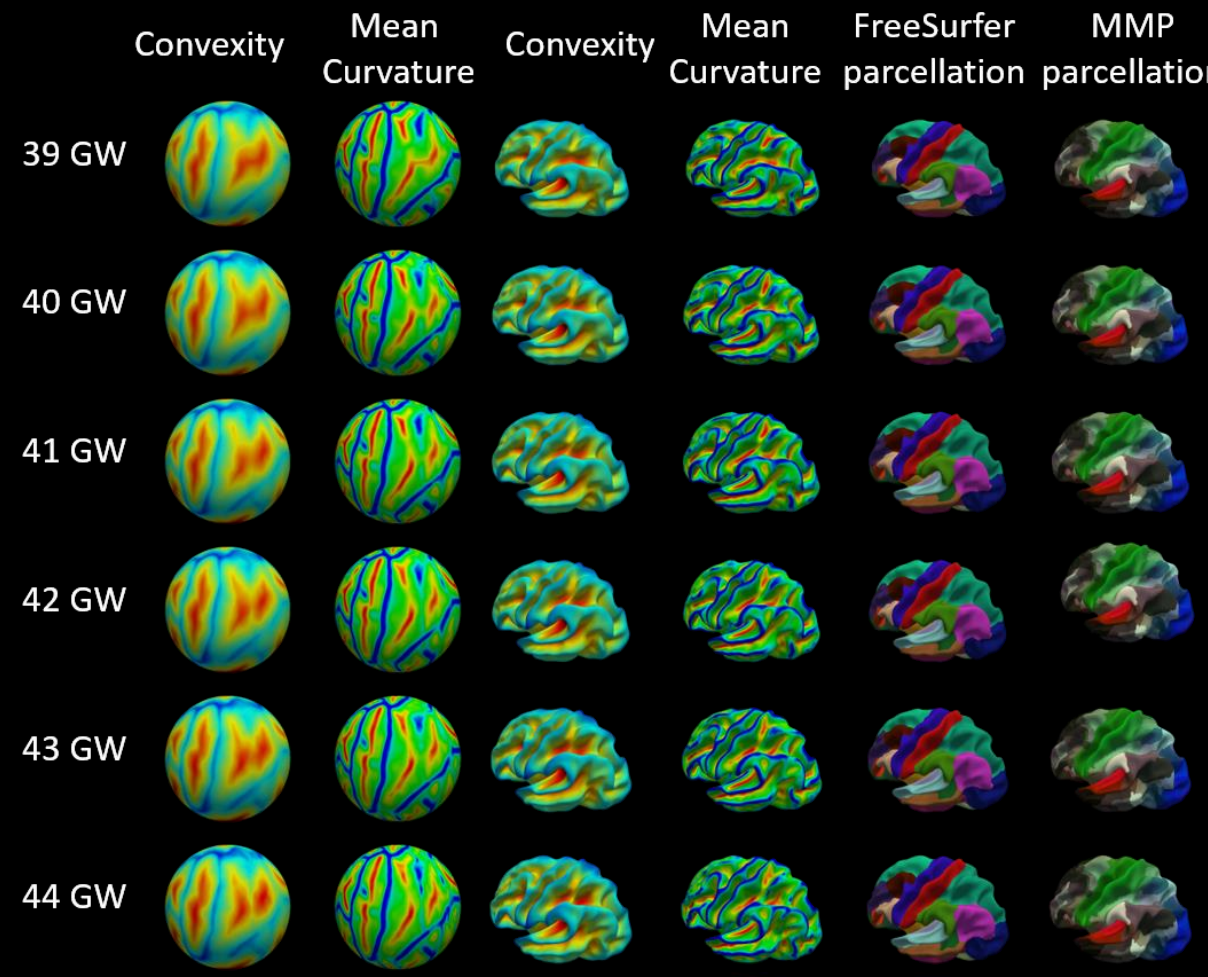
Input

Output

UNC Baby Cortical Surface Atlas Sets



4D *infant* cortical surface atlases



Spatiotemporal *neonatal* cortical surface atlases

Based on **700+** neonates

Li et al., Med Image Anal 2015; Wu et al., MICCAI 2017, ISBI 2018

All freely available on NITRC: https://www.nitrc.org/projects/infantsurfatlas

Step 6. Cortical Surface Parcellation

Command: SphericalParcellation

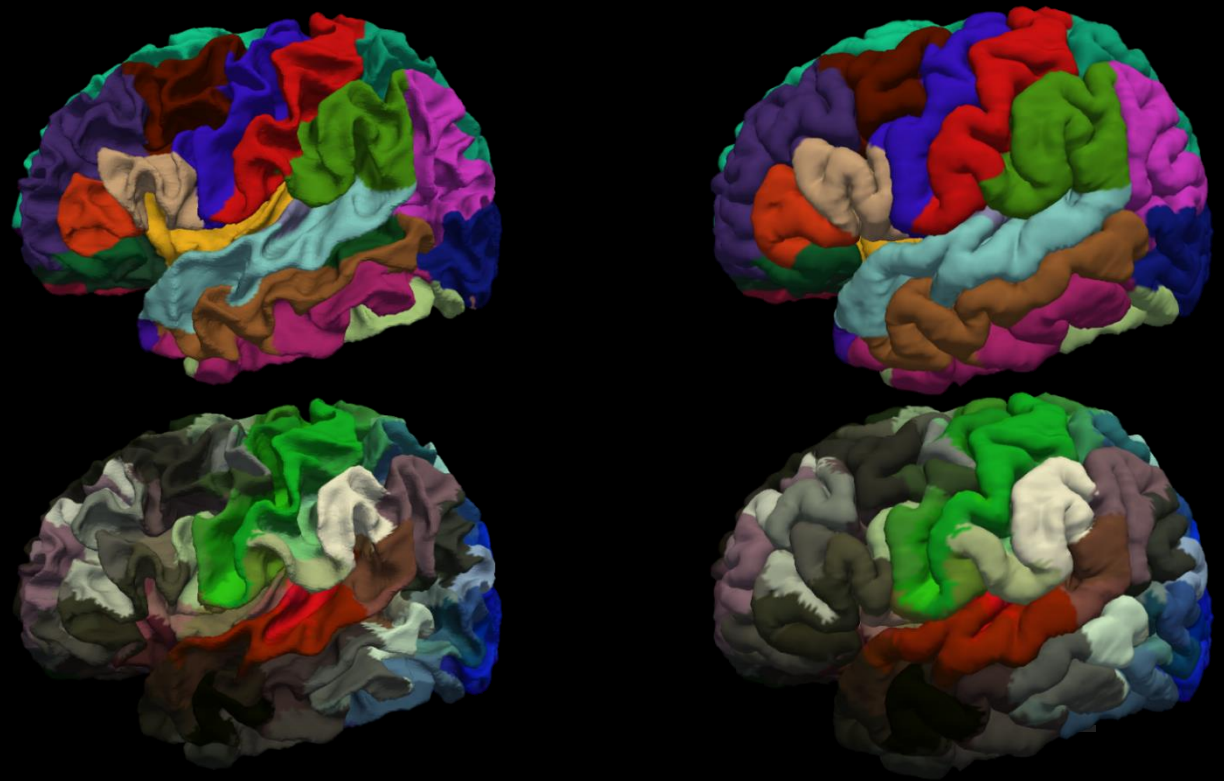
Inputs: SubjectSurface, AlignedSubjectSphere, AtlasSphere

Outputs: ParcellatedSubjectSurface

Rationale: Propagation of atlas parcellation to subject surface and refinement

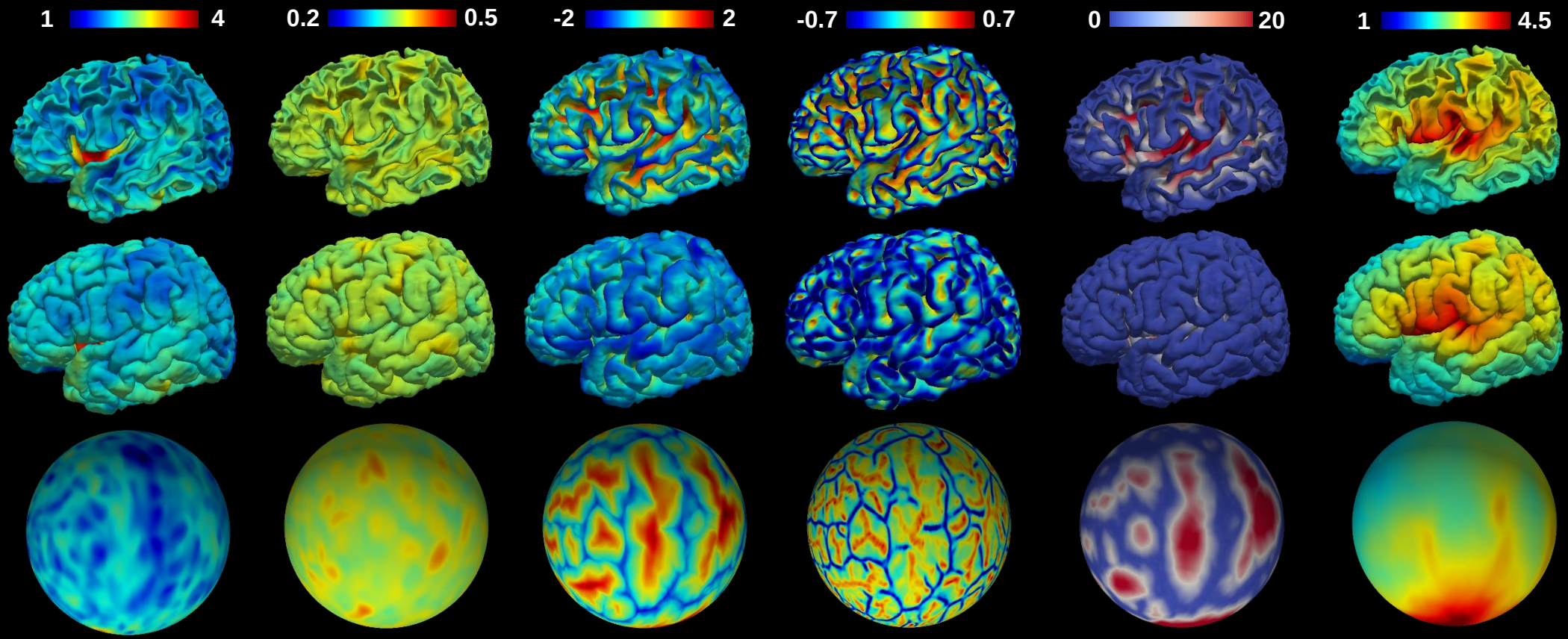
Freesurfer parcellation

HCP MMP parcellation



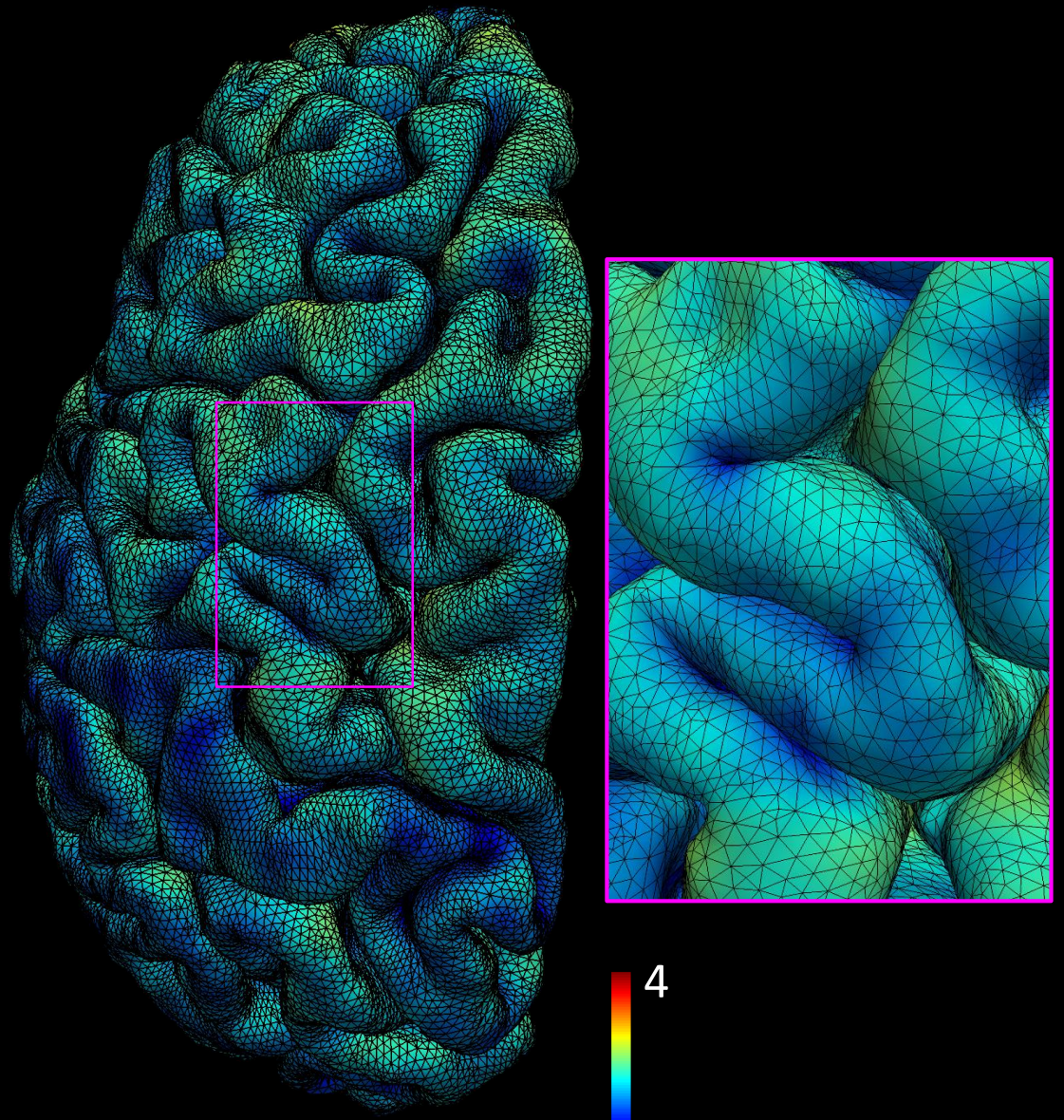
Li et al., Med Image Anal 2014; Wu et al., MICCAI 2017

Step 7. Computing and Resampling Cortical Attributes

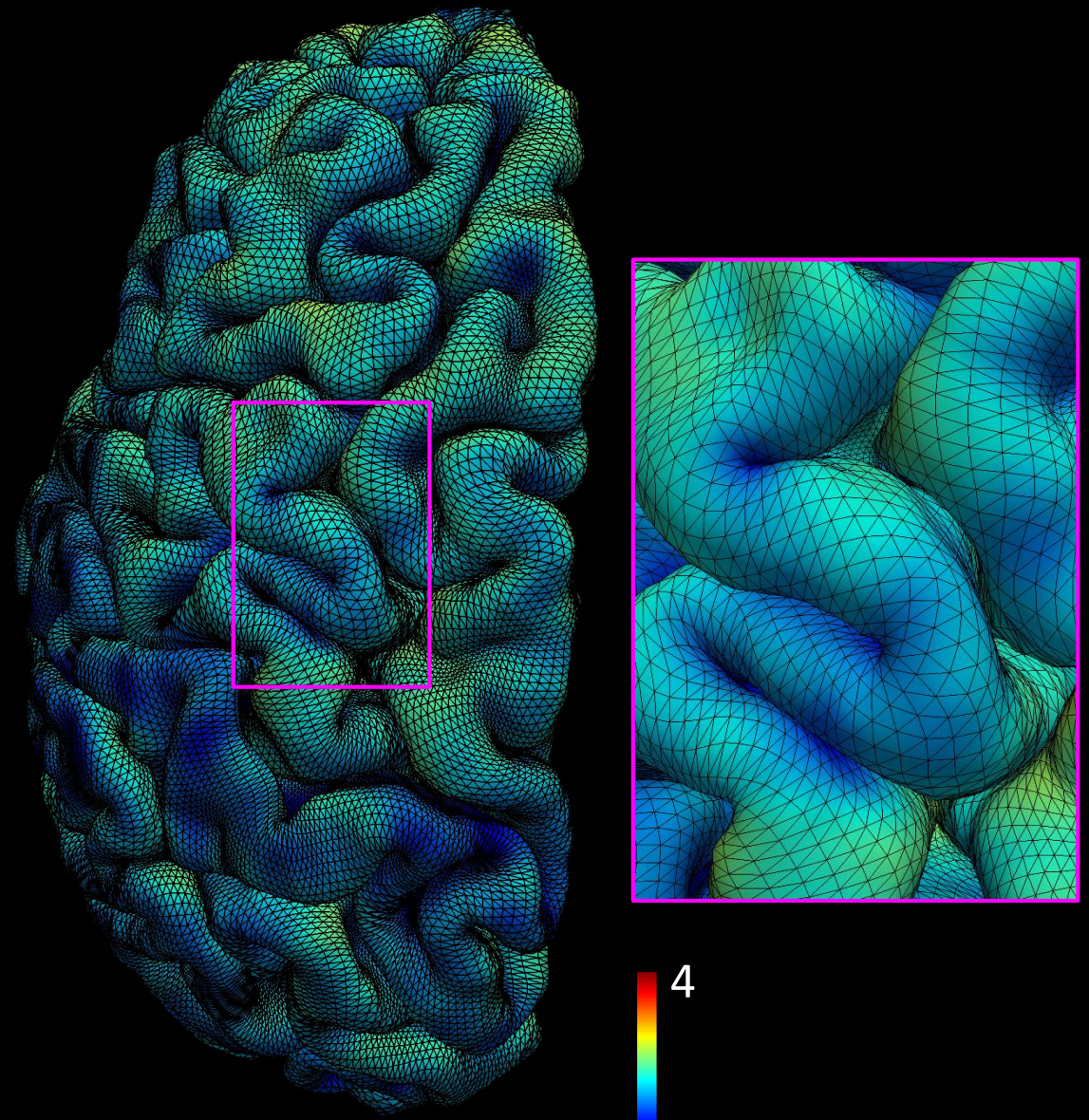


Cortical thickness | Surface area | Average convexity | Mean curvature | Sulcal depth | Local gyrification index

Step 7. Computing and Resampling Cortical Attributes

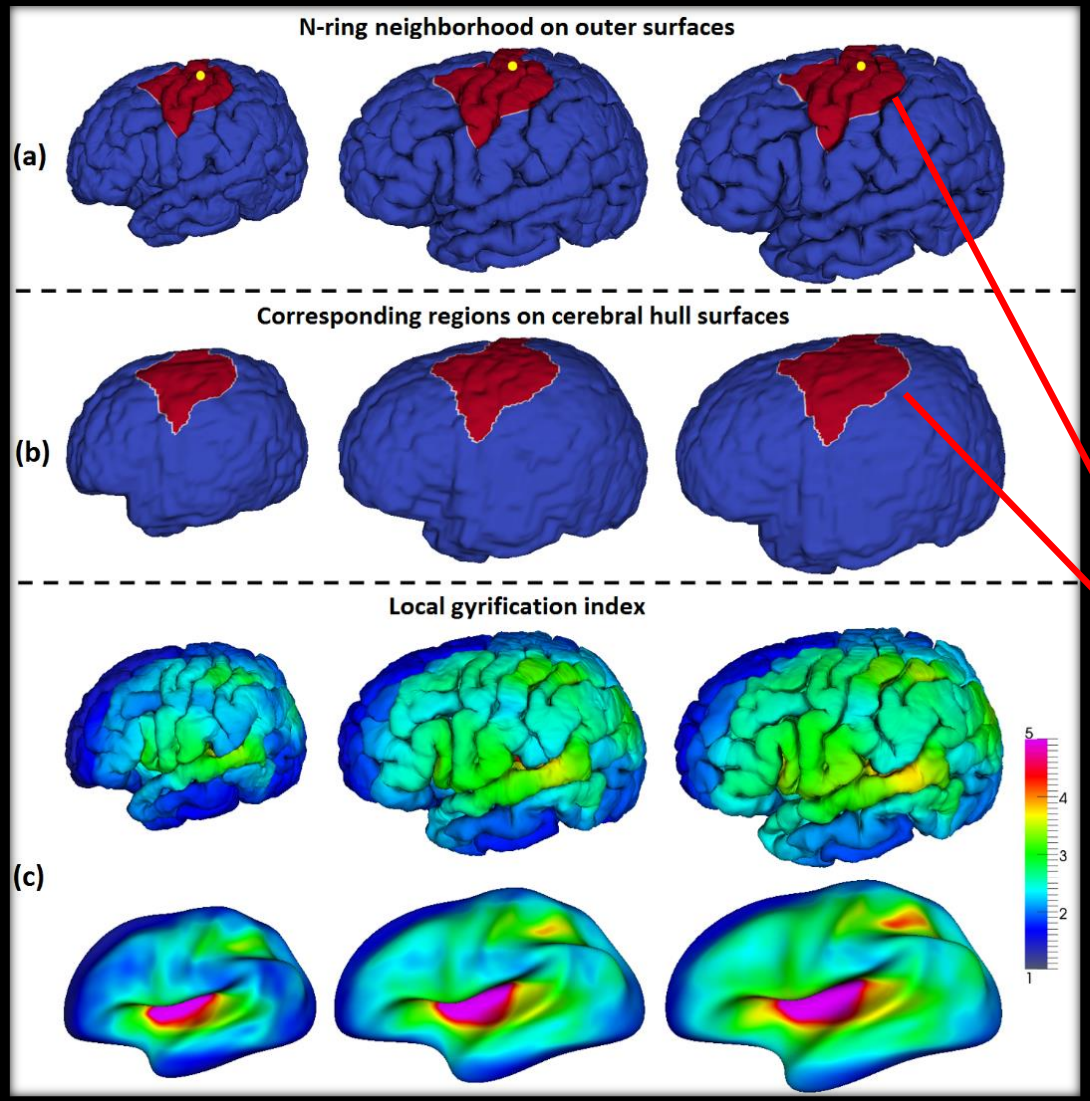


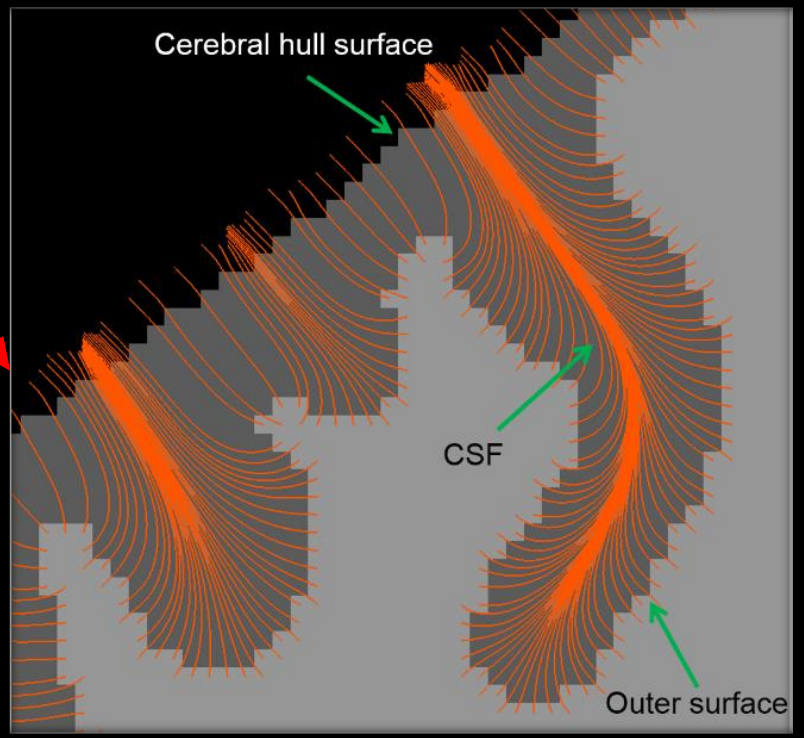
Before resampling



After resampling

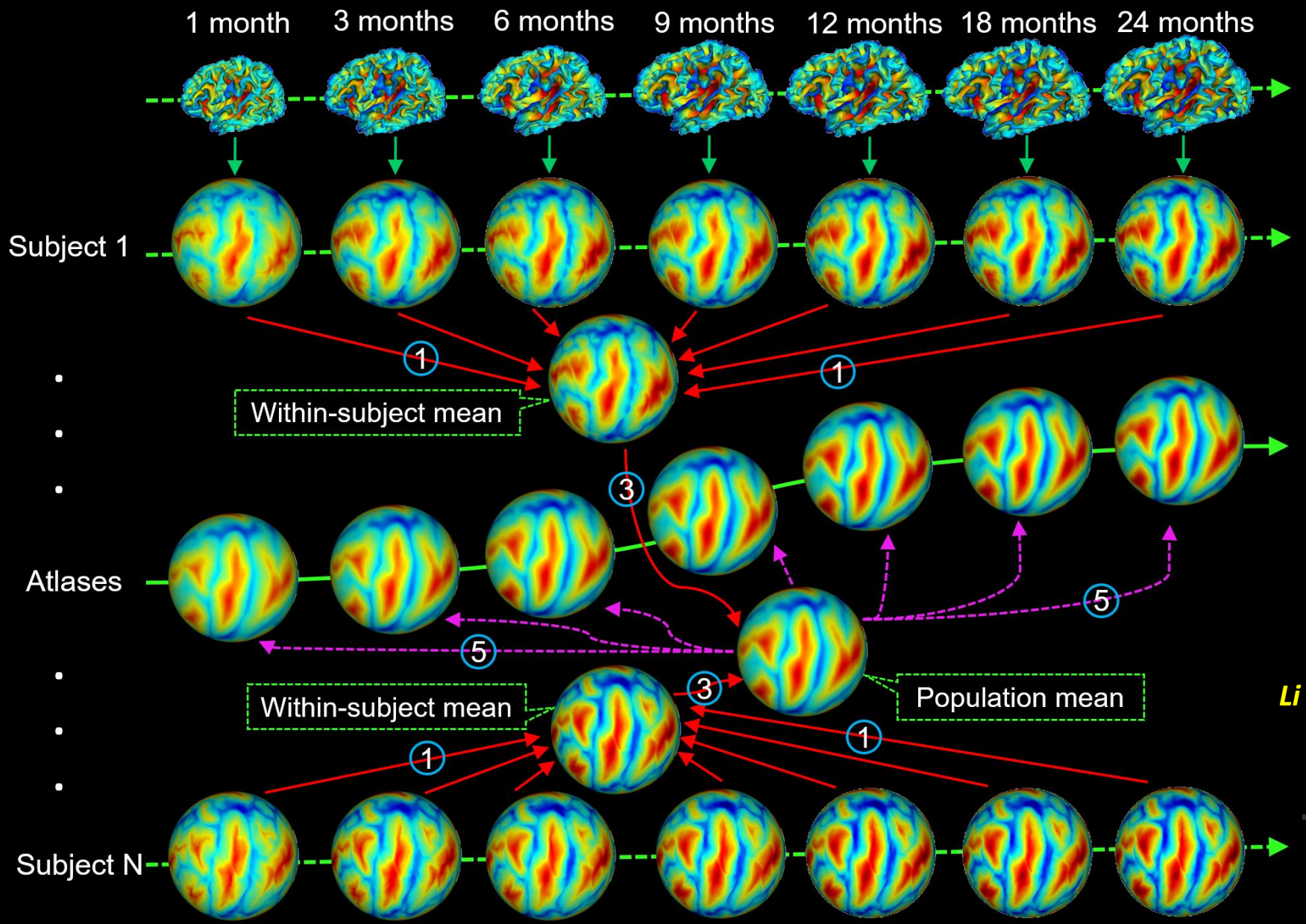
Step 7. Computing and Resampling Cortical Attributes





Li et al., J Neuroscience 2014

Longitudinal Surface Atlas Construction



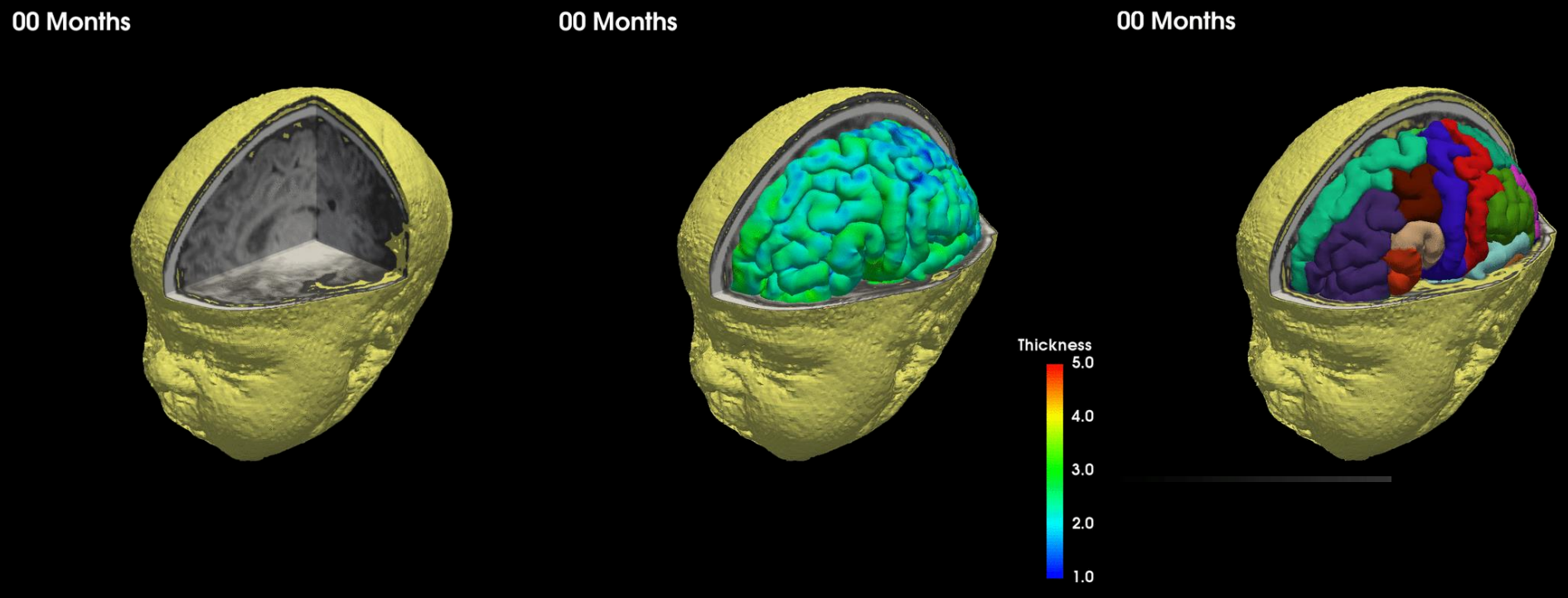
Li et al., Med Image Anal 2015

BRIC BIOMEDICAL RESEARCH IMAGING CENTER

Research Lab IDEA

Conclusion

- **Infant-dedicated tools for surface-based brain mapping**
  - Topology correction, surface reconstruction, registration, parcellation, modeling and atlas building
- **Region-specific growth patterns during early brain development**
  - Surface area, cortical thickness, local gyrification, myelin content, sulcal folding, and their hemispheric asymmetries



上海科技大学
ShanghaiTech University

脑科学与脑疾病
第八节课：研究人类大脑功能的rs-fMRI及其基本分析技术

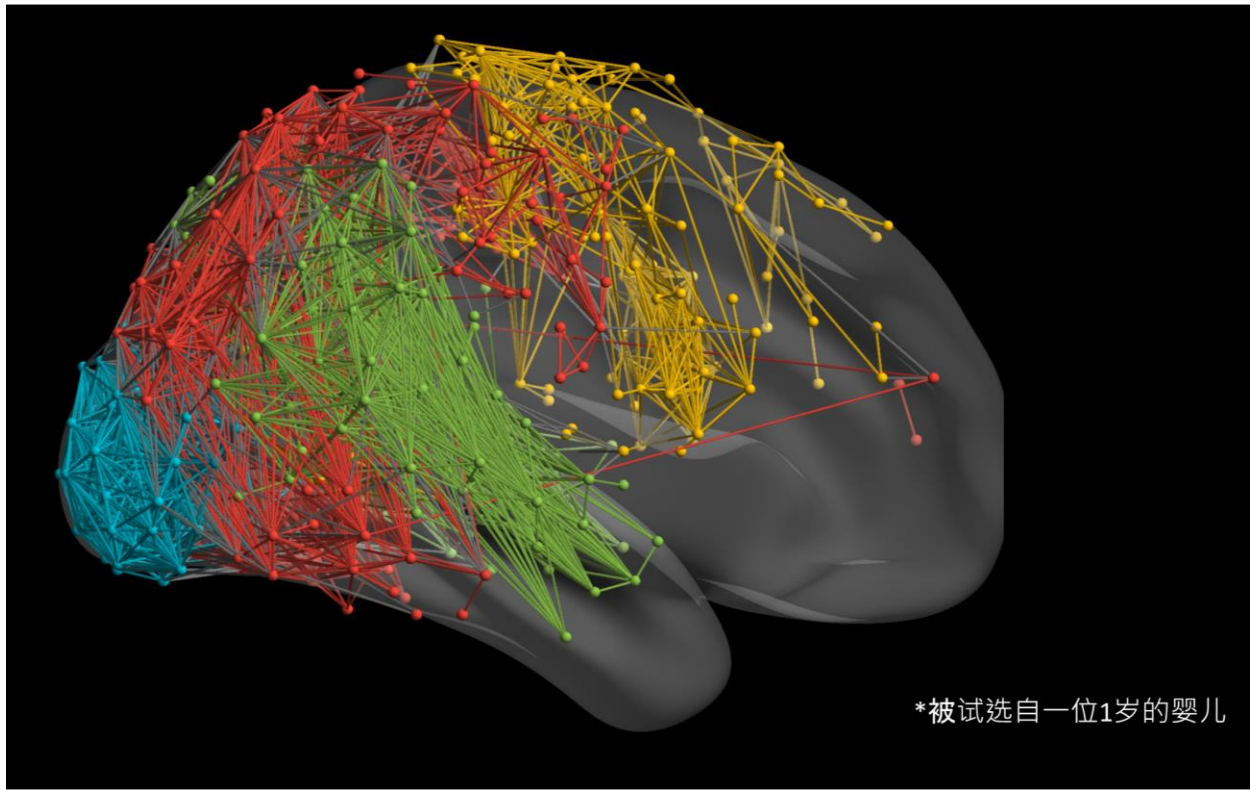
Resting-state fMRI

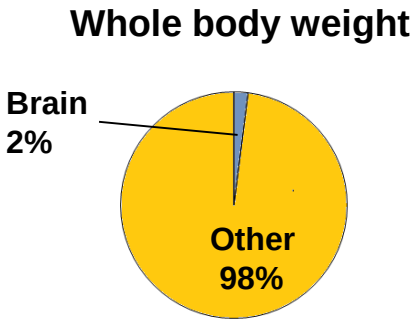
https://www.youtube.com/watch?v=K4tZxrv3iiE

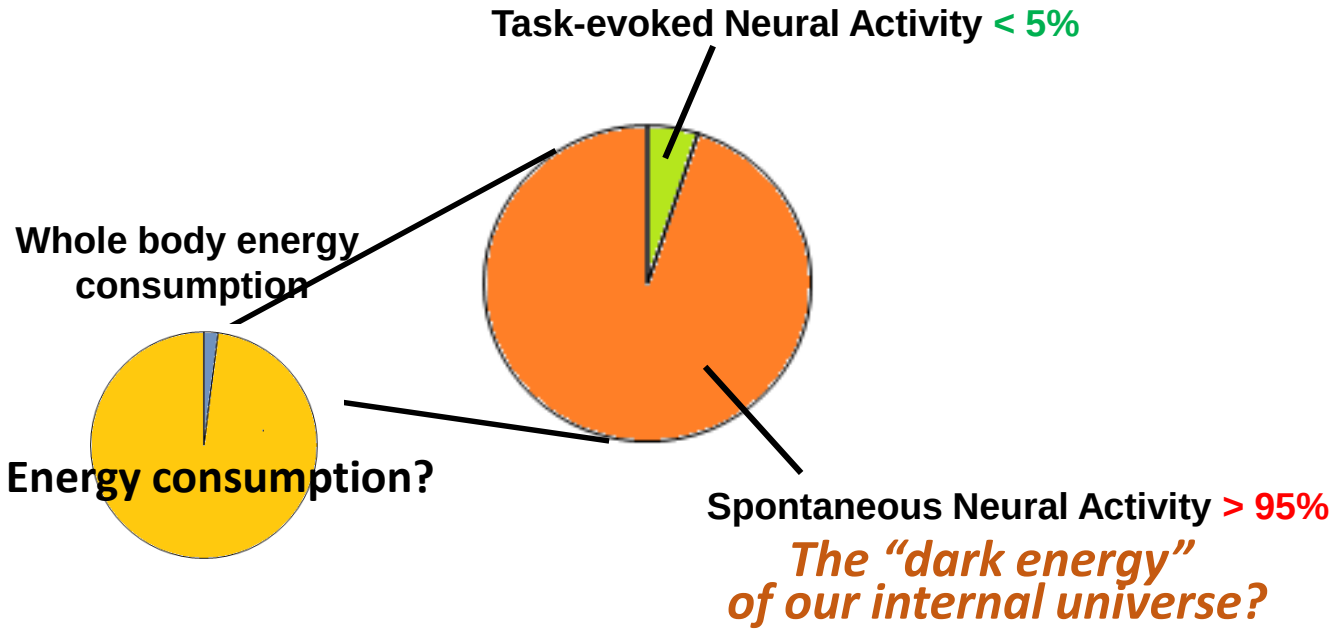
https://www.youtube.com/watch?v=BOysZS0rdFQ

提出问题

- 既然神经元用突触连接在一起，能否在脑区水平（宏观水平）看到脑区和脑区之间的连接?







The "dark energy" of our internal universe?

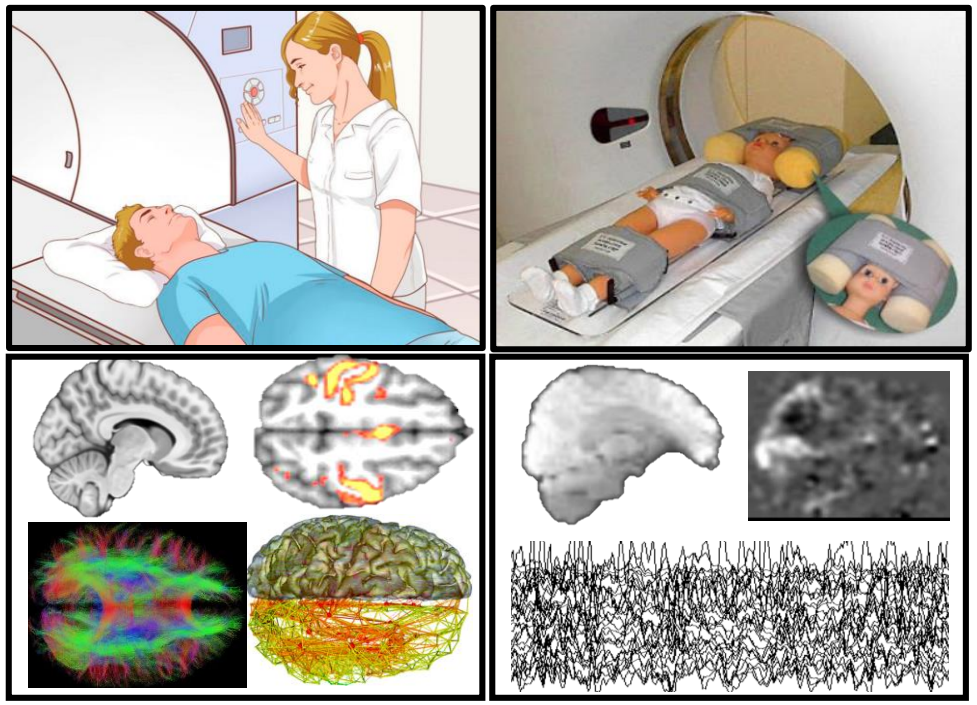
Spontaneous neural activity maybe related to:

- Preparation
- Alertness
- Consciousness
- Self reference
- Monitoring
- Memory consolidation
- Mind wondering/day dreaming …

Resting-state fMRI helps to probe the dark energy while performing no explicit task

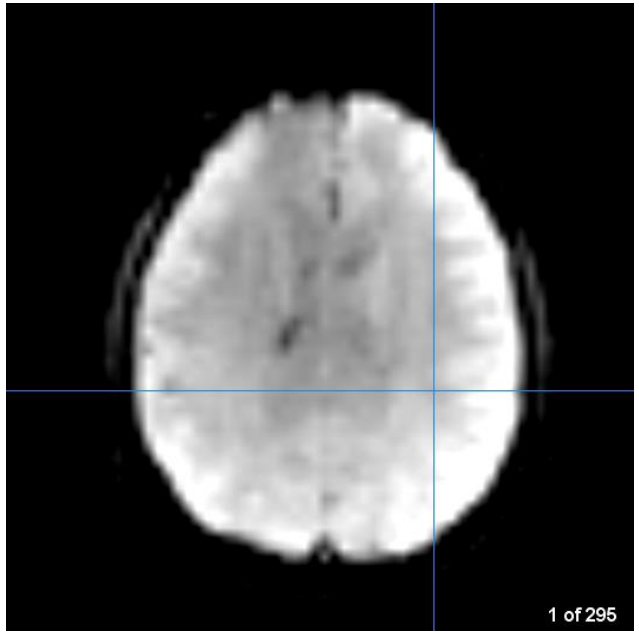
Raichle (2006) Science, Vol. 314, Issue 5803, pp. 1249-1250

What You Think Rs-fMRI Is **What Rs-fMRI Really Is**



Resting-state fMRI

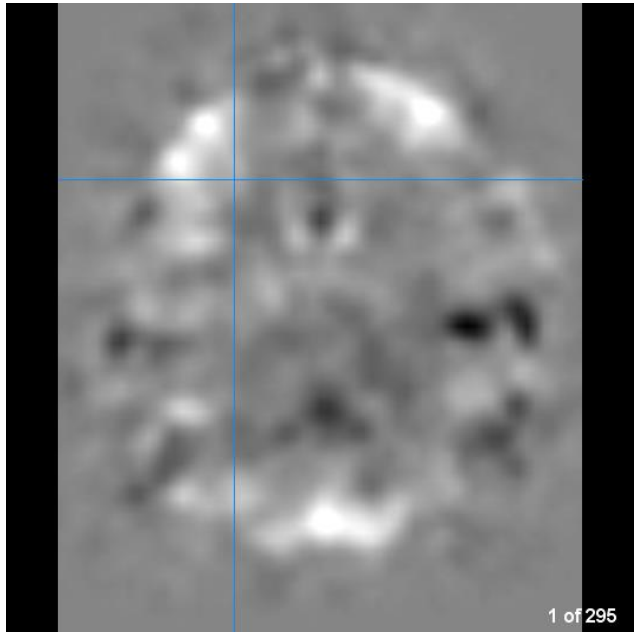
- No explicit task
- Eye closed
- Keep still
- Don't think particularly
- Stay awake



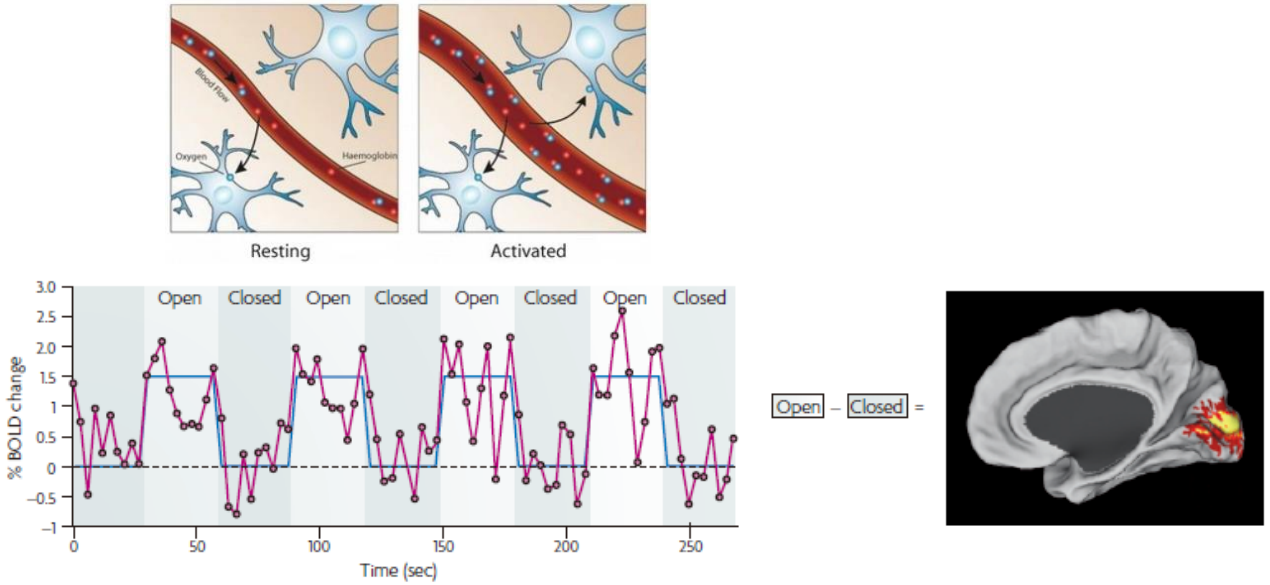
Raw data

Resting-state fMRI data

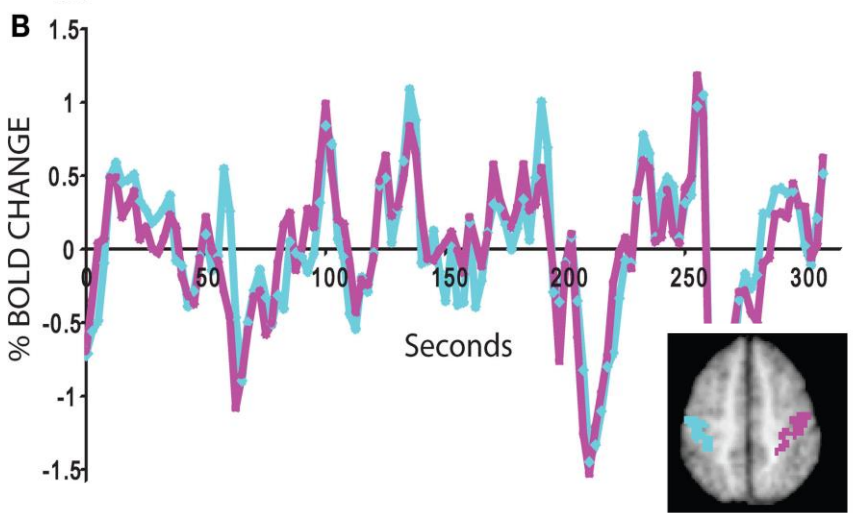
- Blood oxygen level dependent (BOLD) signal
- 4D (3D spatial + 1D temporal)
- Very noisy
- Physiological artifacts
- Real BOLD signal only varies 1-3%



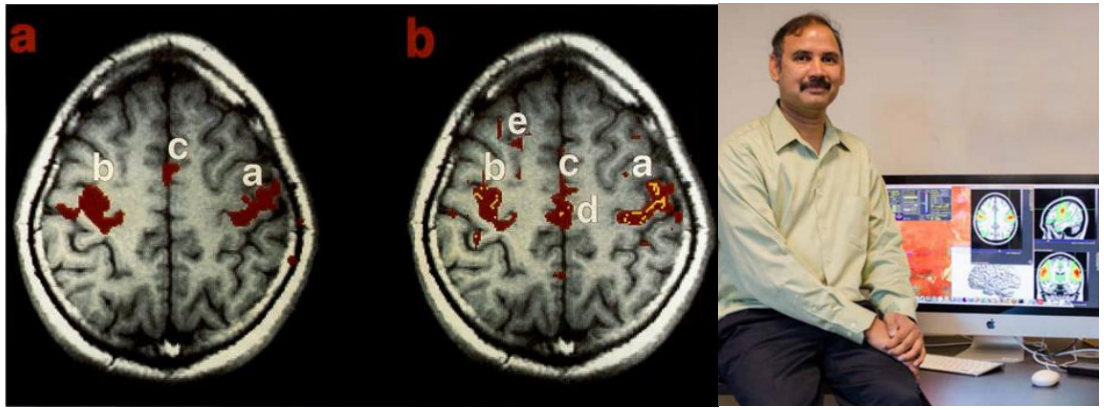
After pre-processed



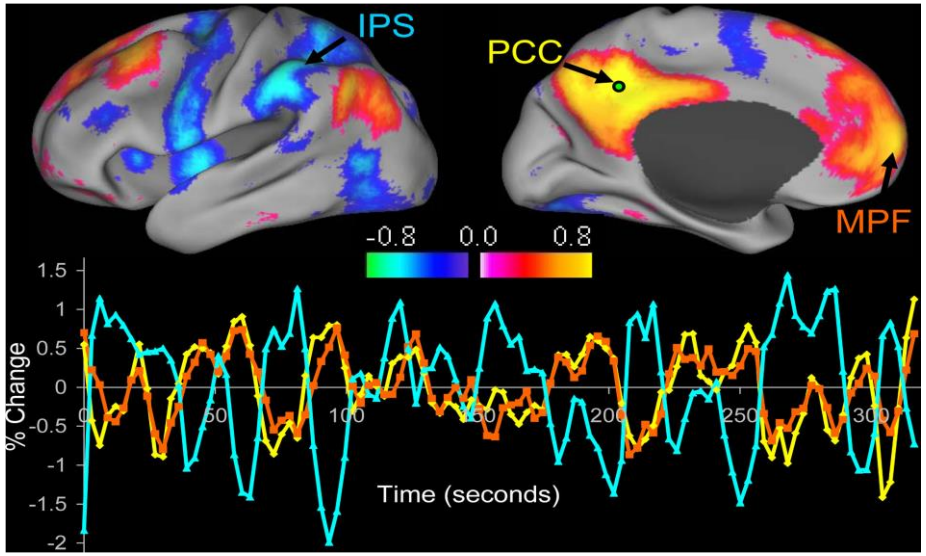
Visual task, Fox et al., Nature, 2007



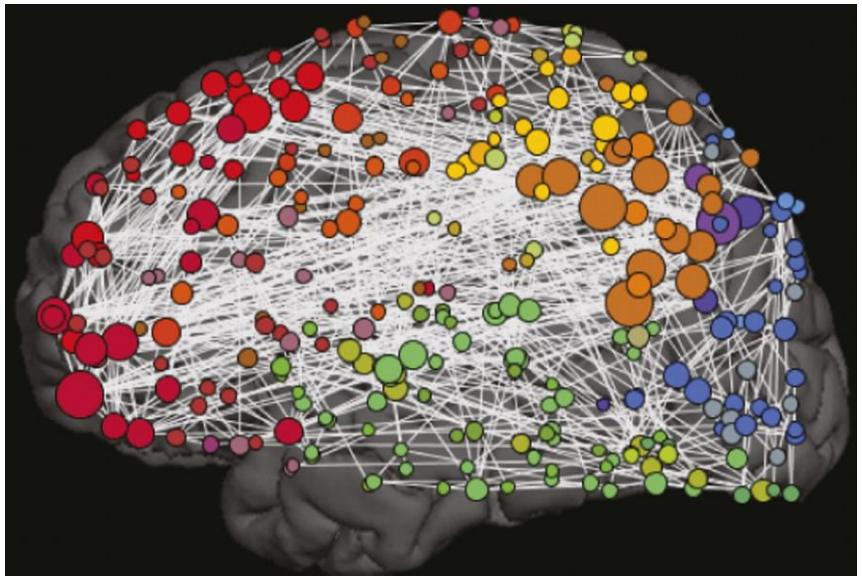
Fox et al., 2006, Nat. Neurosci



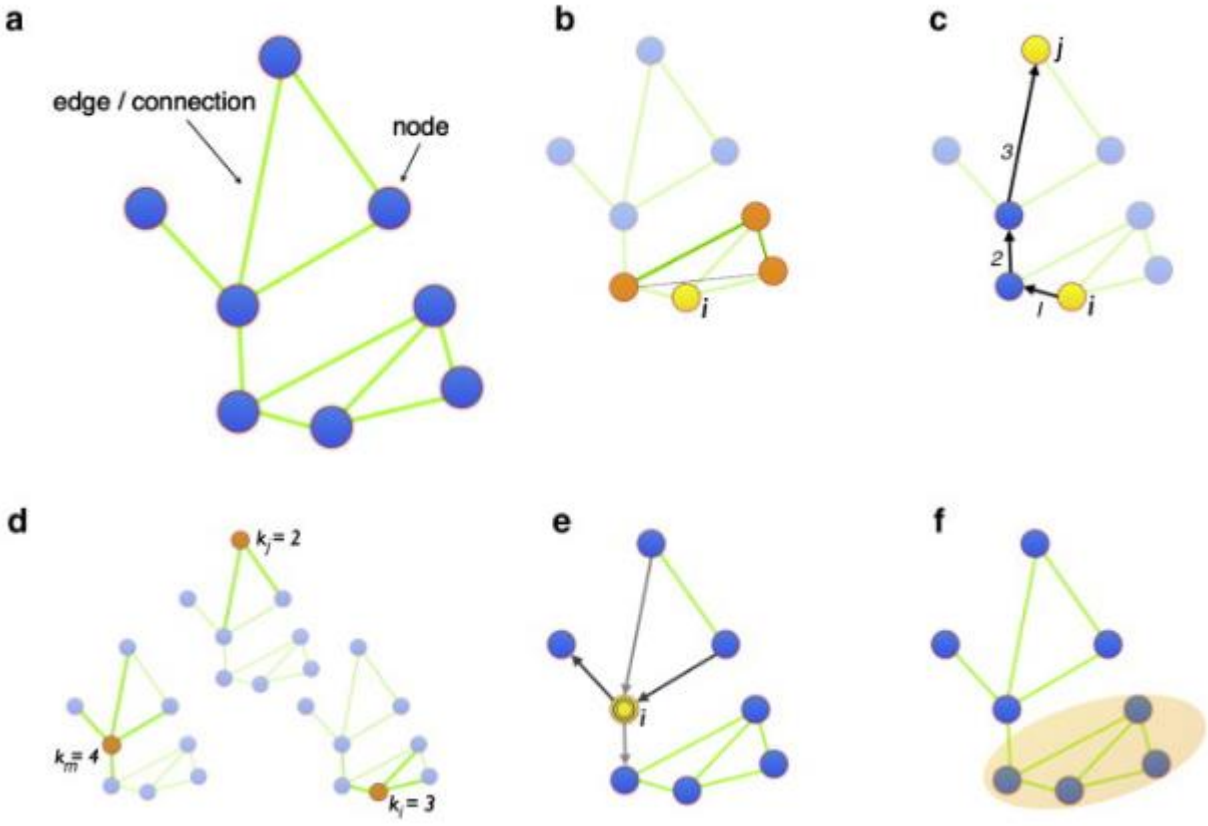
Bharat Biswal 1995



Fox et al., 2005, PNAS



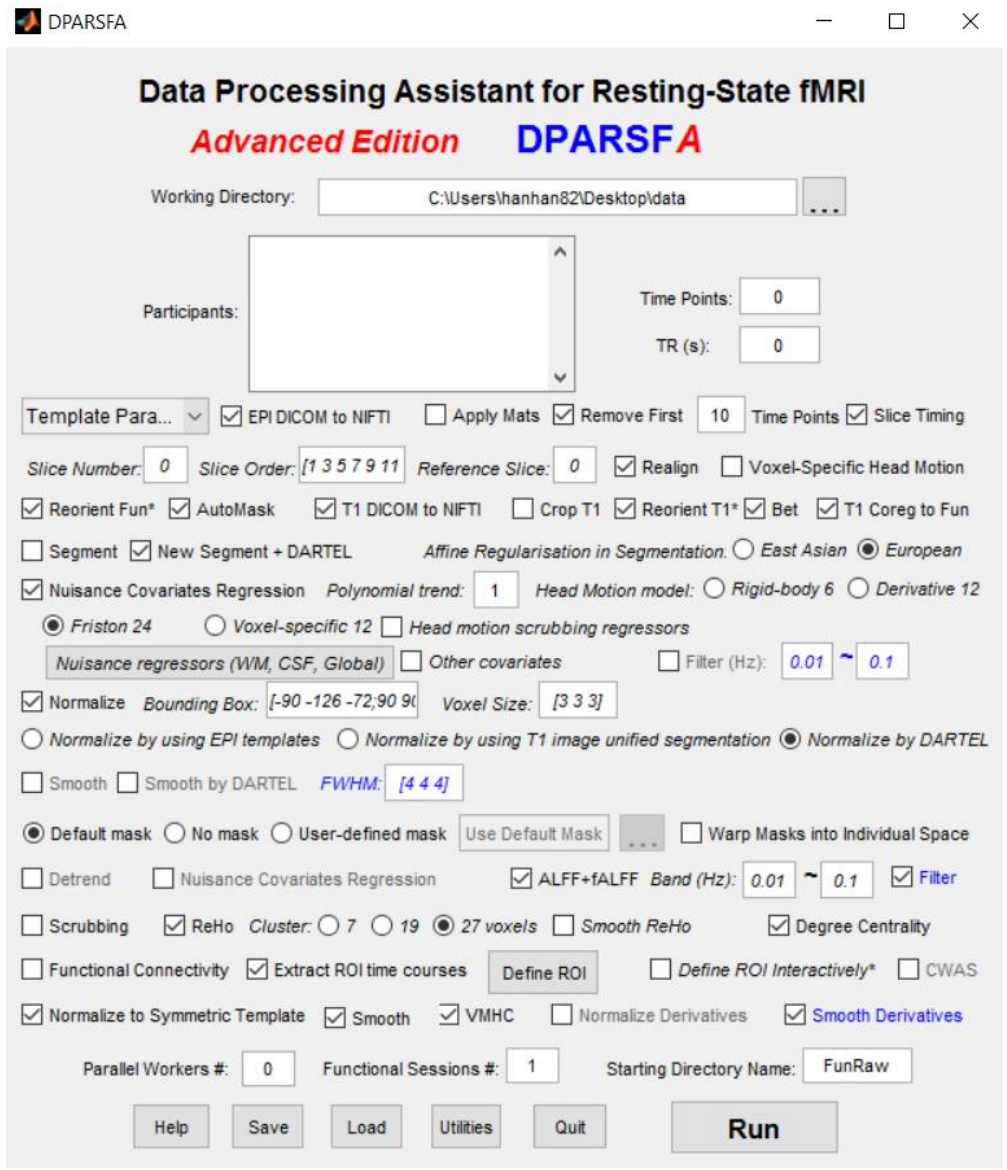
PNAS, vol. 109 no. 15, pp. 5549–5550



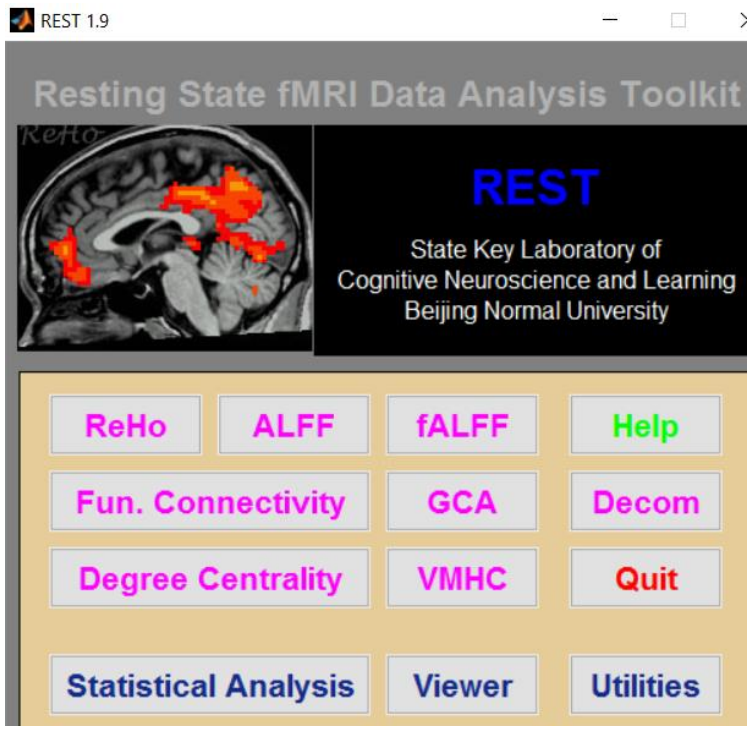
Van Den Heuvel, 2010

Software & Toolboxes & Datasets

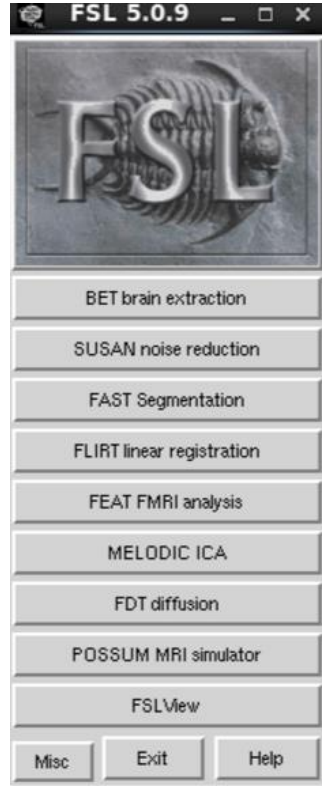
http://fsl.fmrib.ox.ac.uk/fsl/fslwiki/



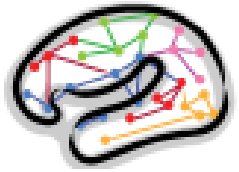
http://rfmri.org/



http://restfmri.net/forum/index.php



https://afni.nimh.nih.gov/afni/



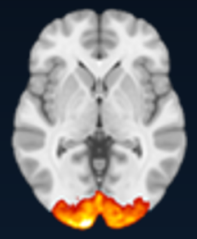
CONNECTOME
COORDINATION FACILITY

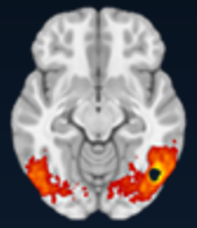
https://www.humanconnectome.org/

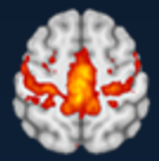
婴幼儿脑影像智能分析：功能

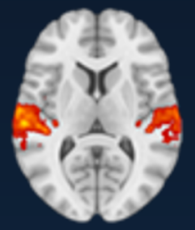
上海科技大学
ShanghaiTech University

初级视觉

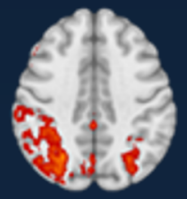

次级视觉网络

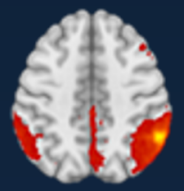

视觉联合网络


感觉运动


听觉网络

默认网络1


额顶网络1


额顶网络2

默认网络2

婴儿的9个大尺度脑功能网络的空间模式已和成人非常相似

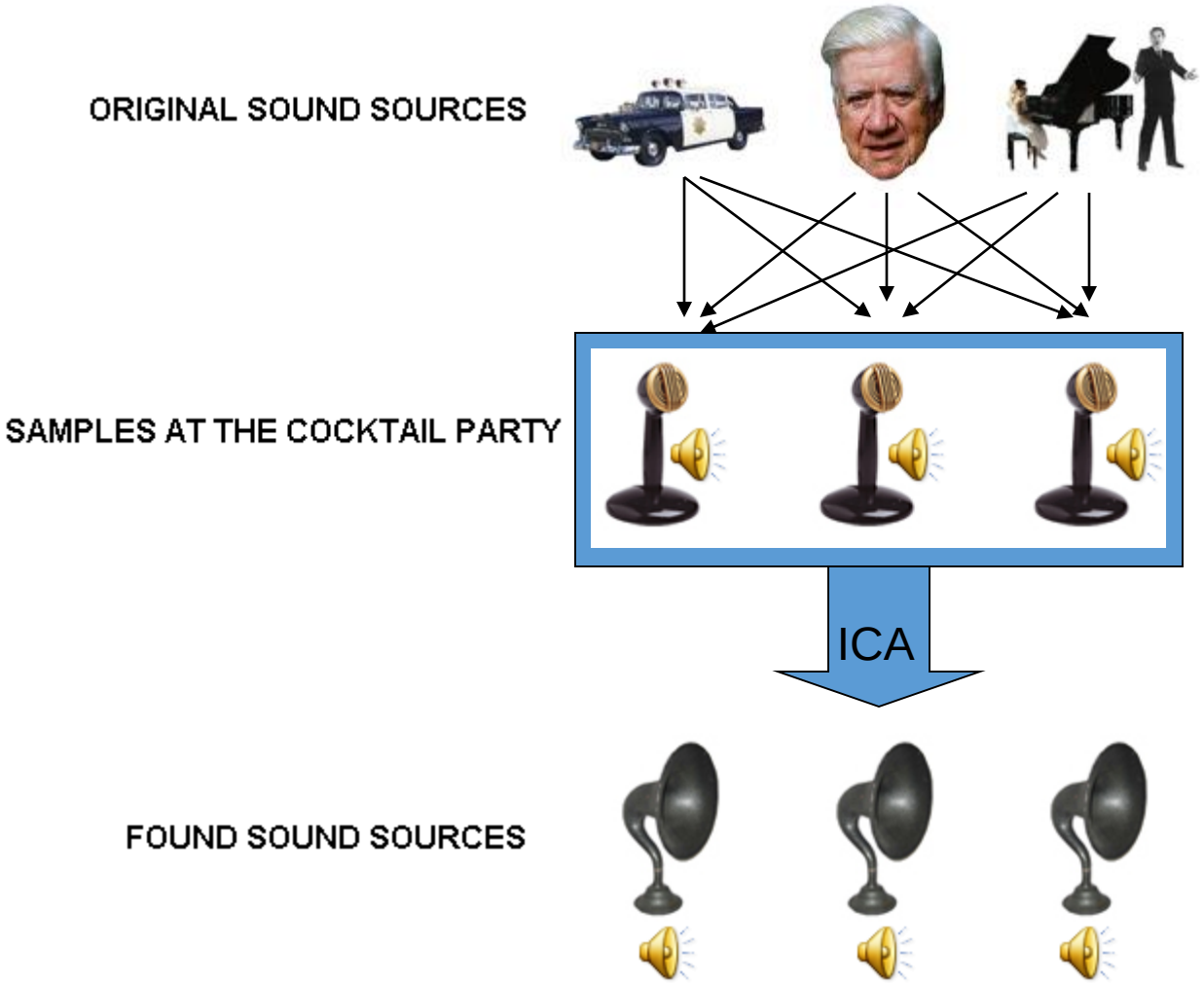
提出问题

- 如果想从数据中自发的“寻找”或“读出”这种同步性该如何做?

设想思路：信号处理中的ICA（鸡尾酒会）

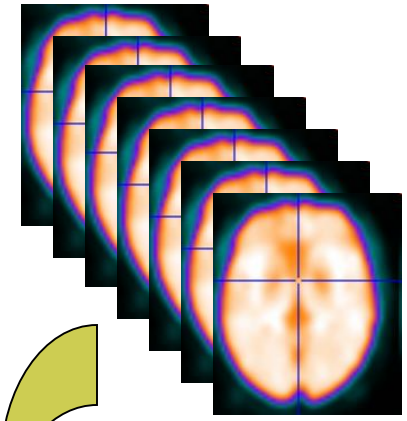
上海科技大学
ShanghaiTech University

“Cocktail party” problem: can I clearly hear each person’s voice?



Jaakko Särelä, Patrik Hoyer and Ella Bingham, graphic design by Petri Saarikko.

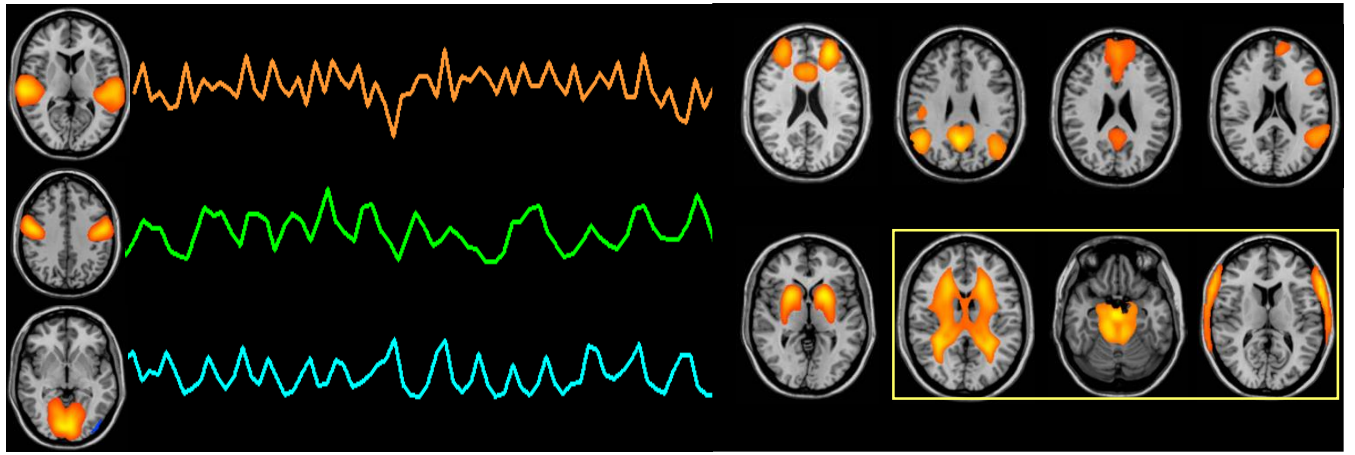
设想思路：静息态fMRI研究中的ICA



One of the "two most commonly used methods in functional connectivity MRI studies". (Zhang & Raichle, 2010, *Nature Review*)

"The second most popular technique for analysing spontaneous BOLD data" with "tremendous promise". (Fox & Raichle, 2007, *Nature Review Neurosci*)

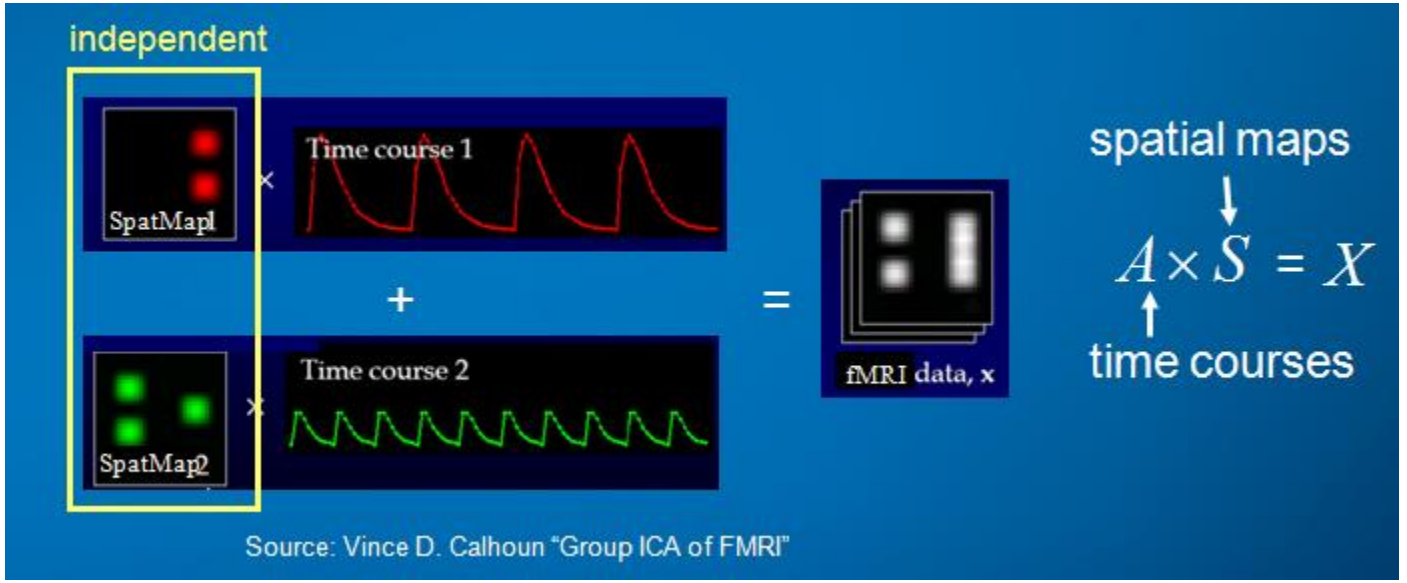
ICA

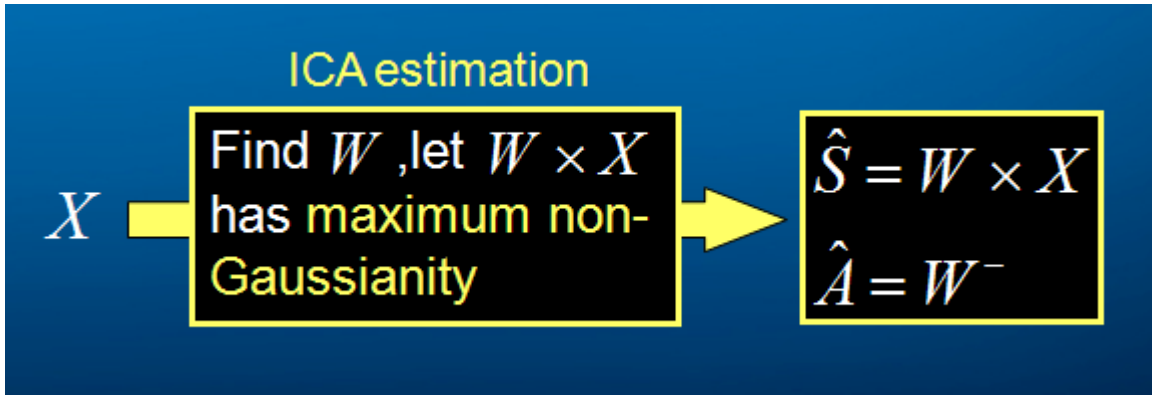


van de Ven et al., 2004
Kiviniemi et al., 2003
Jafri et al., 2008
Damoiseaux et al., 2006
Calhoun et al., 2009
Damoiseaux et al., 2008
Greicius et al., 2004
Calhoun et al., 2008
Calhoun et al., 2004
…

ICA derived resting-state functional connectivity (RSFC) networks

ICA原理：假设和计算


independent
SpatMap1
Time course 1
+
SpatMap2
Time course 2
=
fMRI data, x
spatial maps
$A \times S = X$
time courses
Source: Vince D. Calhoun "Group ICA of FMRI"

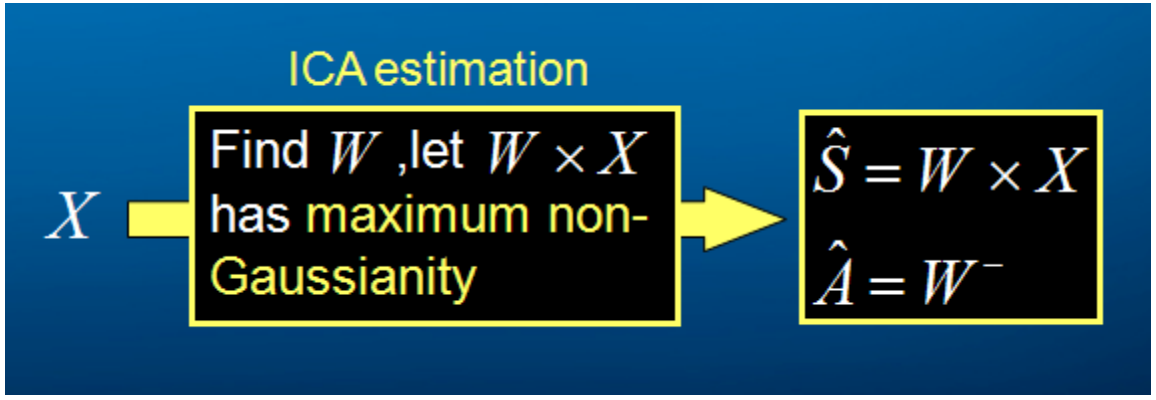

ICA estimation
X
Find W ,let $W \times X$ has maximum non-Gaussianity
$\hat{S} = W \times X$
$\hat{A} = W^{-}$

ICA计算：如何解混（求W）？

中心极限定理告诉我们：多个独立的随机变量组合成的新变量的分布将更加趋向高斯分布。

假设x1, x2为两个独立的变量， 根据中心极限定理，他们的组合a*x1+b*x2将趋向高斯分布。

但如果我们遍历所有的a和b，终于找到一对a和b，使得a*x1+b*x2的分布非常远离高斯分布（非高斯性最大）了；这时为了不违背中心极限定理，x1和x2就必定不是两个独立的变量，故a*x1+b*x2只可能是一个“源”变量s1，或者s2。这里的a，b构成了W。



相比GLM，ICA的优缺点

上海科技大学 ShanghaiTech University

General Linear Model

Pros: a known model
easy to interpret

Cons: unknown HRF
unable if without hypothetical model
ignore relationship between voxels

ICA

Pros: without need of HRF or hypothesis
count for relation between voxels
find underlying data structure

Cons: hard to interpret
instability problems

In many cases we need ICA

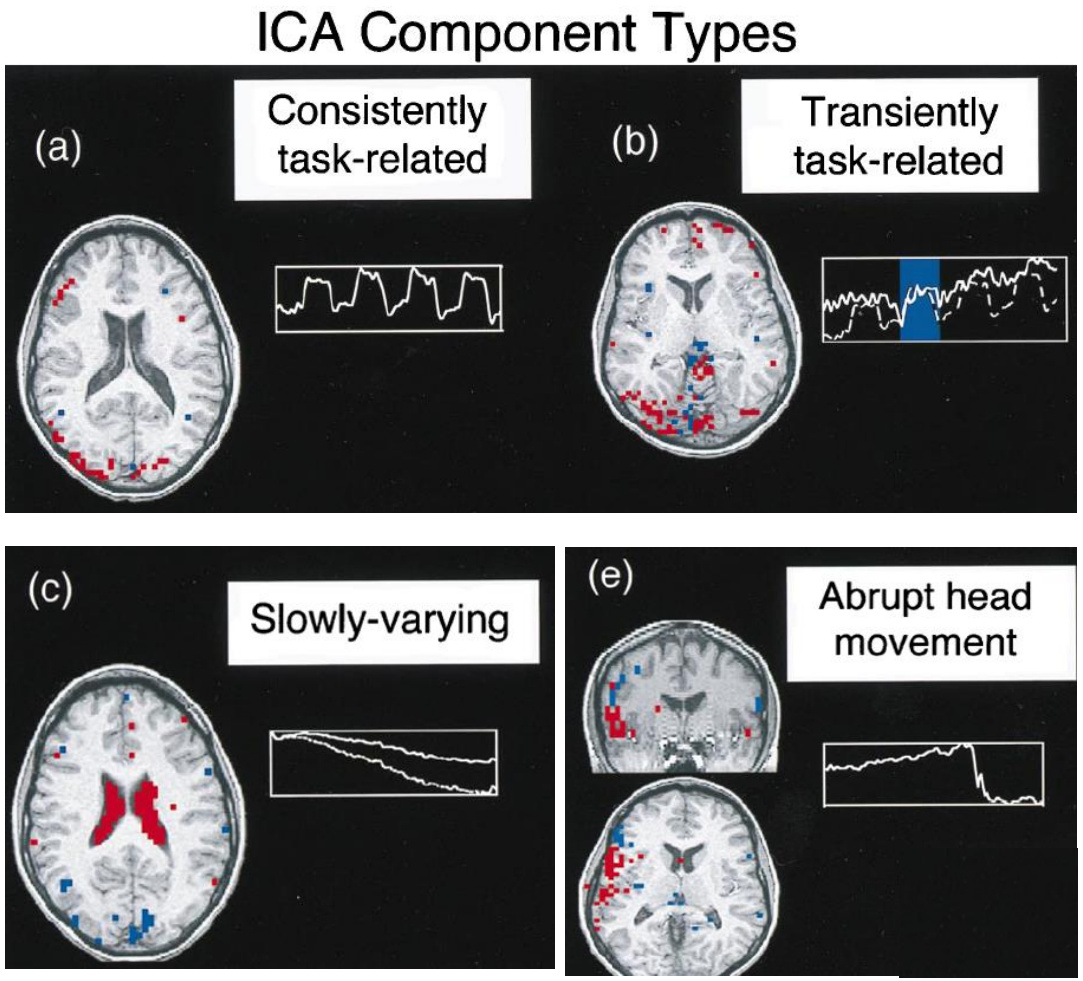
1. Different brain areas/subjects have different HRF;
2. Special tasks: listening music, watching movie;
3. Resting-state data;
4. Data with heavy noise and artifacts;
5. You are not only interested with task-evoked effect,
   but also other brain networks, e.g., transient effect;
6. … … …

McKoewn et. al., 1998, Hum.Brain.Mapp.

ICA的应用1：任务态fMRI数据分析

上海科技大学 ShanghaiTech University

Can we find task activations instead of using SPM & GLM?

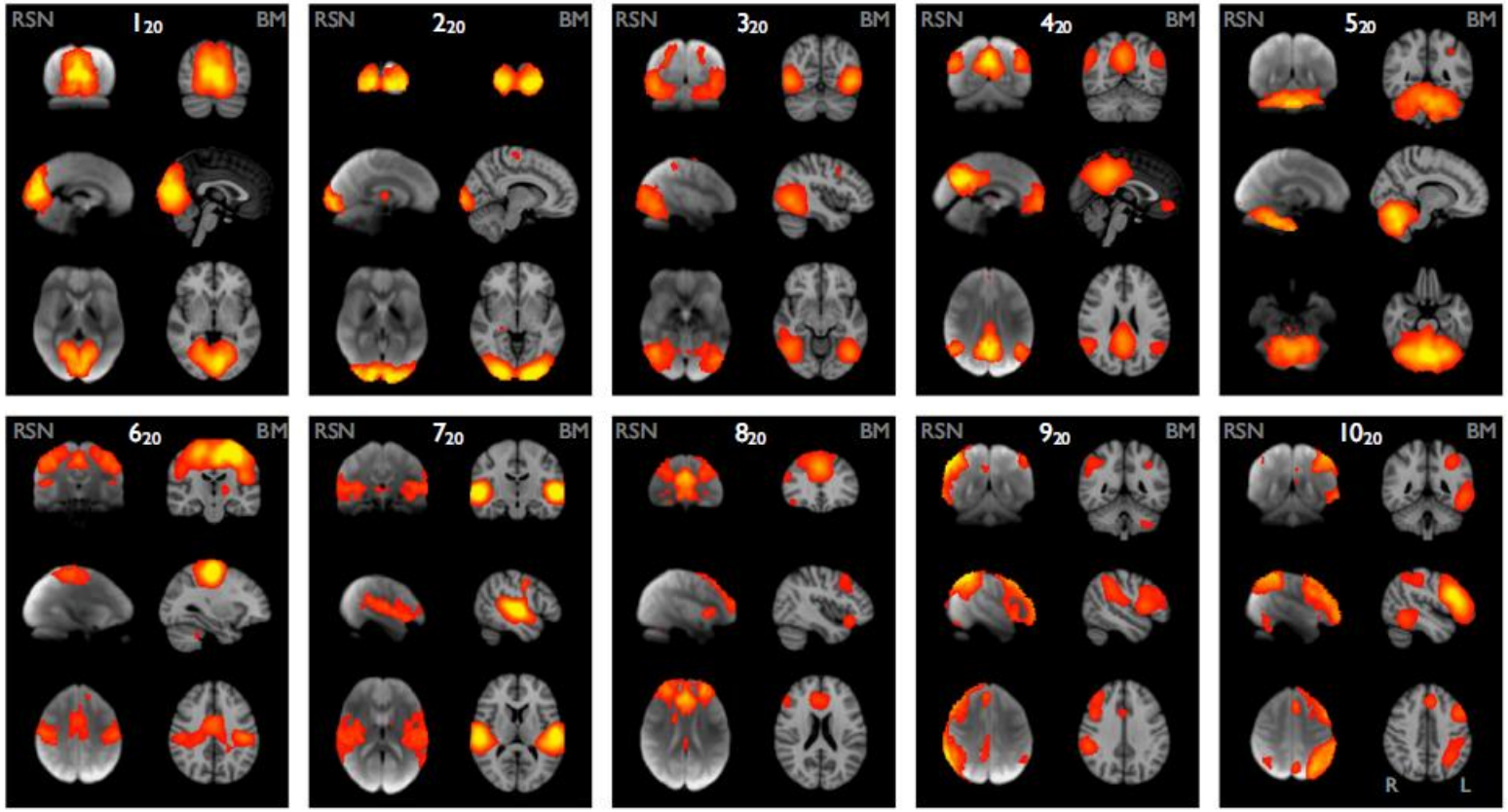


Mckeown et al., 1998,PNAS, “Stroop color-naming task”

ICA的应用2：静息态功能网络

上海科技大学
ShanghaiTech University

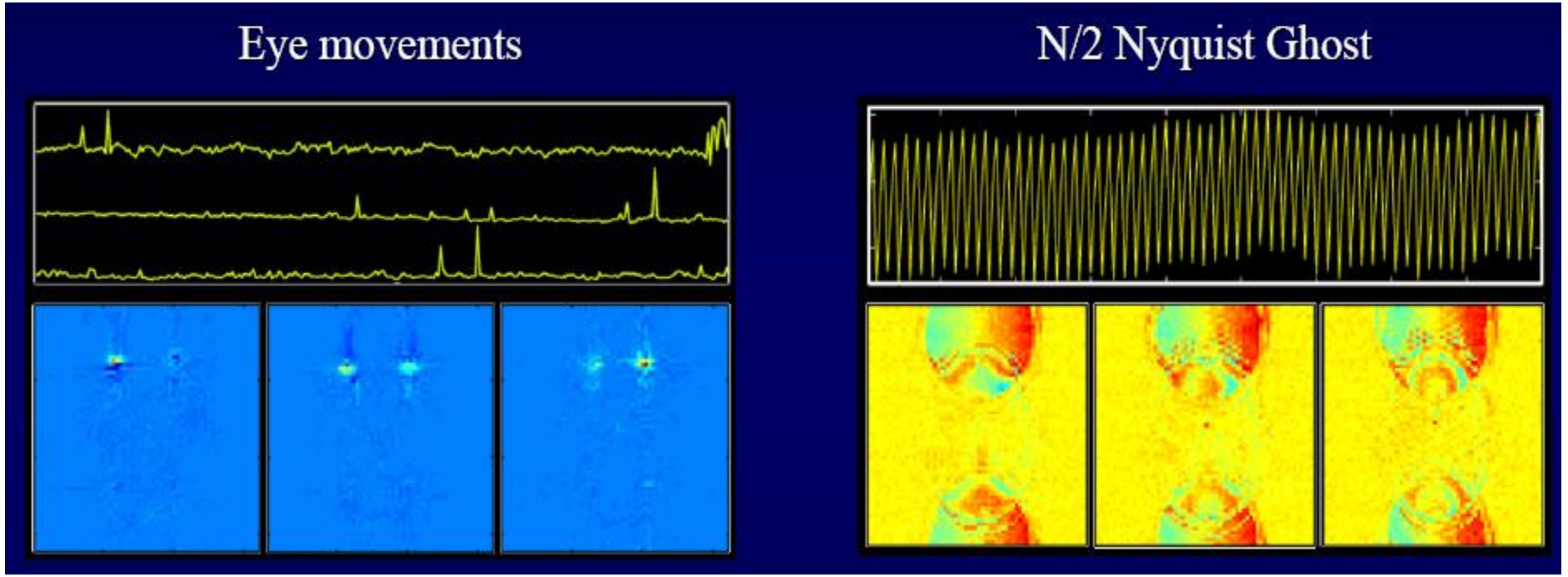
Can we find various resting-state functional networks?



Smith et al., 2009, PNAS, “Correspondence of the brain’s functional architecture during activation and rest”

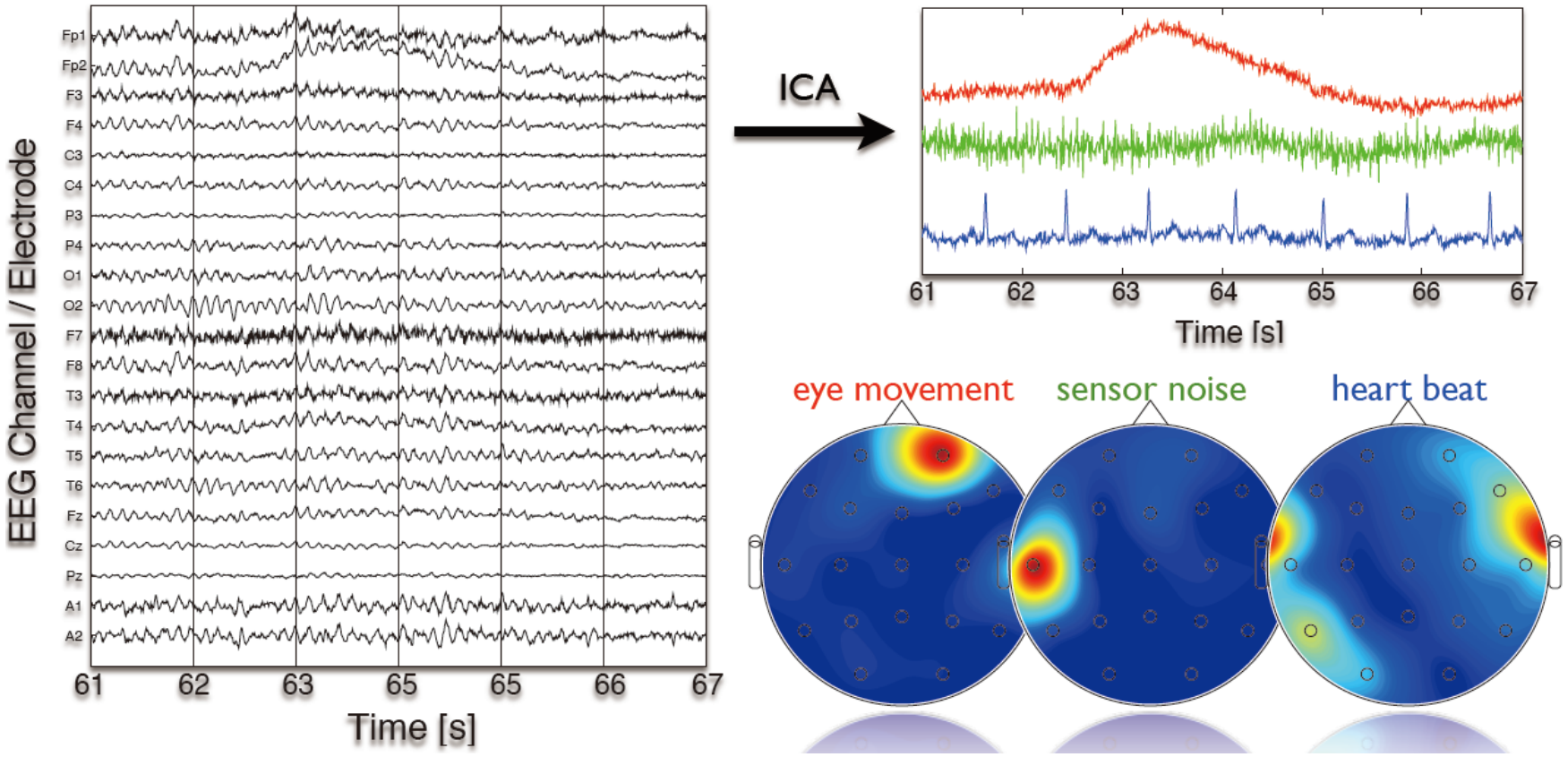
ICA的应用3：fMRI数据去噪

Can we detect and remove artifacts with no a priori knowledge?



Source: Christian Beckmann's "Little shop of fMRI Horrors"

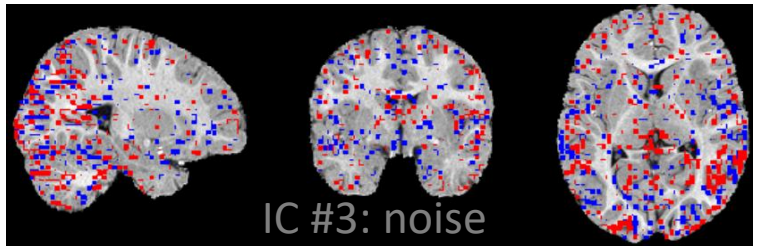
ICA的应用：EEG数据分析

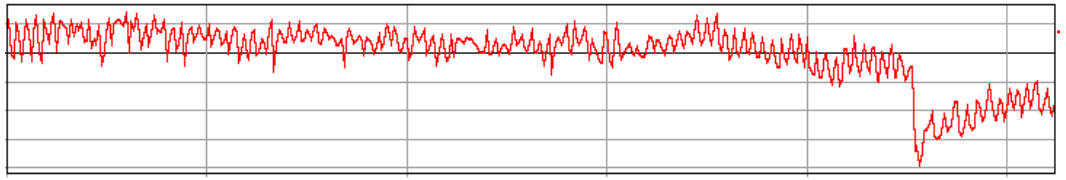


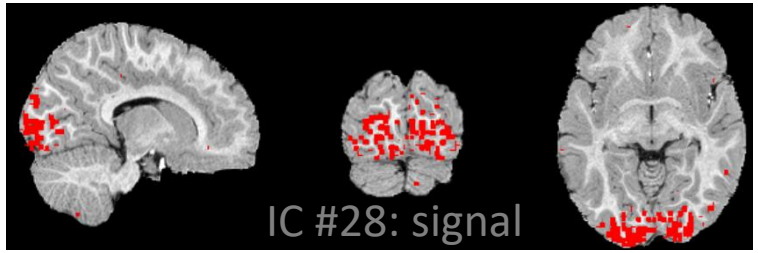
Fabian Theis, From source to network: multiscale biostatistics, APP 2007

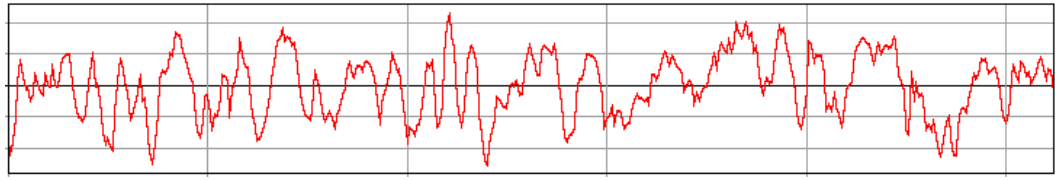
婴幼儿扫描

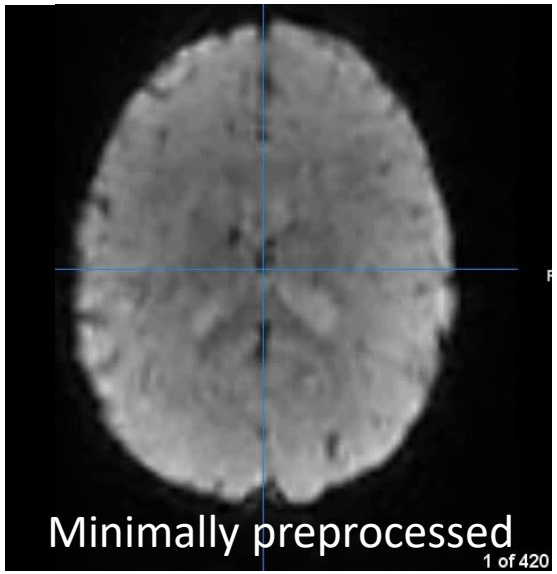
婴幼儿fMRI去噪

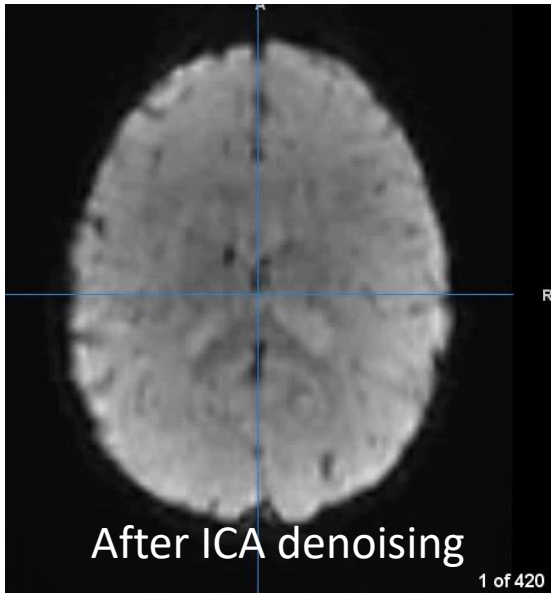

IC #3: noise



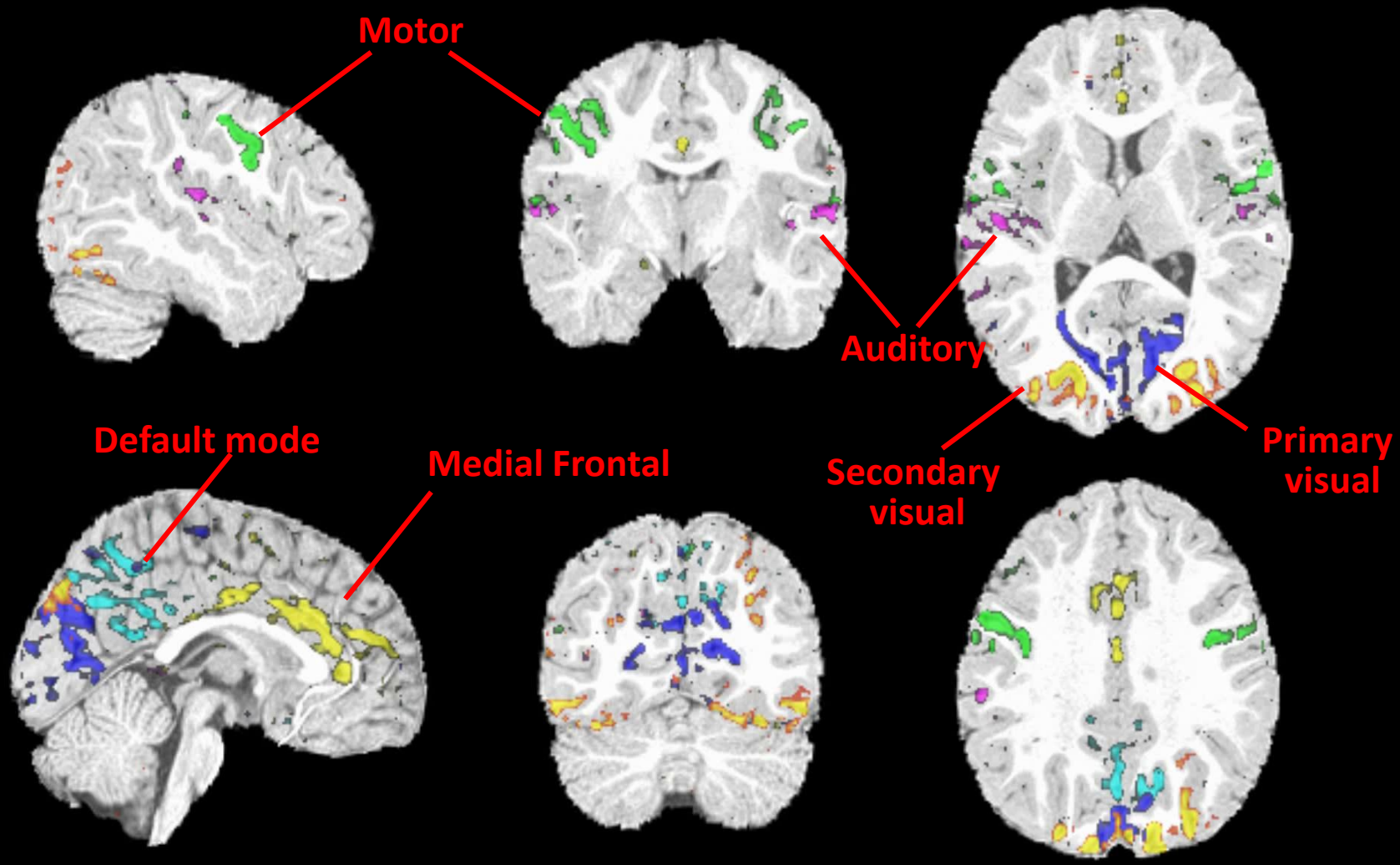

IC #28: signal




Minimally preprocessed
1 of 420

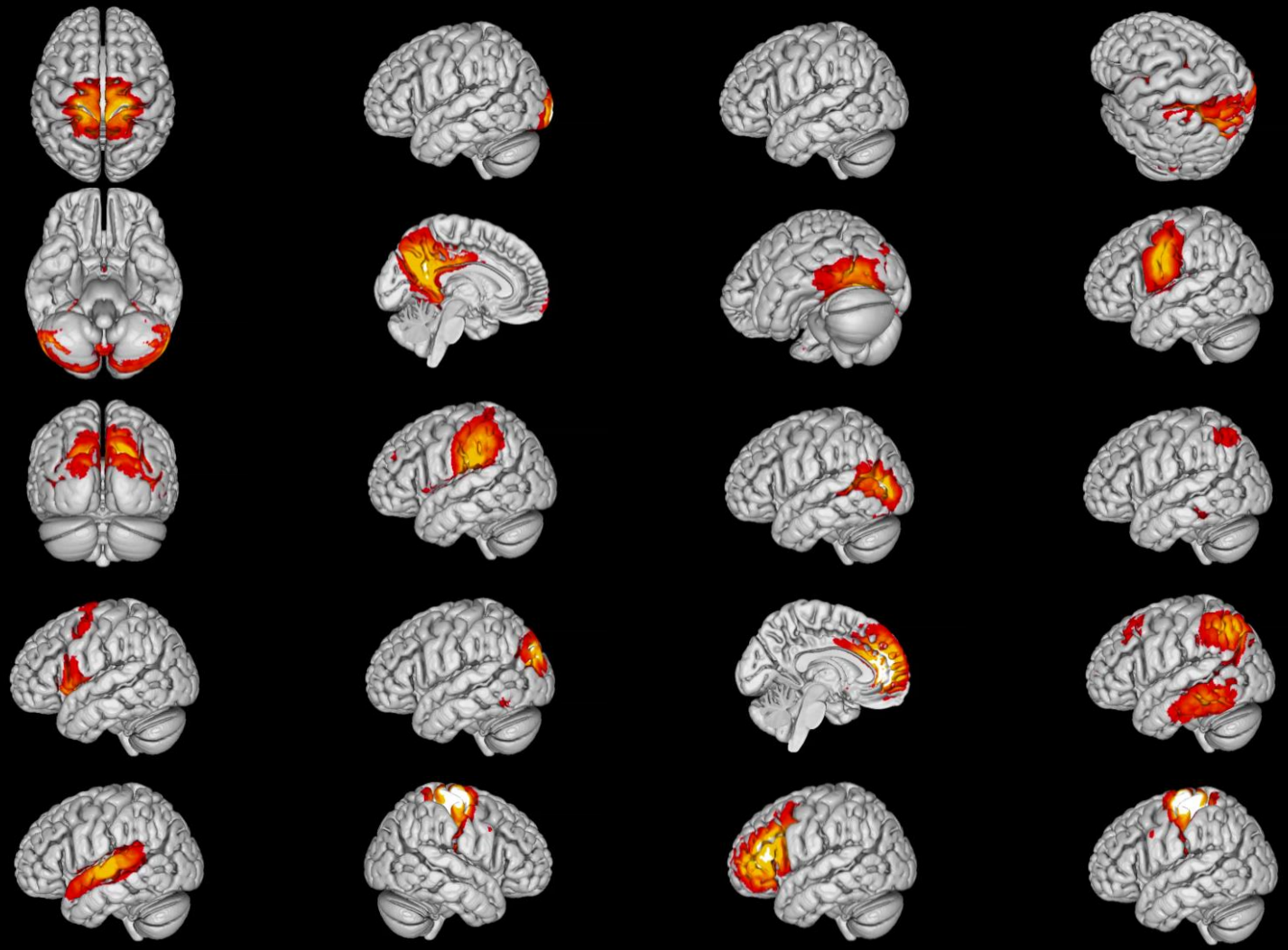

After ICA denoising
1 of 420

美国BCP：个体化分析（功能网络）



ICA: Independent Component Analysis
The data is from one BCP subject at ~1 year old

婴幼儿脑影像智能分析：功能图谱（0-2岁）

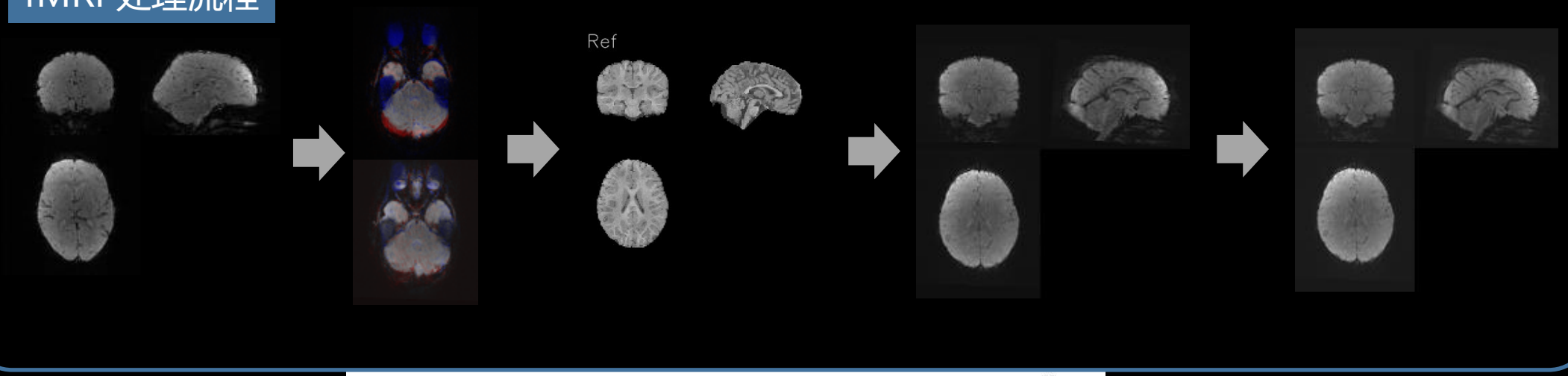


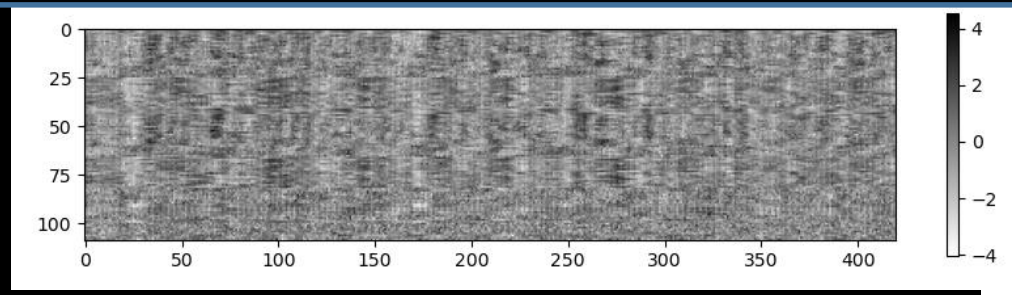
The data was from a longitudinal infant brain development project “Multi-visit Advanced Pediatric Brain Imaging Study (MAPS) for Characterizing Structural and Functional Development”.

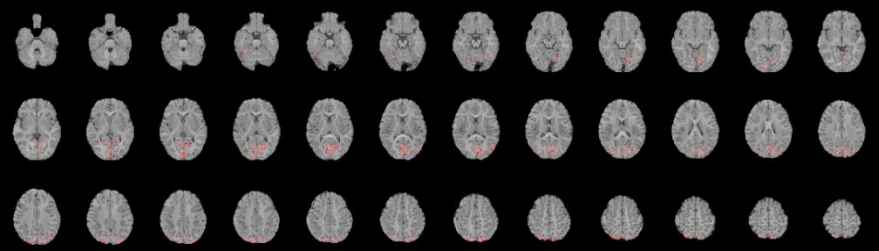
脑疾病与智能计算研究组
Brain Disorder and Intelligent Analytics Laboratory (DIAL)

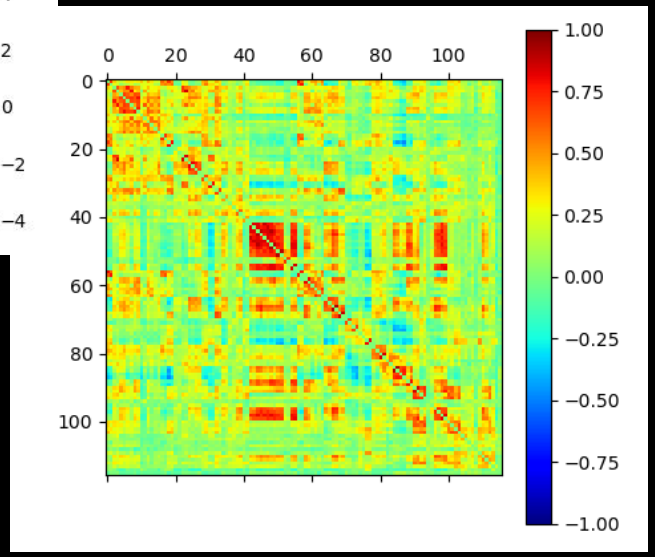
婴幼儿功能MRI数据分析技术研发

fMRI 处理流程









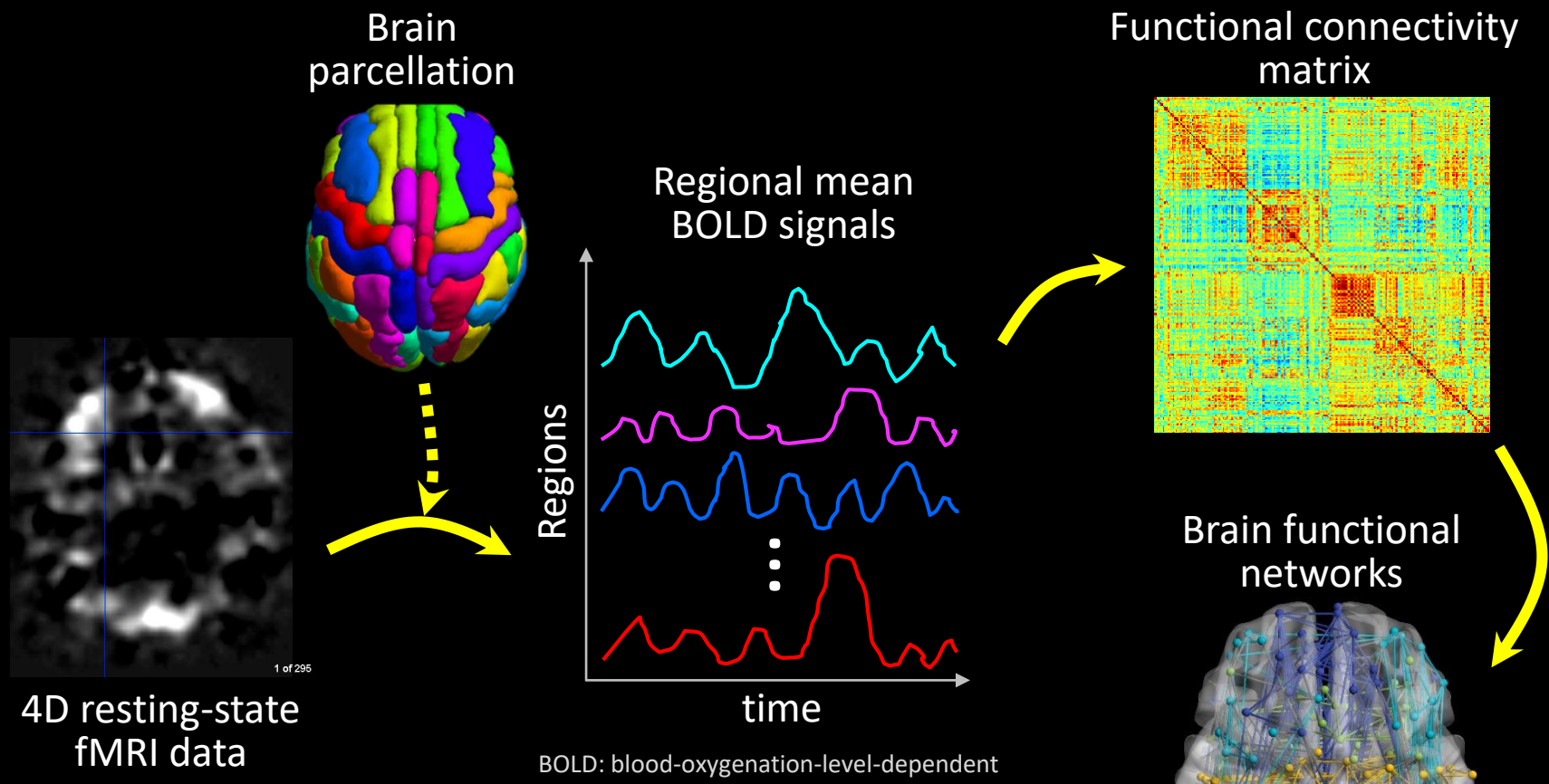
解决问题

- 模型驱动：定种子点→全脑相关分析 (Pearson correlation)
- 数据驱动：独立成分分析

问题延伸

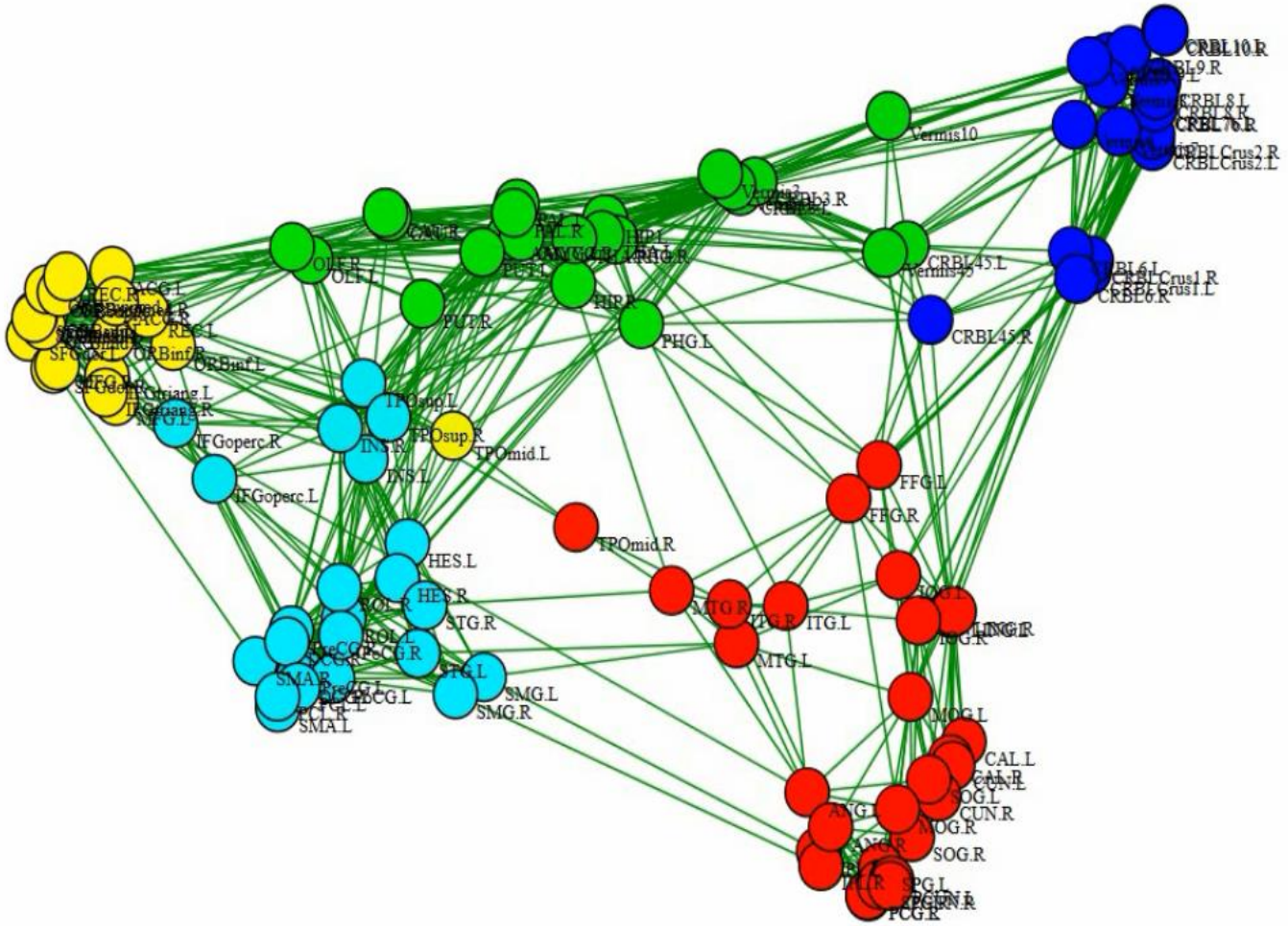
- 想做模型驱动但是不知道种子点怎么定怎么办？

Brain functional connectivity network



Most previous studies investigated static brain functional connectome

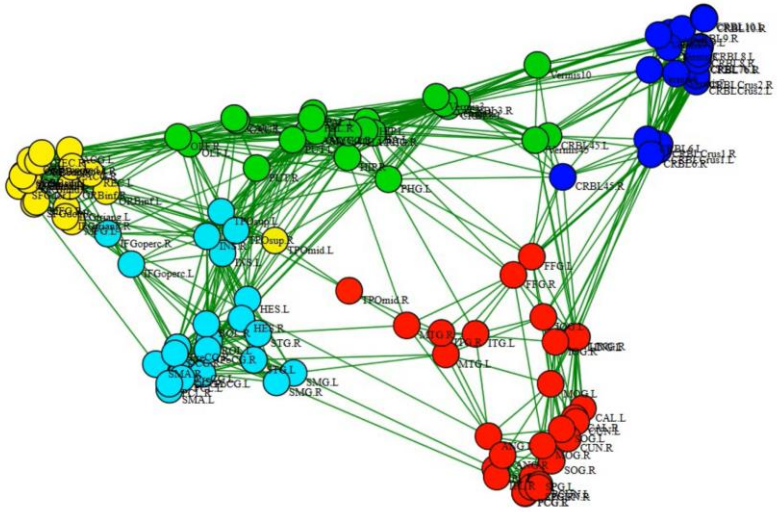
婴幼儿脑网络



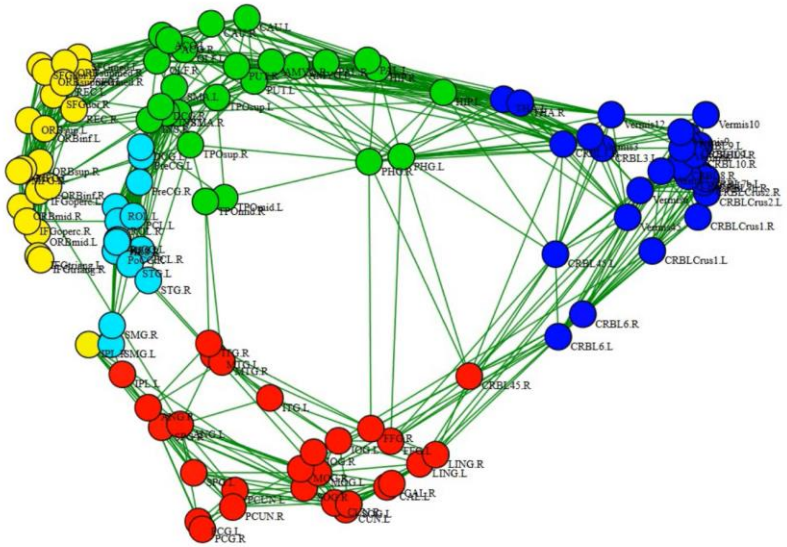
婴幼儿0-2岁全脑功能网络的拓扑结构发育

婴幼儿脑网络

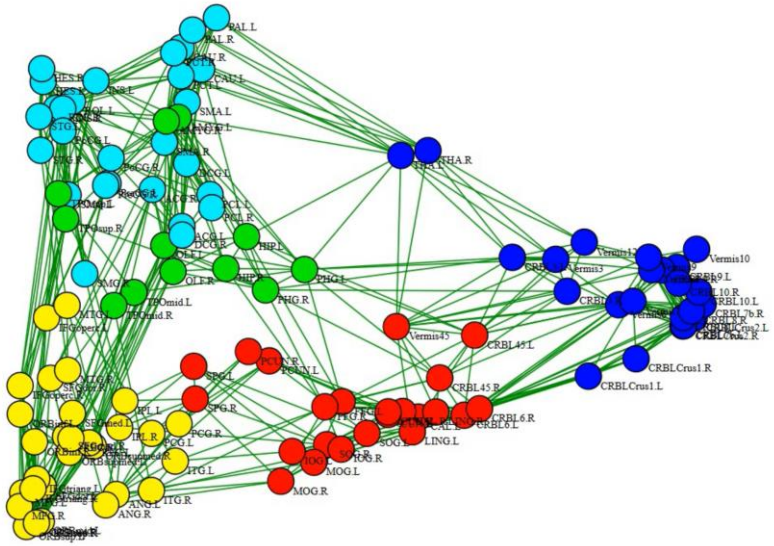
上海科技大学
ShanghaiTech University



新生儿（0岁）



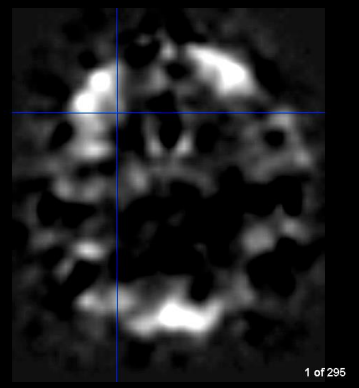
婴儿（0.5岁）



幼儿（2岁）

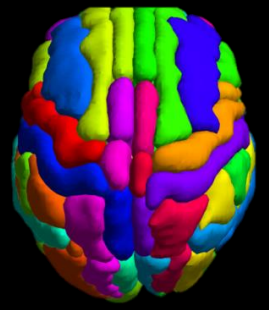
随年龄增长，功能网络从相对分离，持续强化，到网络之间出现“混叠”

Brain dynamic functional connectivity network

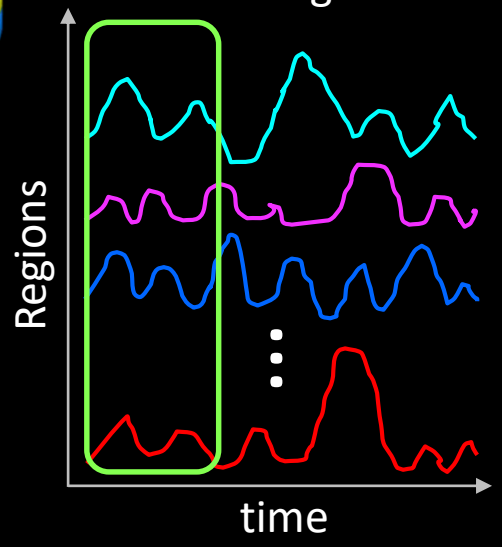


4D resting-state fMRI data

Brain parcellation

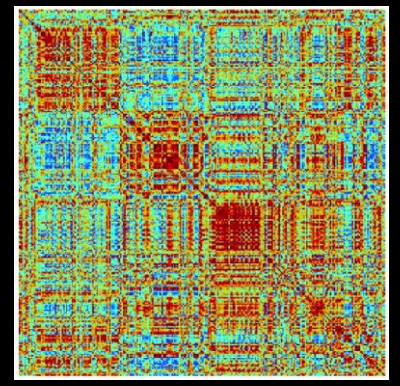


Regional mean BOLD signals

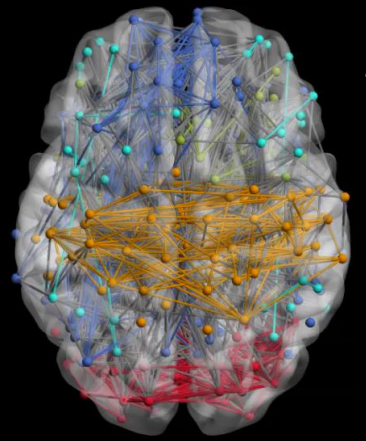


BOLD: blood-oxygenation-level-dependent

"Dynamic" functional connectivity matrix



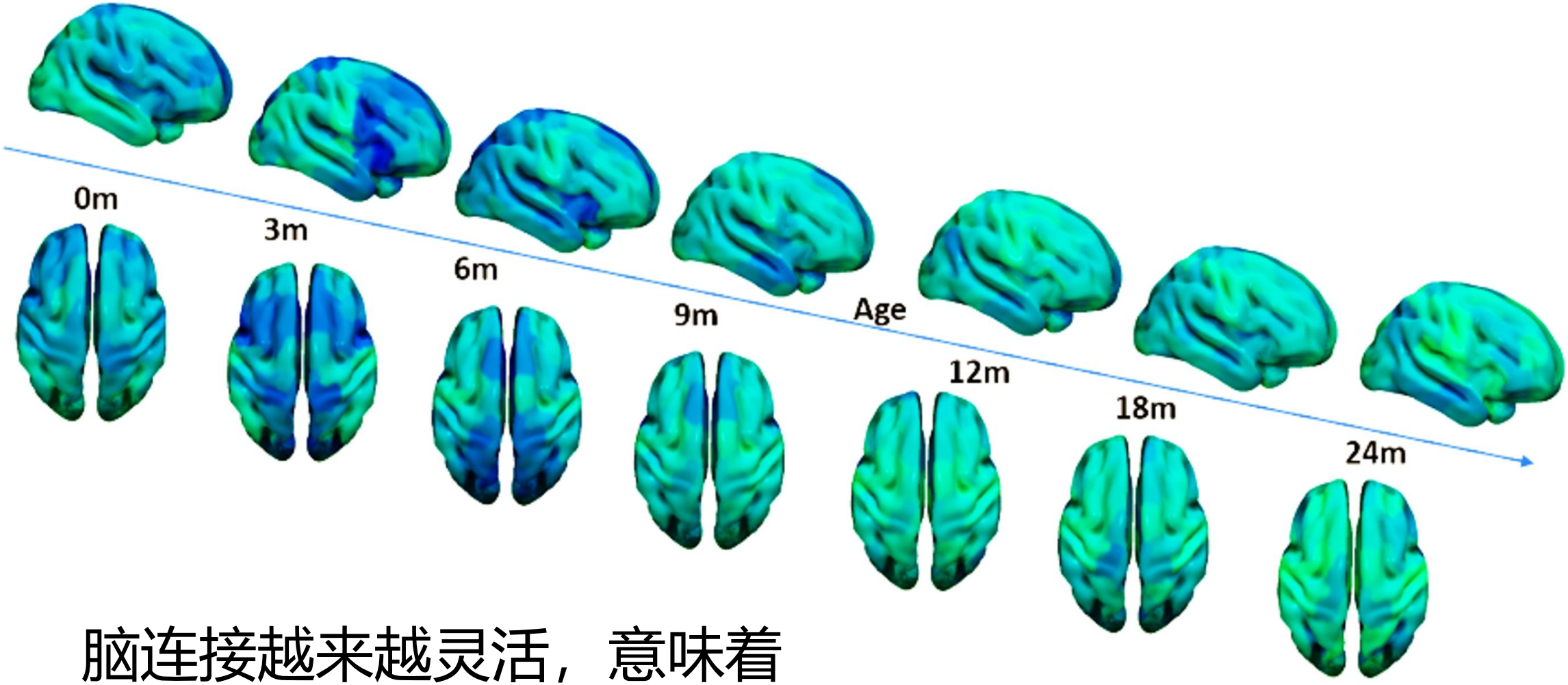
"Dynamic" brain functional networks



Brain FC is dynamic, time varying, adaptive, biologically meaningful

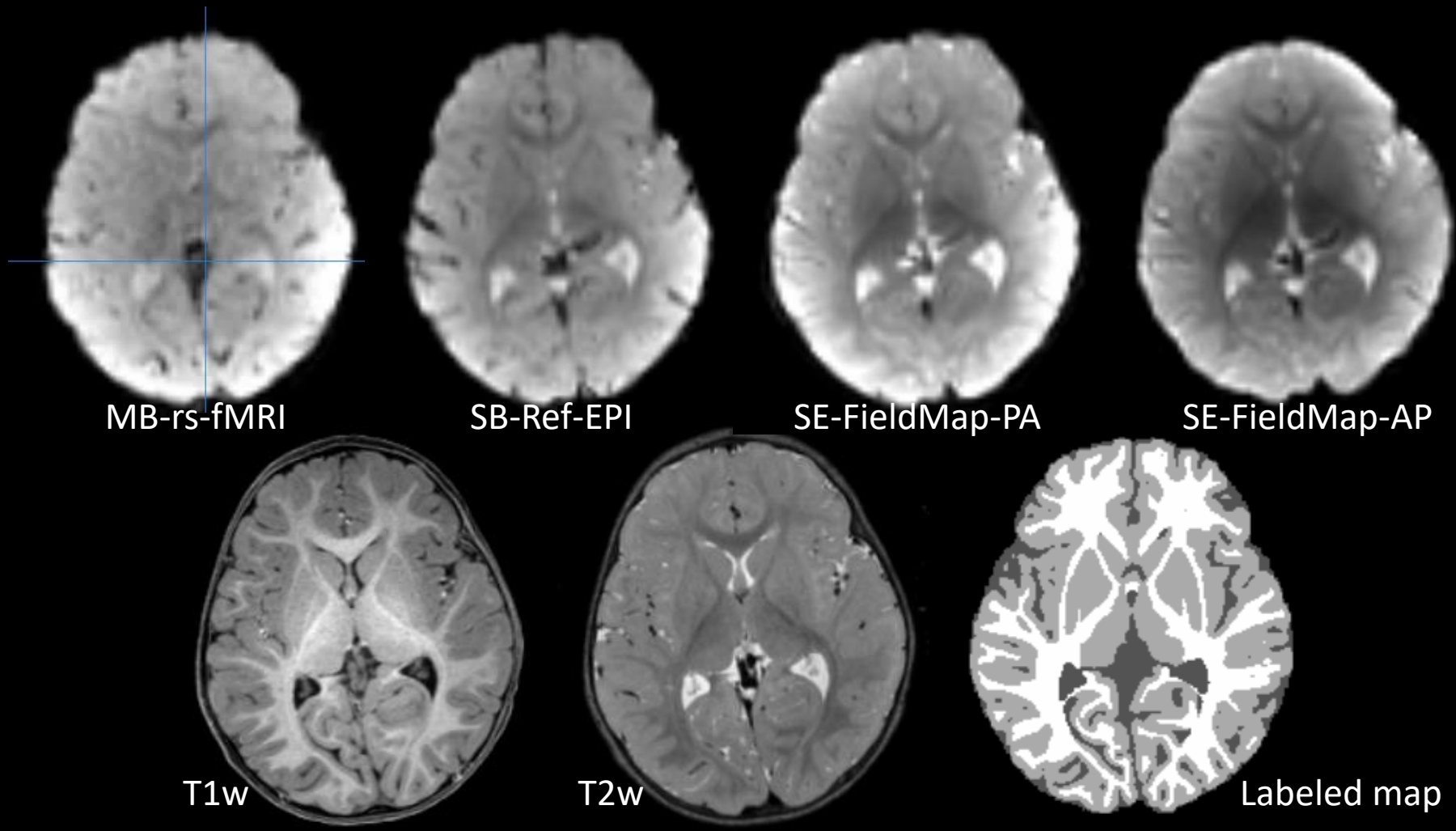
婴幼儿脑动态功能连接分析：功能灵活度

上海科技大学
ShanghaiTech University



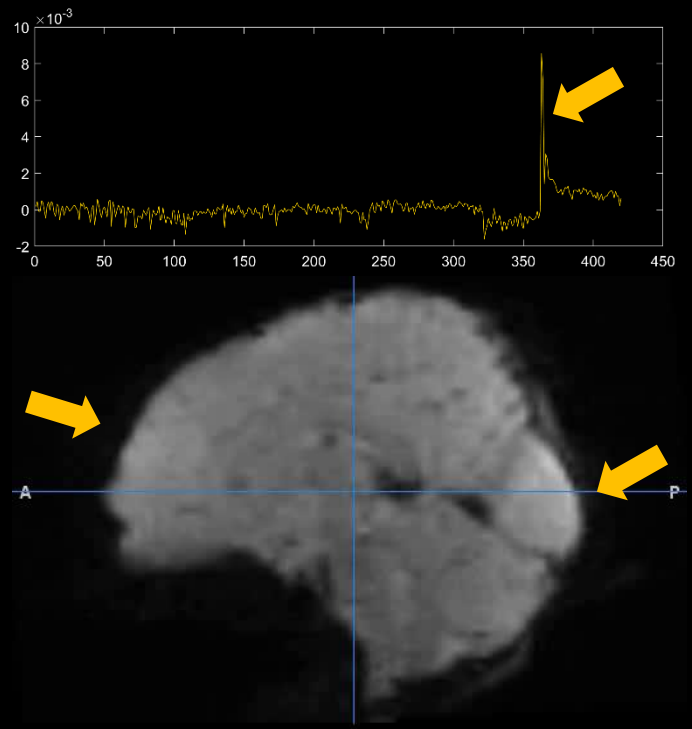
脑连接越来越灵活，意味着
可以做越来越复杂，需要高度协调的运动和认知活动

Sample data using BCP protocol

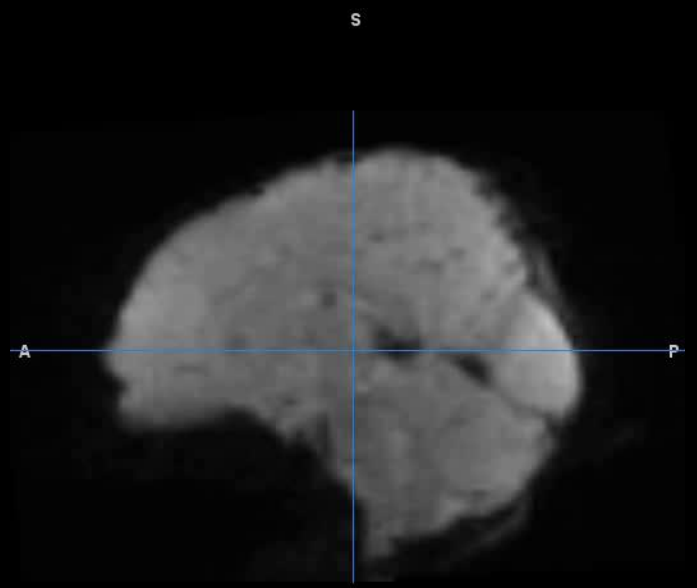


SB: single-band EPI, MB: multiband EPI, EPI: echo-planar image,
SE: spin-echo imaging, Ref: high-contrast single volume using SB-EPI
PA/AP: phase-encoding directions, P: posterior, A: anterior

Head motion correction

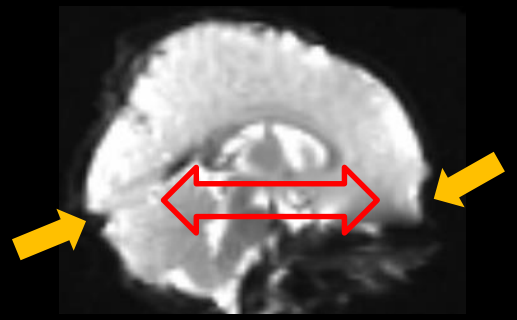


Before correction

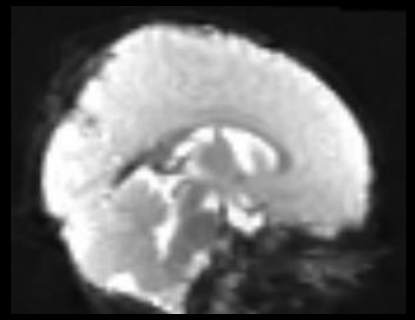


After correction

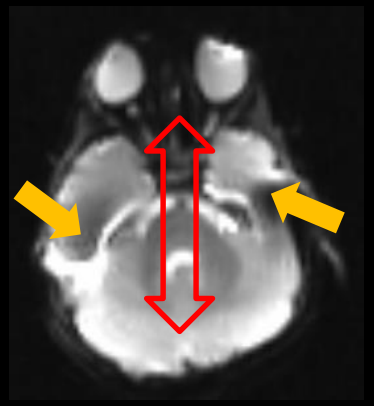
Echo planar imaging distortion and correction

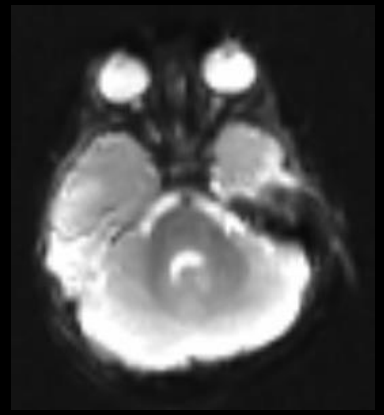


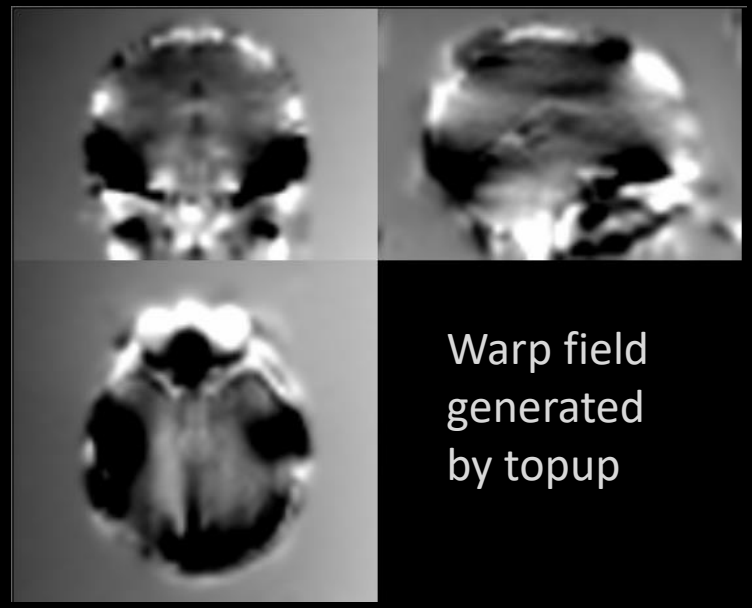
Before correction



After correction

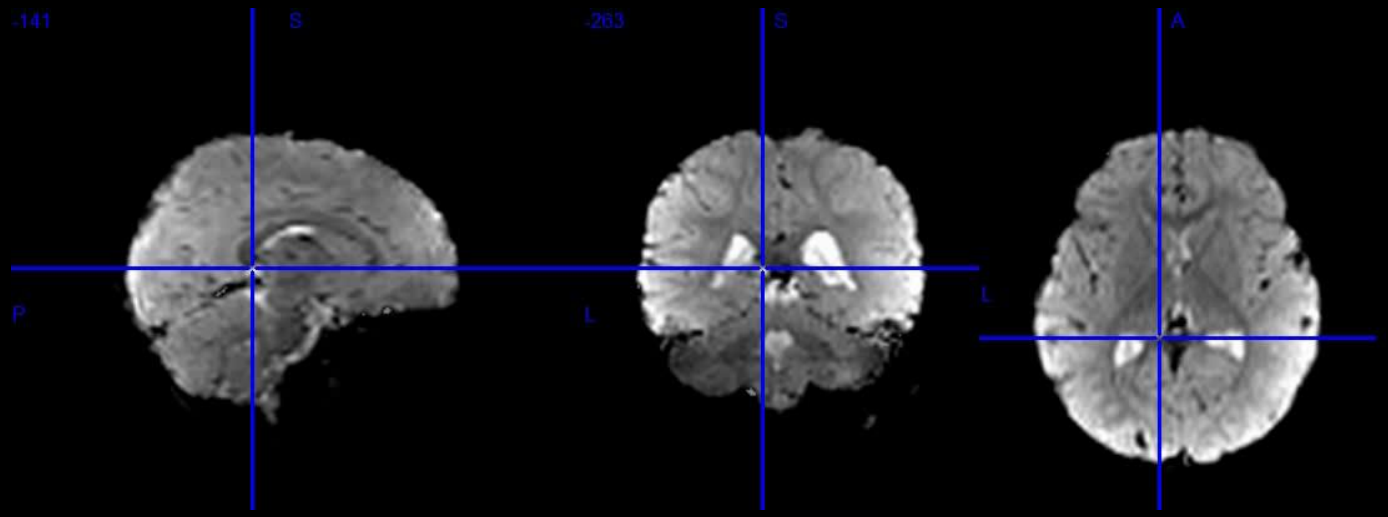






Warp field generated by topup

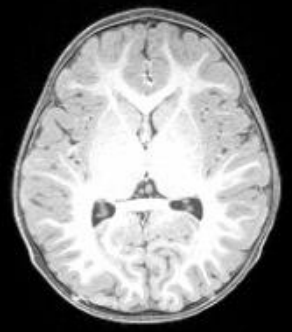
EPI to Anat-Image (T1w, T2w) registration



SB-ref image overlaid on the labeled map (generated by segmentation)

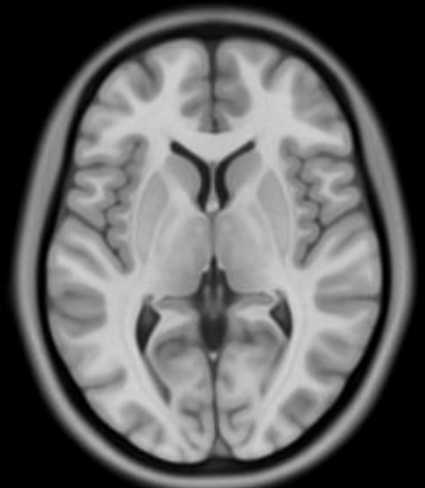
Tissue label image-based anat-func registration

Individual space



T1w image

Difficult



T1w image

MNI space (ICBM152-T1-asym)

Individual space

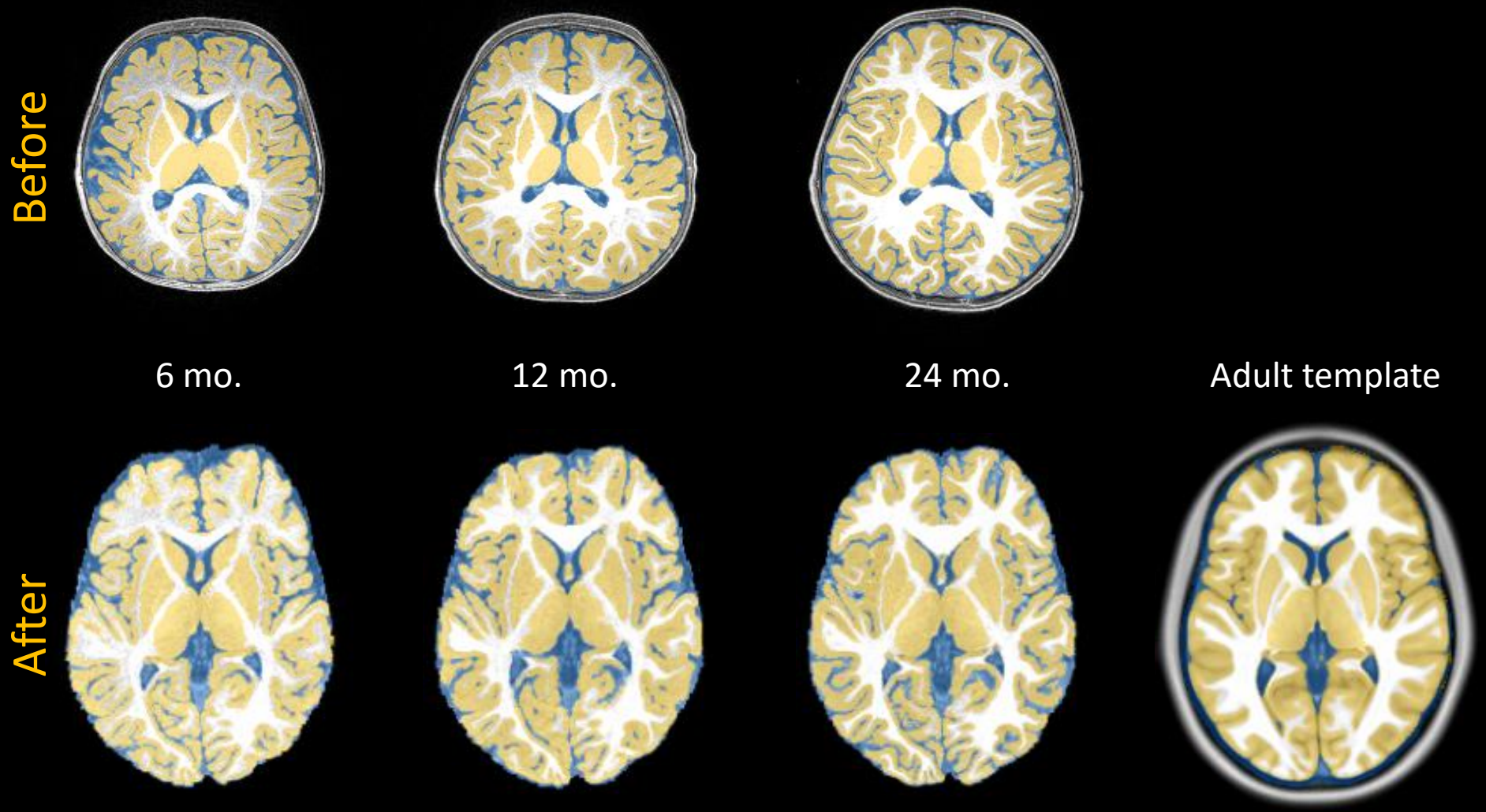
Labeled image

Easy

Labeled image

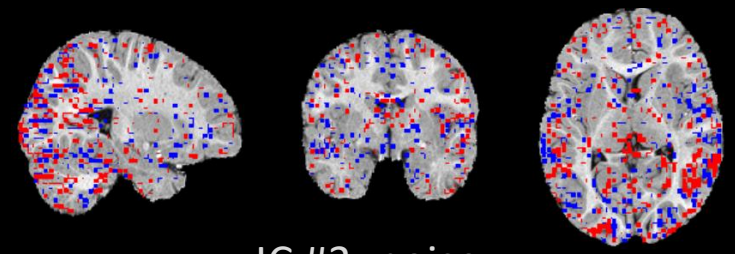
MNI space (ICBM152-T1-asym)

Result: well aligned longitudinal scans of a subject

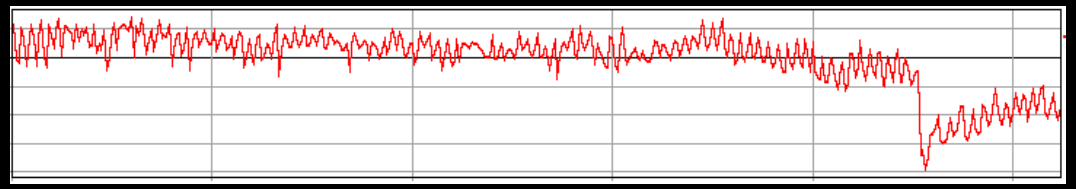


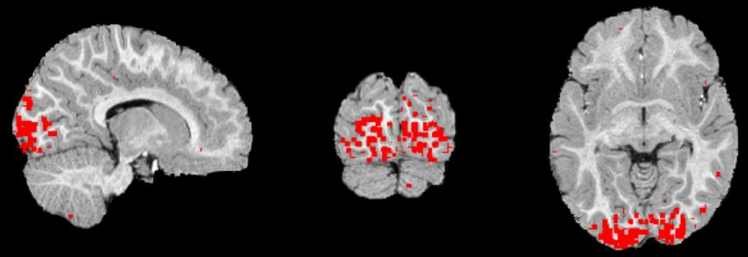
Result: well aligned images from different subjects

Extensive preprocessing (denoising)

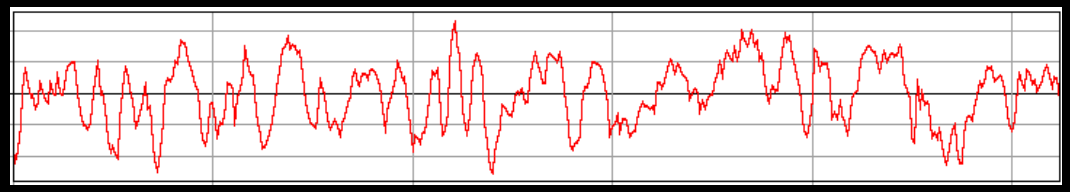


IC #3: noise



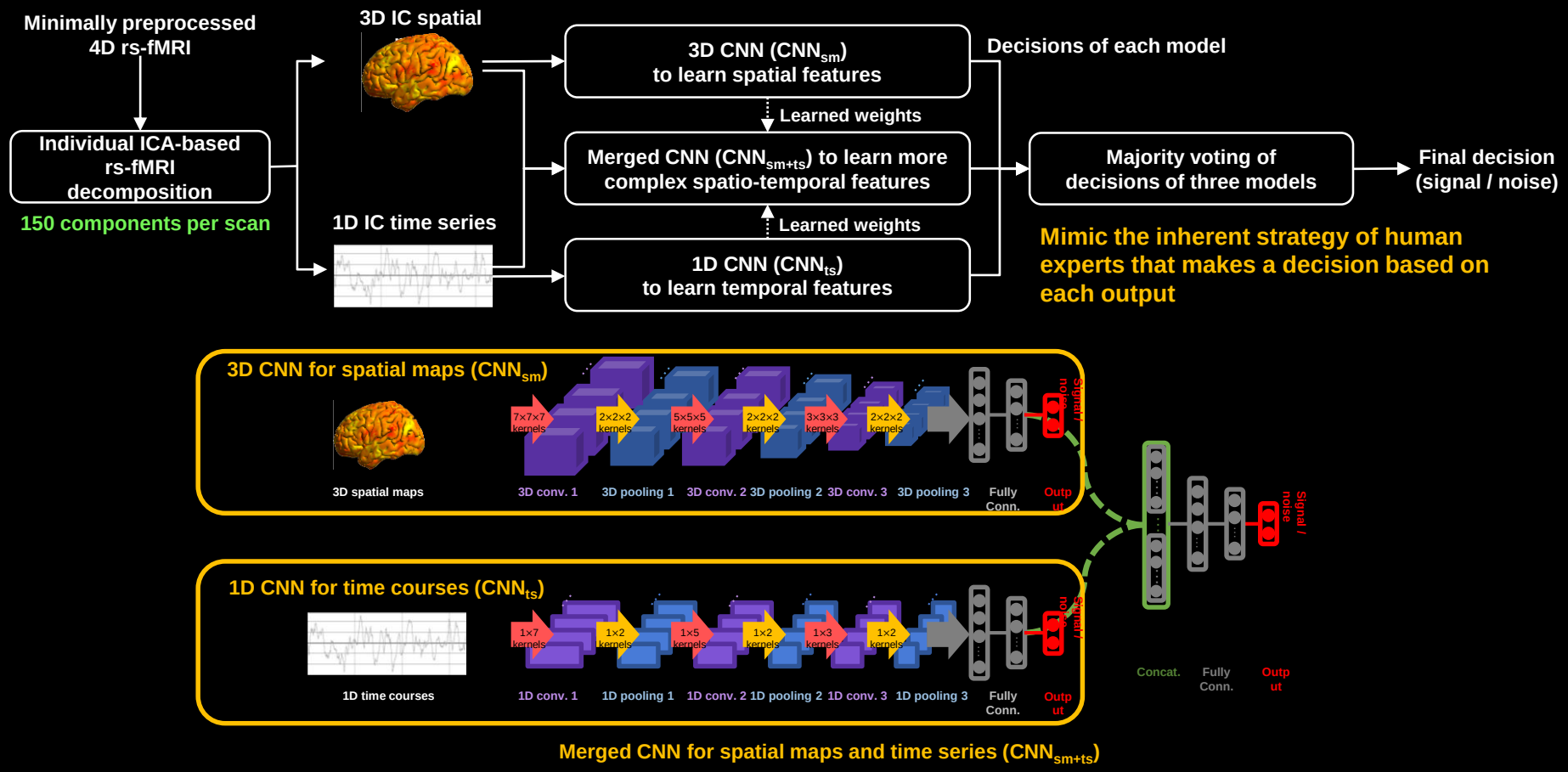


IC #28: signal



Extensive preprocessing (deep-learning-based denoising)

Deep learning-based Automatic fMRI Denoising



(Kam, Zhang et al., OHBM 2019, MICCAI 2019)

Accuracy = 97.12 % ~20 s per subject

脑疾病与智能计算研究组
Brain Disorder and Intelligent Analytics Laboratory (DIAL)

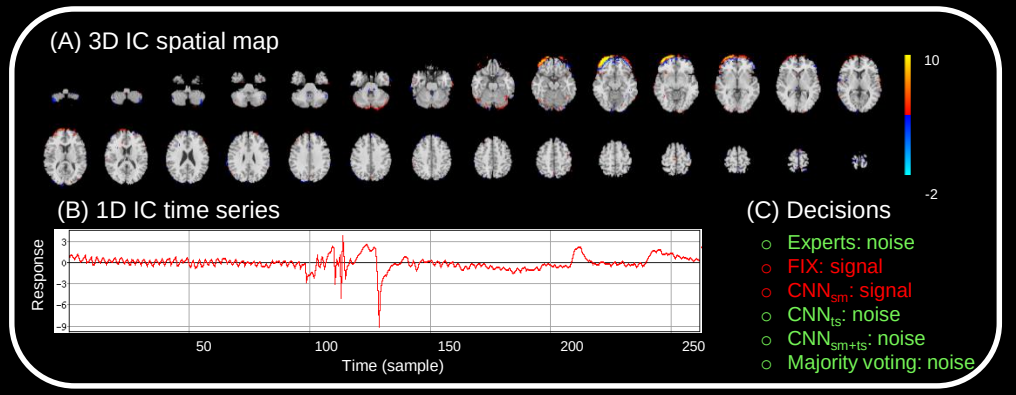
Deep learning-based fMRI denoising: result

Method	ACC (%)	SEN (%)	SPEC (%)
FIX	96.19 ± 3.78	94.48 ± 12.52	96.54 ± 3.42
CNN_{sm}	96.52 ± 2.02	93.47 ± 7.59	97.35 ± 2.18
CNN_{ts}	95.34 ± 3.61	94.31 ± 8.19	95.49 ± 4.43
CNN_{sm+ts}	96.98 ± 1.92	94.33 ± 7.56	97.66 ± 1.93
Majority voting	97.12 ± 1.89	94.50 ± 7.96	97.77 ± 1.96

Our deep learning-based method accurately classified 128 components more compared to the FIX

Method	Time spent for a subject (for a component) (s)	
	CPU	GPU
FIX	2,402 ± 78.84 (~16.01)	-
CNN_{sm}	1,061 ± 28.12 (~7.07)	12.12 ± 0.33 (~0.08)
CNN_{ts}	43 ± 0.24 (~0.29)	3.01 ± 0.22 (~0.02)
CNN_{sm+ts}	1,146 ± 39.68 (~7.64)	12.46 ± 0.26 (~0.08)
Majority voting	2,291 ± 79.33 (~15.27)	28.77 ± 0.94 (~0.19)

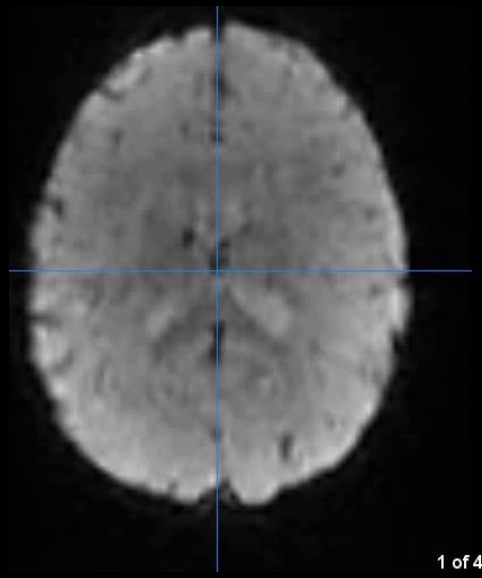
Our method is significantly fast and takes less than 1*s* to label one component using GPU computing



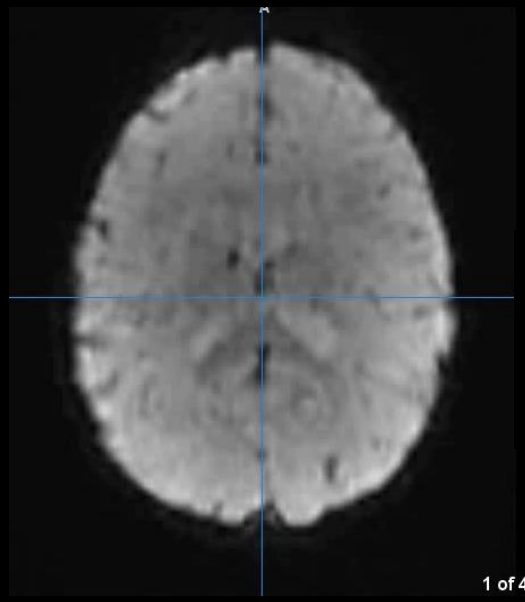
An example of decision making by each model

- The proposed method is more accurate (>97%) and much faster (<1s)
- It succeeds in application to all the Brain Connectome Project data (N=630)
- It is successively integrated into an automatic infant fMRI pipeline

Extensive preprocessing (denoising)

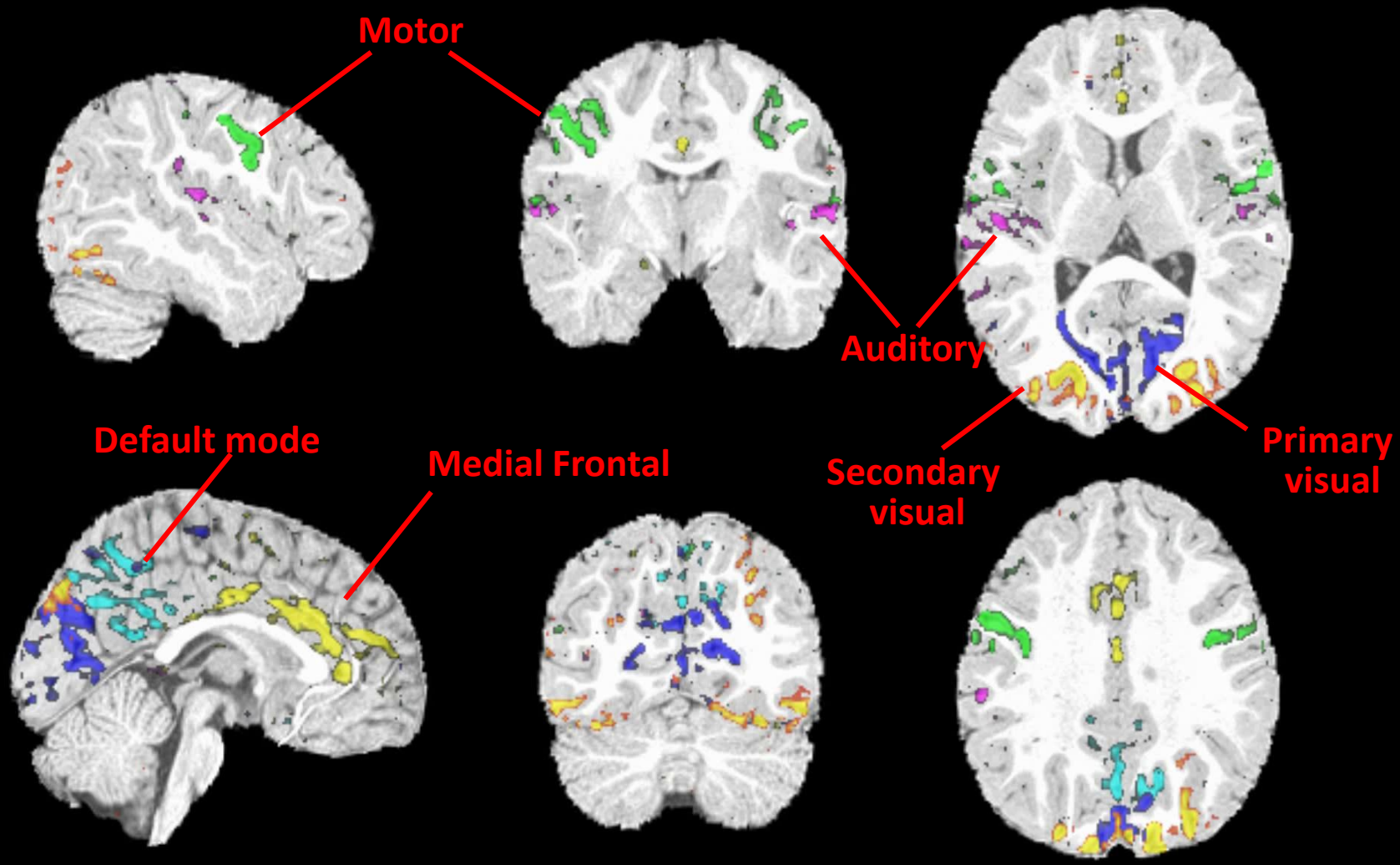


Minimally preprocessed



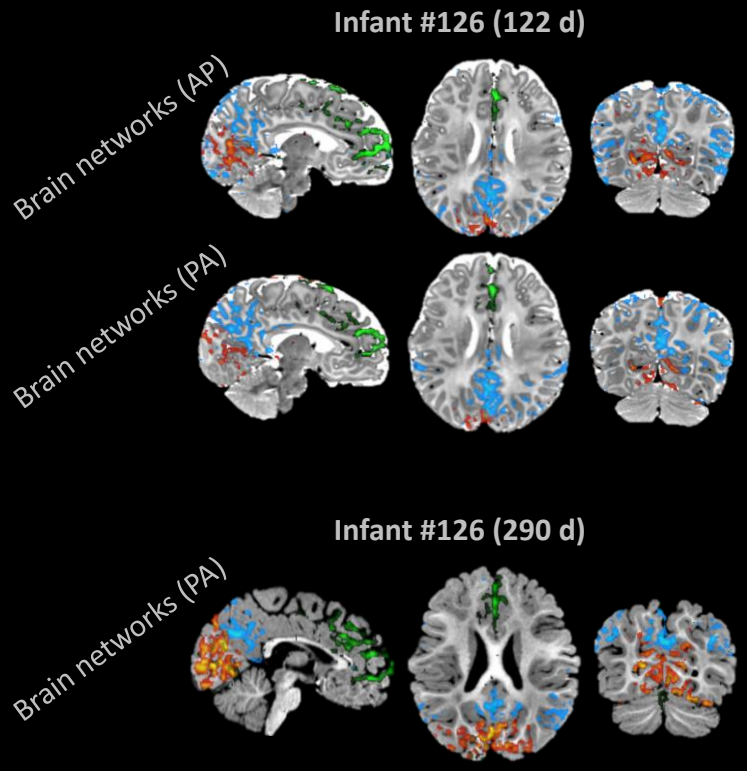
After denoising

Functional network extracted using ICA

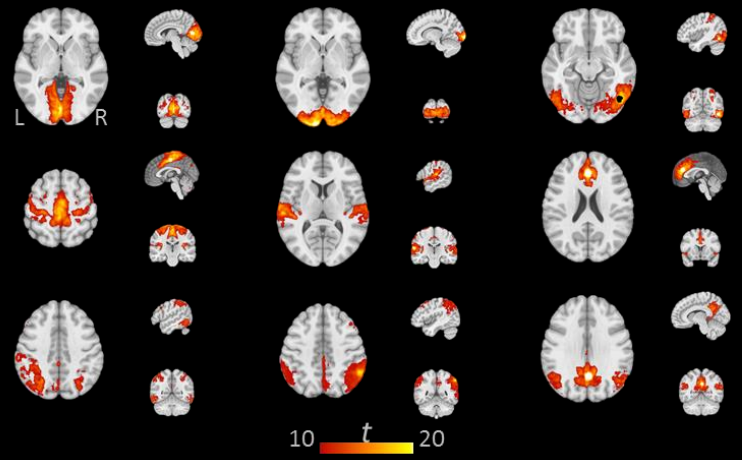


ICA: Independent Component Analysis
The data is from one BCP subject at ~1 year old

Initial results

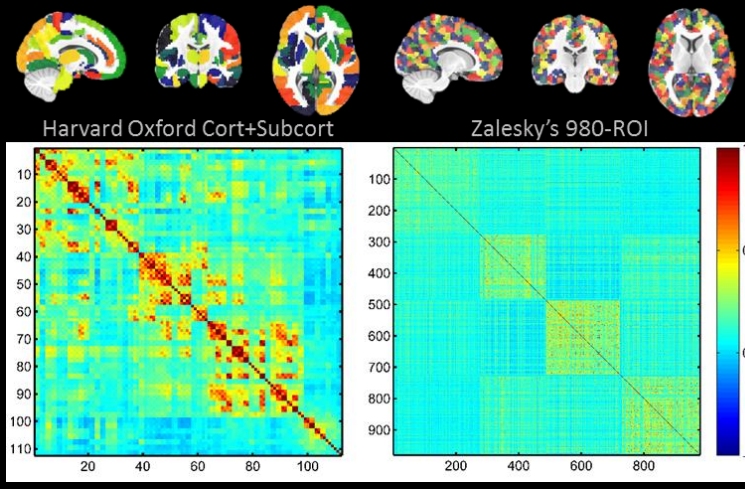


Seed-based correlation



(9 seeds are derived from HCP)

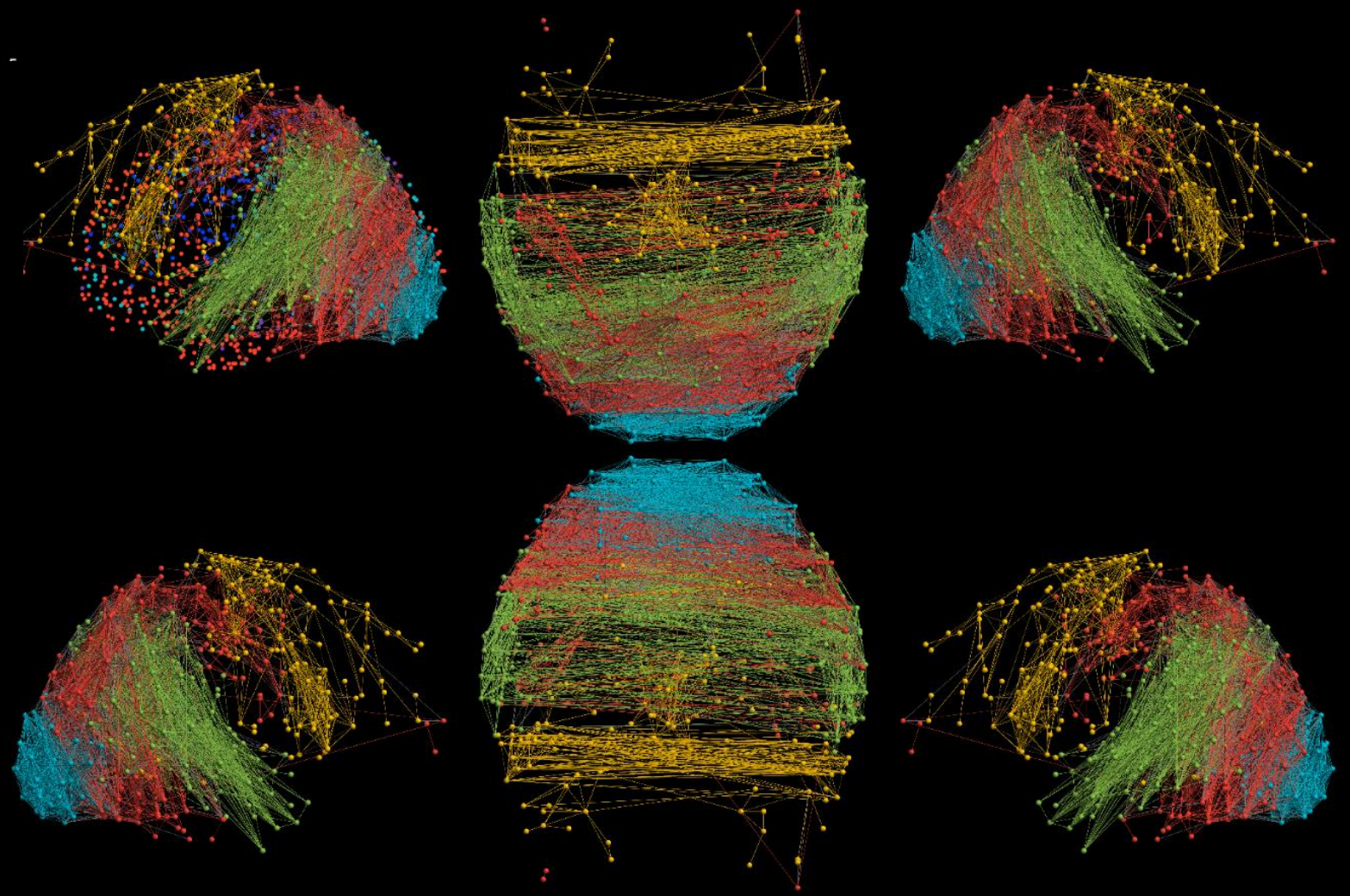
Pair-wise ROI-based correlation



(Atlas-1: 112 regions, atlas-2: 980 regions, matrix rearraged based on the detected modules, 20% density, 148 fMRI data from 60 infants aged 270-360 days)

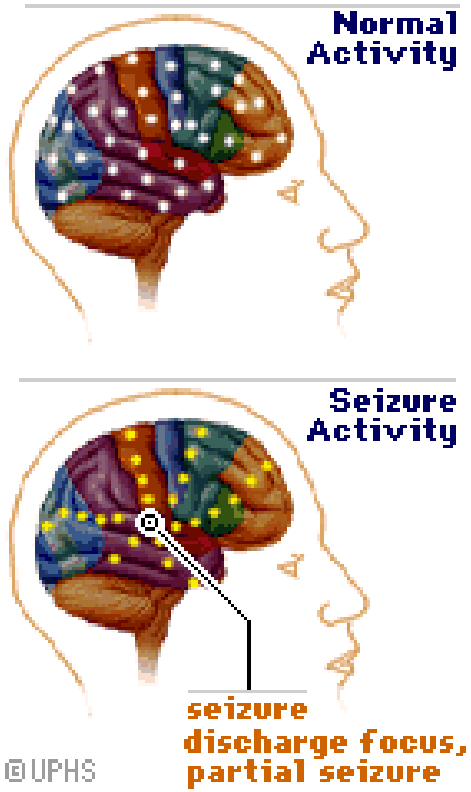
(Han Zhang et al., OHBM 2019)

Large-scale brain functional connectome

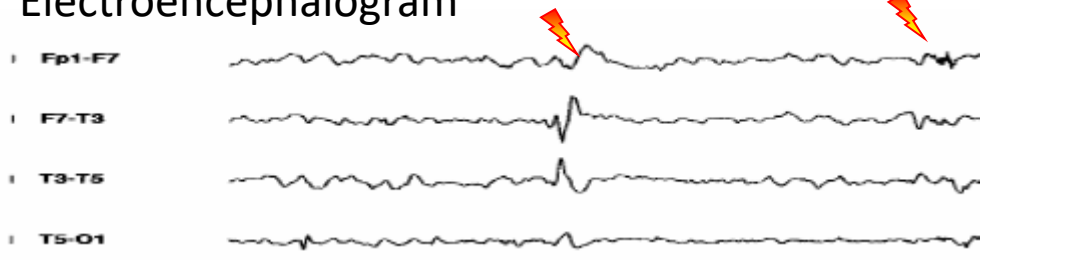


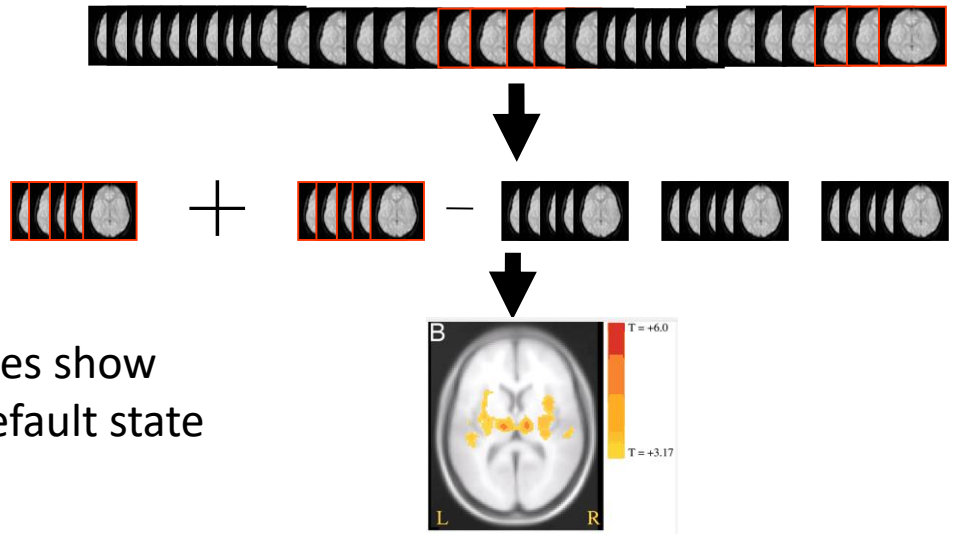
FC network (Pearson's correlation), sparsity = 1%
The data is from one BCP subject at ~1 year old

Partial seizure discharge focus localization



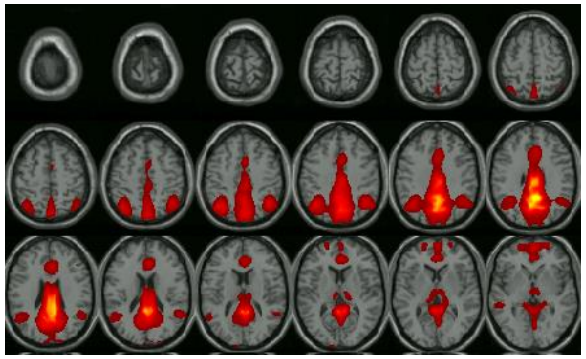
Electroencephalogram



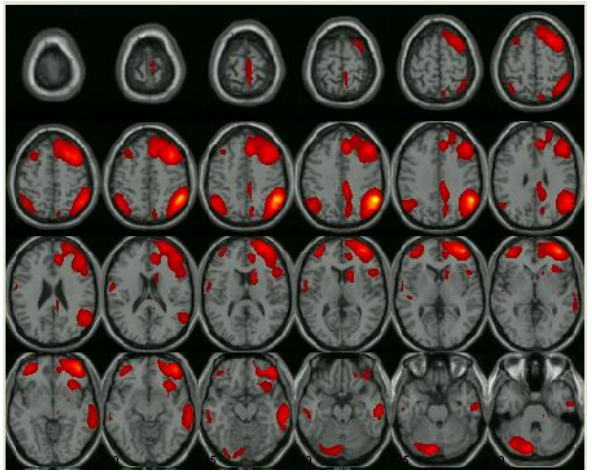


Gotman et al. (2005) Generalized epileptic discharges show thalamocortical activation and suspension of the default state of the brain. PNAS 102 (42) 15236-15240

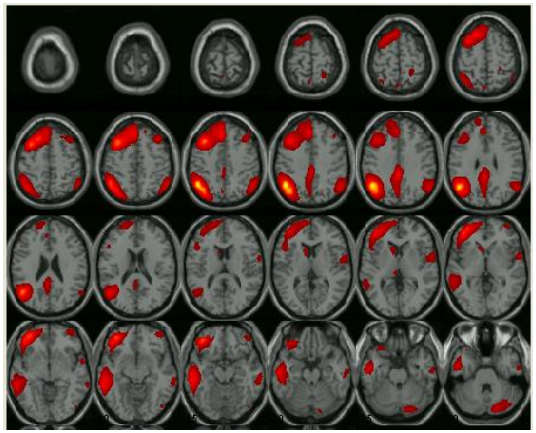
Various brain functional networks



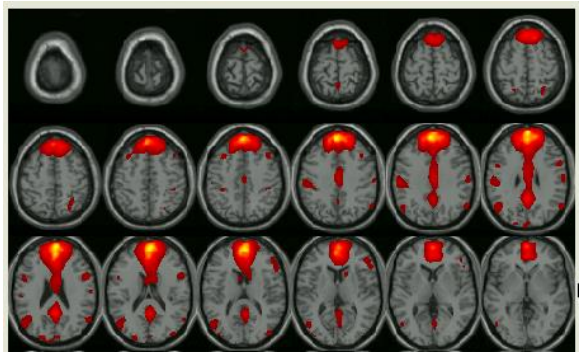
pDMN: higher cognition



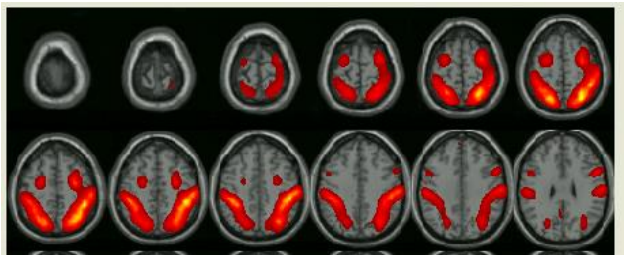
right FPN: control, attention, memory



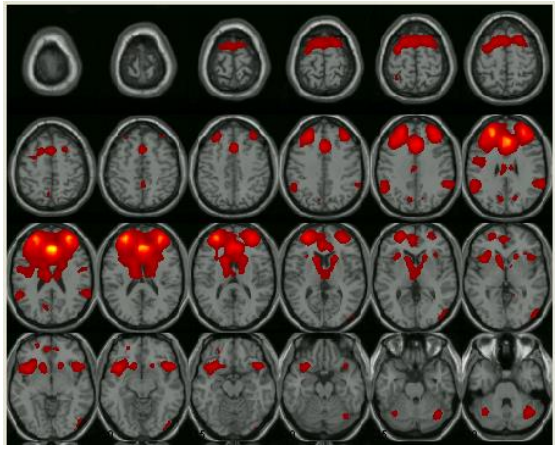
left FPN: control, attention, memory



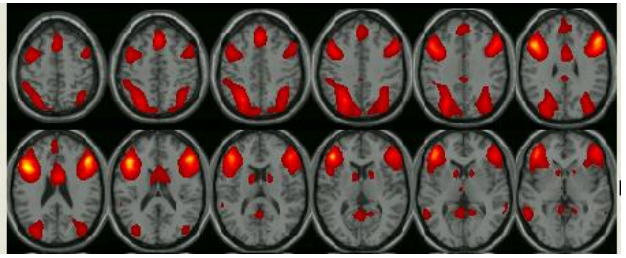
aDMN: self referential



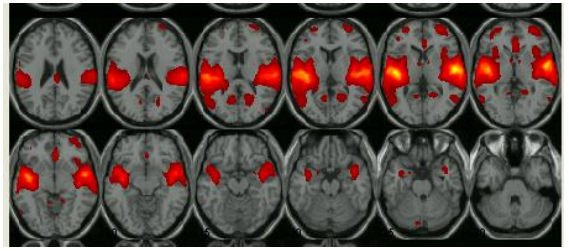
dorsal attention network



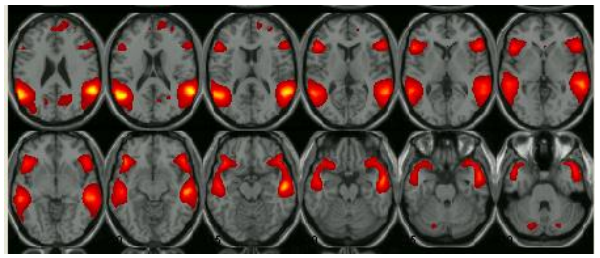
salience network: emotion



executive control network

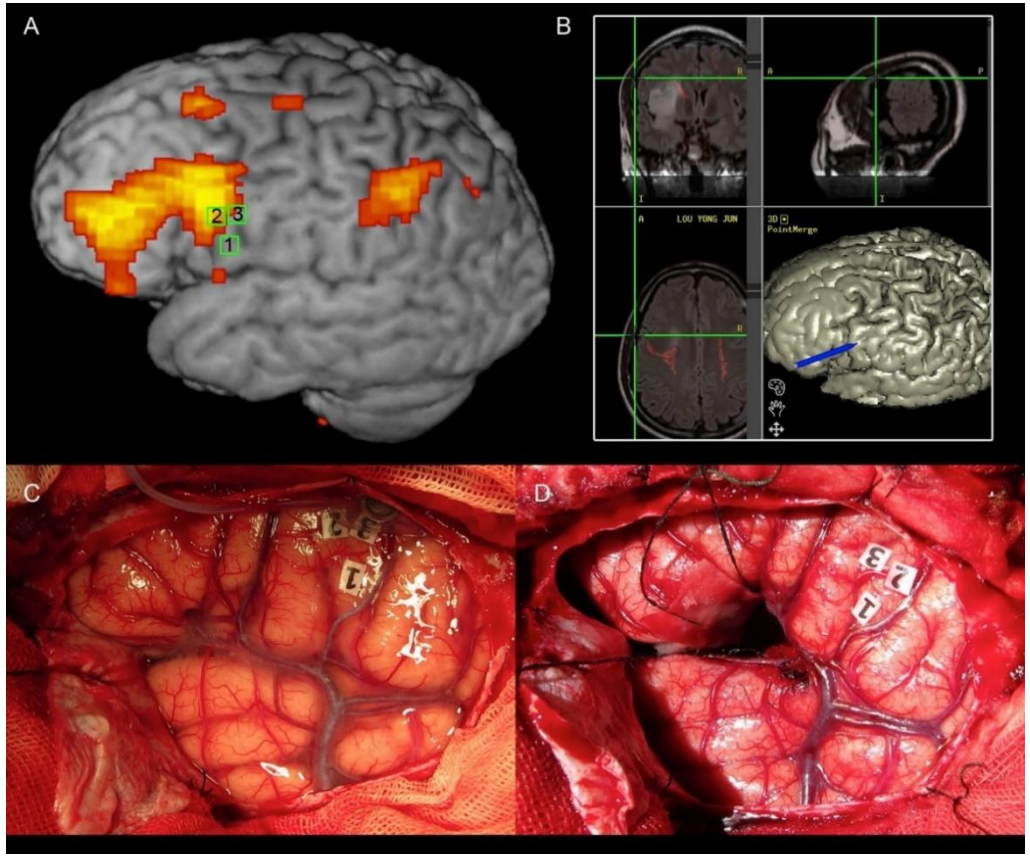


auditory network

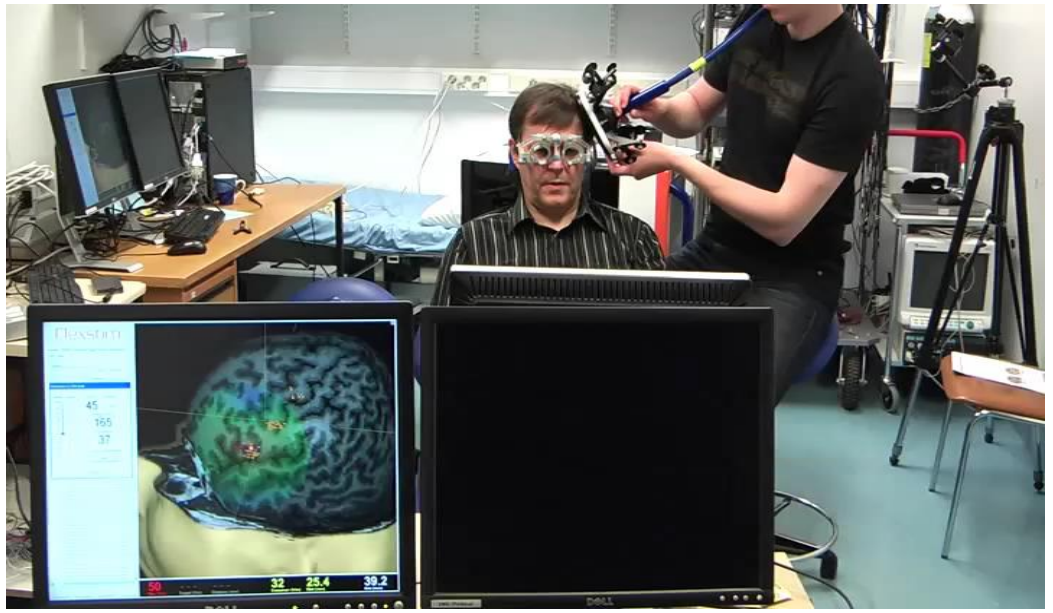


language network

FC based language network localization for presurgical planning (brain tumor patient)



Lu, Zhang et al., 2015, HBM



Sollmann et al., 2013

TMS: Transcranial magnetic stimulation
Like Intraoperative electrocortical stimulation

上海科技大学
ShanghaiTech University

谢谢大家

加入我们！

致谢

基金委、科技部、上海市科委、中科院上海分院项目
UNC Chapel Hill的同事们（Li, Gang, Yap, Mingxia ...）
上科大IDEA Lab和脑疾病与智能计算研究组 团队
中国0-6岁婴幼儿大队列项目 团队

张寒

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上海科技大学
ShanghaiTech University

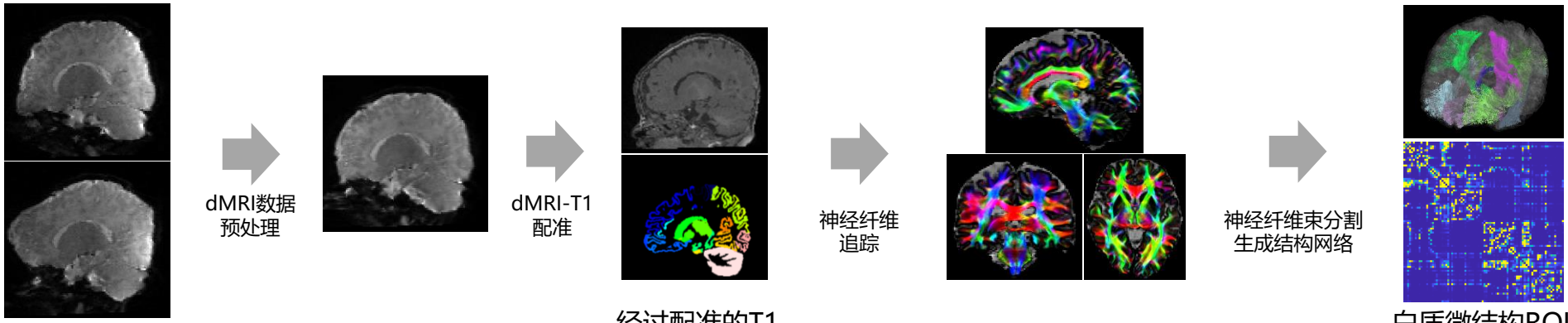
脑科学与脑疾病

第九节课：研究人类大脑白质结构的dMRI及其基本分析技术

婴幼儿弥散MRI数据分析技术研发

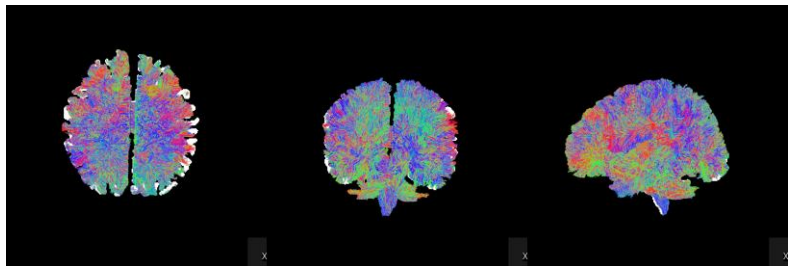
ShanghaiTech University

dMRI 处理流程

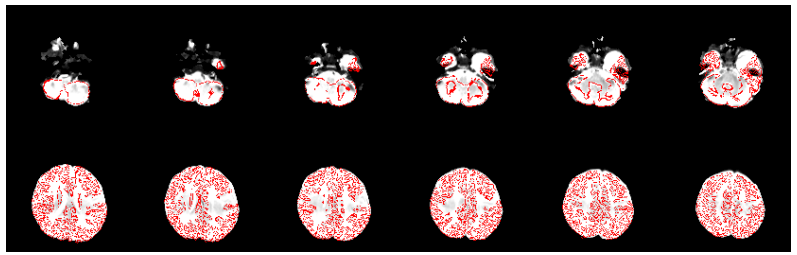


原始dMRI数据　经预处理的dMRI数据　经过配准的T1及脑区ROIs　全脑神经纤维　白质微结构ROIs结构网络分析

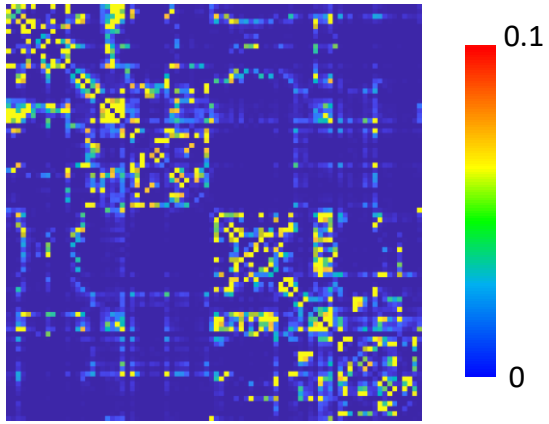
数据预处理质量检验



神经纤维铺满白质区域



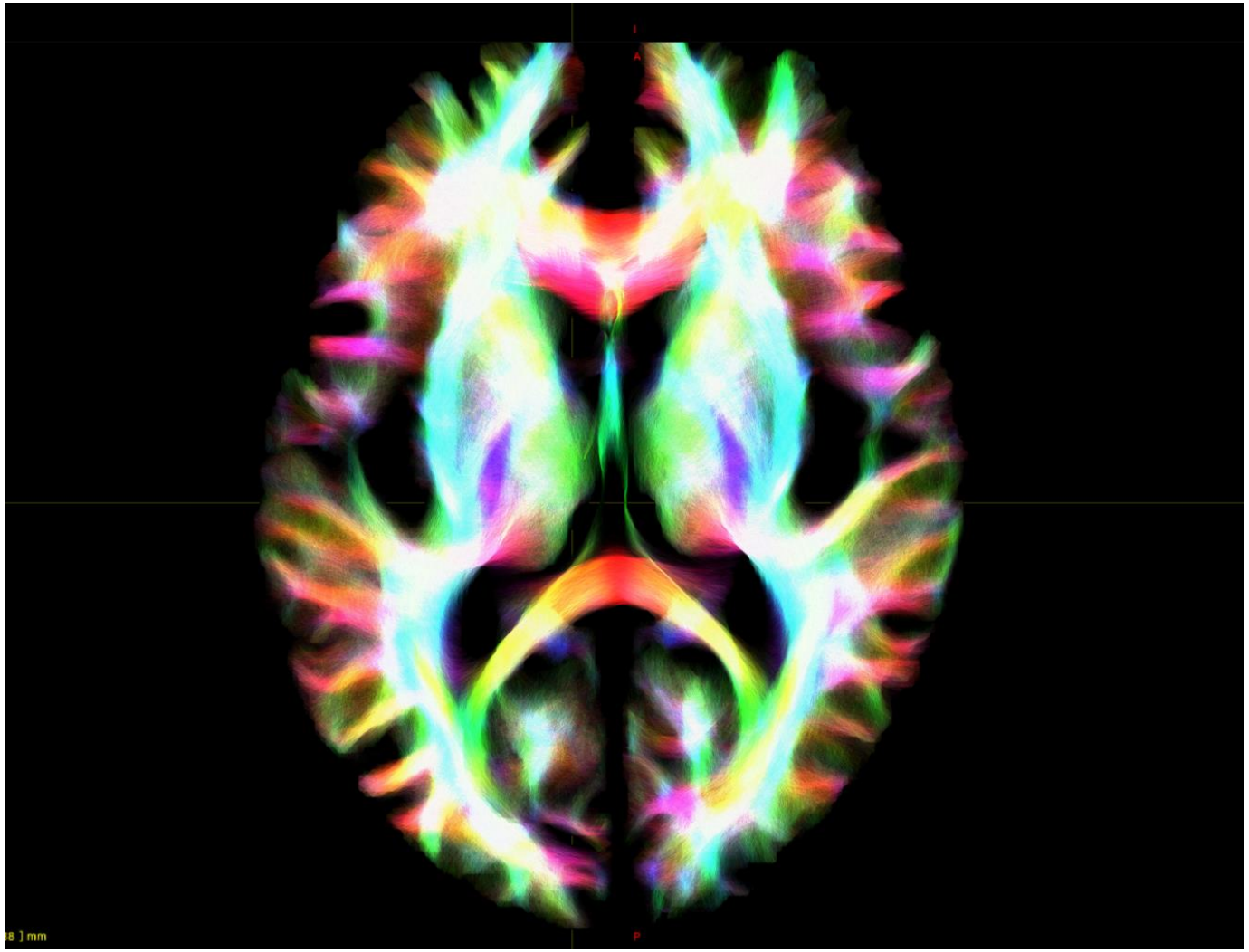
ROIs边缘和dMRI数据的灰白质交界面重合



结构连接矩阵团块结构明显，值域正常

多球壳多成分全脑纤维跟踪和提取

上海科技大学
ShanghaiTech University



UNC SCHOOL OF MEDICINE

Computational Diffusion MRI

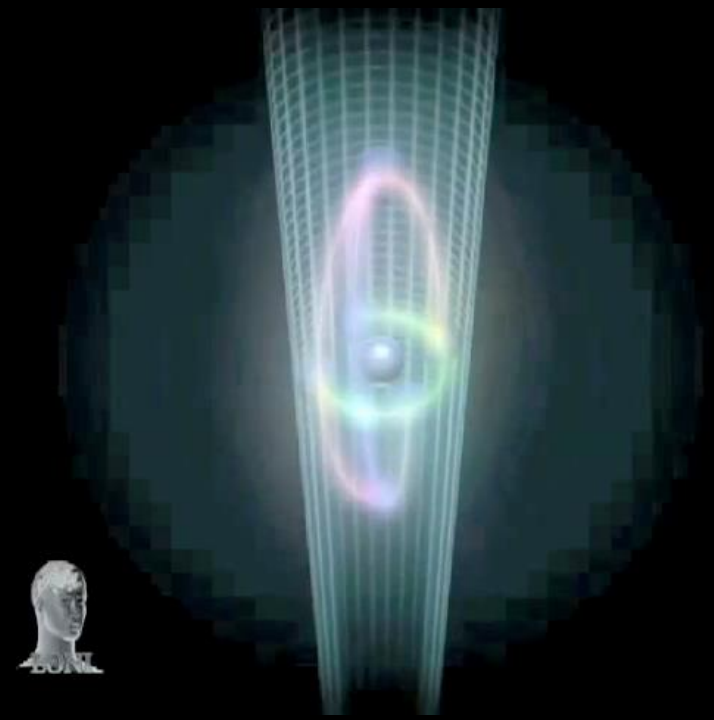
Baby Connectome Project

Pew-Thian Yap, Ph.D.

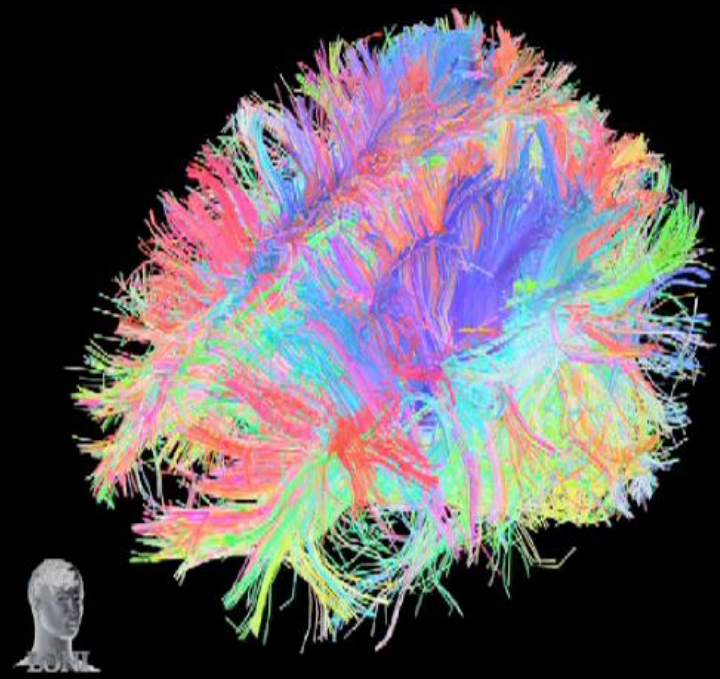
BRIC BIOMEDICAL REBEARCH IMAGING CENTER

Research Lab IDEA

Diffusion MRI Basics

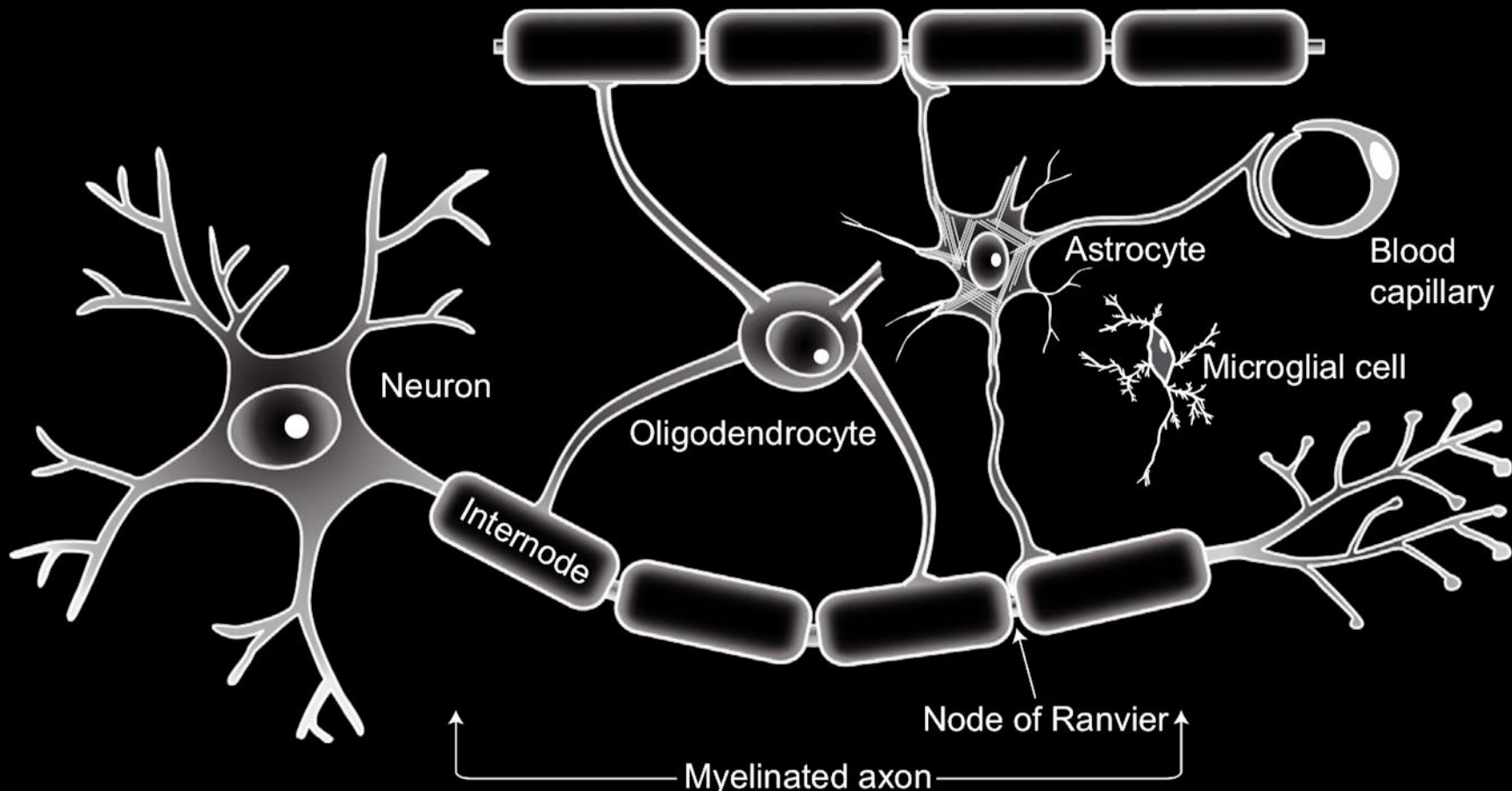


Anisotropic
Diffusion



Tractography

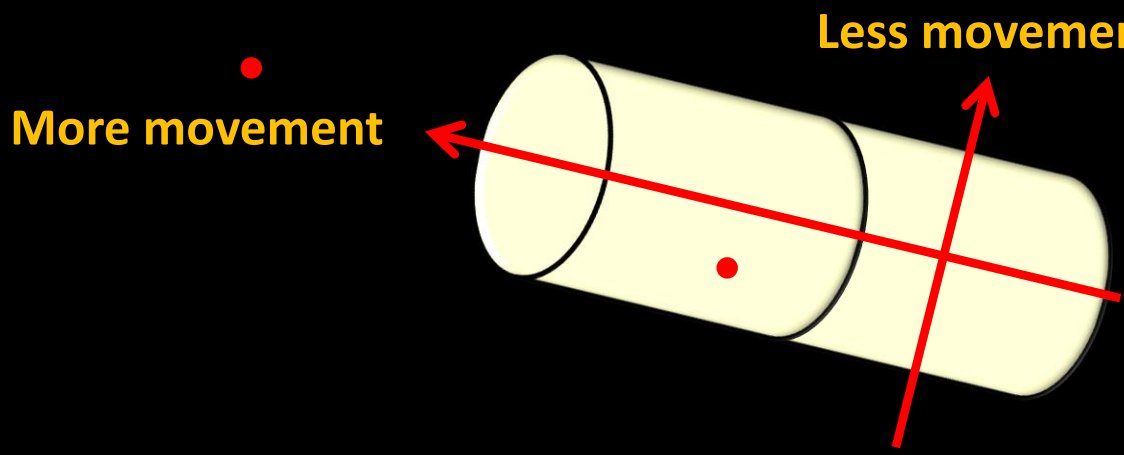
Major Cellular Elements of the White Matter

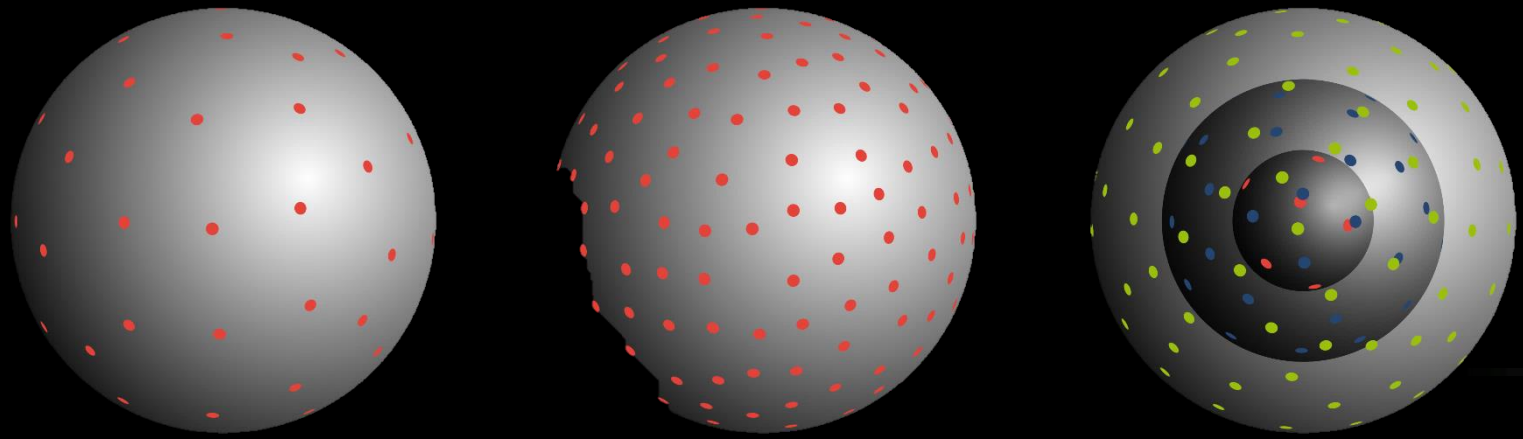


Free and Restricted Diffusion

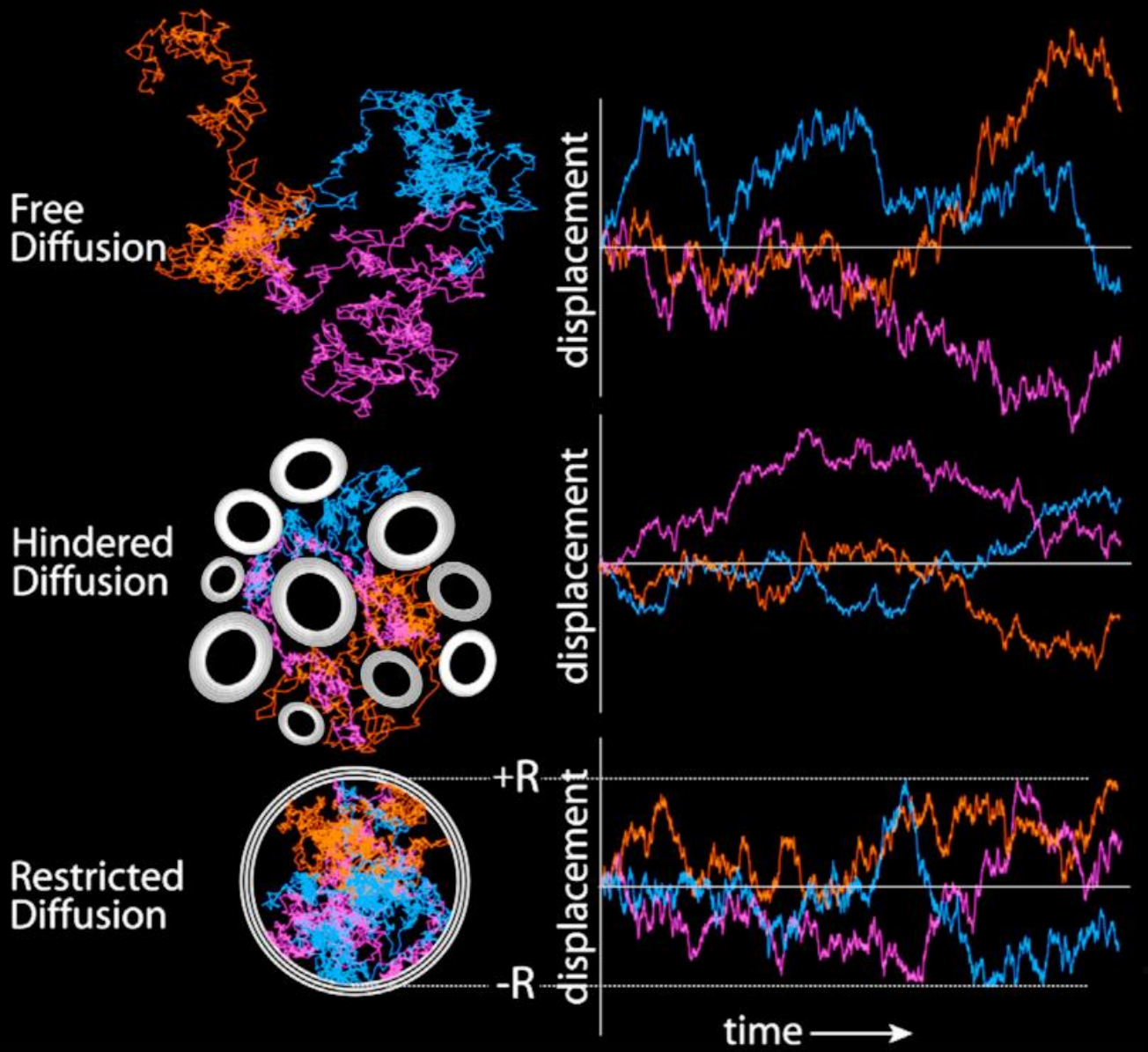
Free diffusion

Restricted diffusion





Free, Hindered, and Restriction Diffusion



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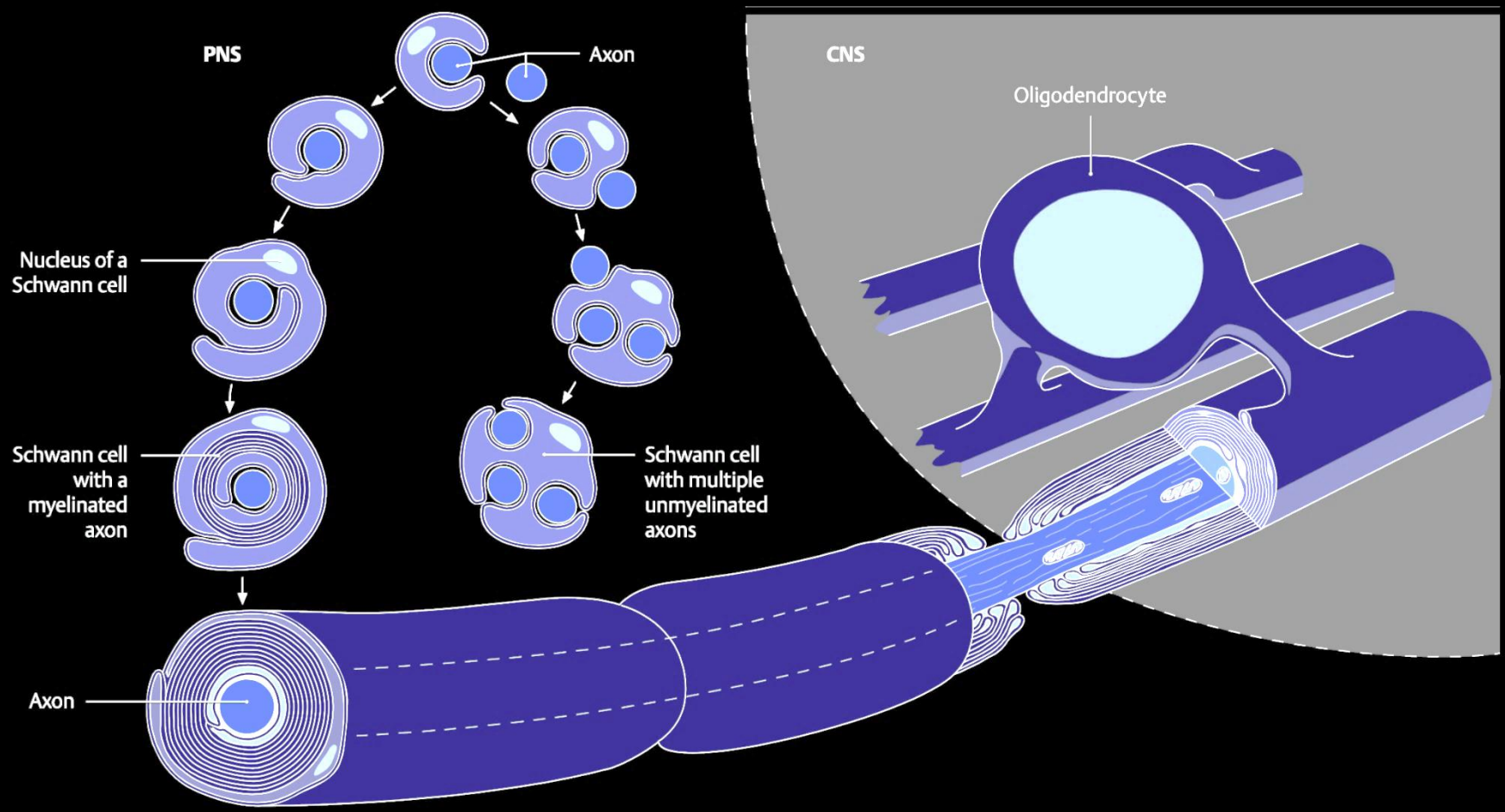
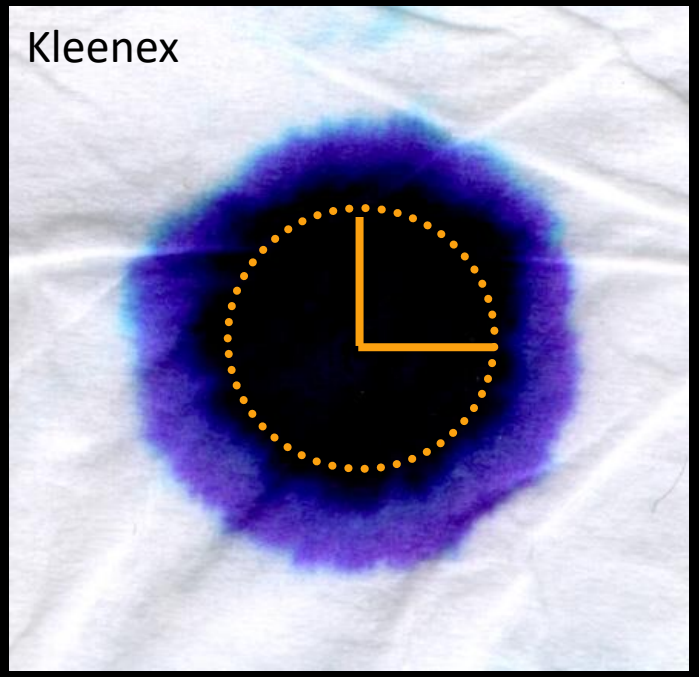
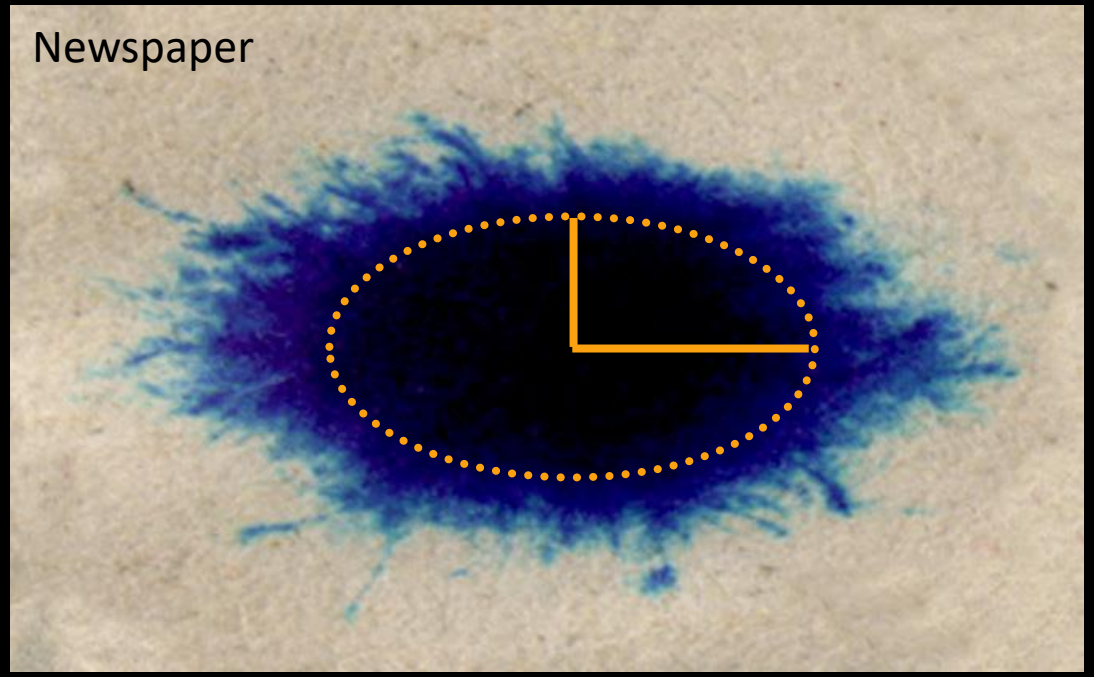
Myelination in the PNS and CNS



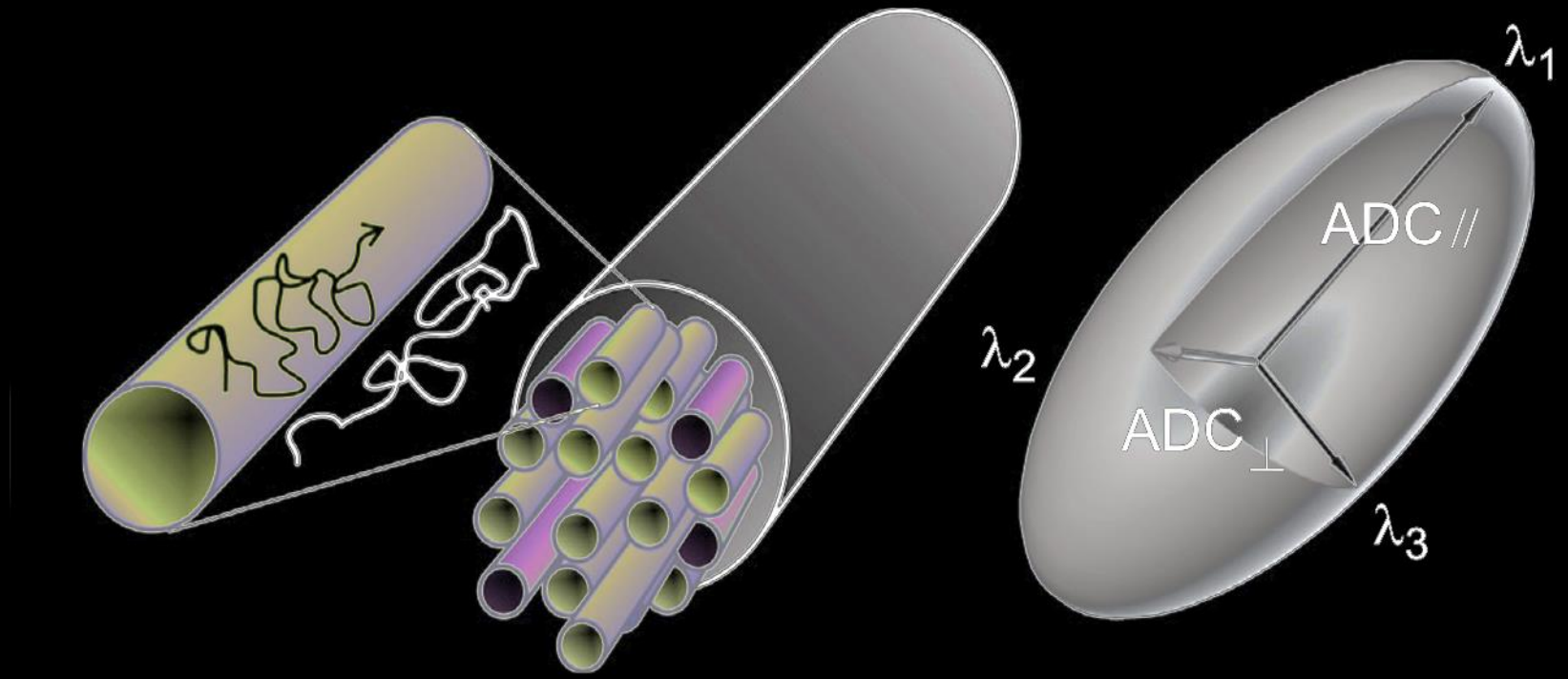
Diffusion Anisotropy





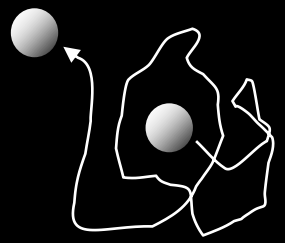
Anisotropy: Diffusion rate depends on direction.

Diffusion Tensor Imaging (DTI)

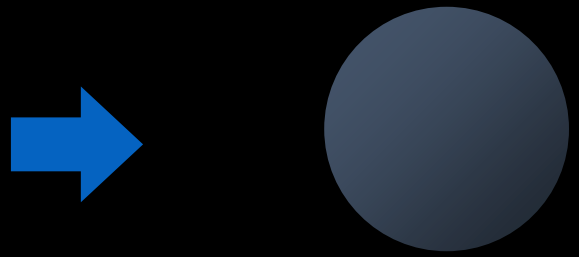


(An)isotropic Diffusion

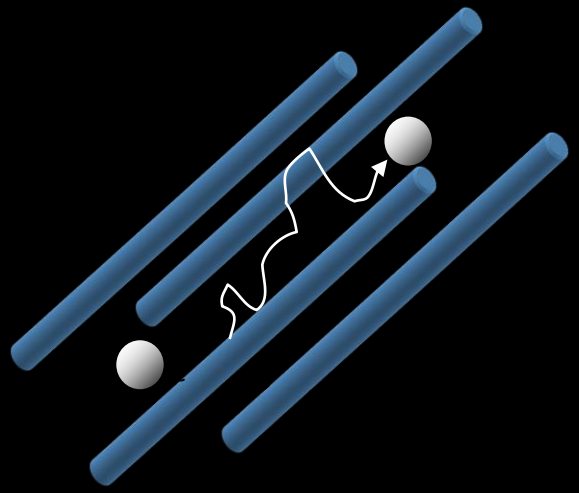
Free diffusion



Isotropic diffusion



Restricted diffusion



Anisotropic diffusion

Fractional Anisotropy (FA)

FA=0.12
λ_1=0.80
λ_2=λ_3=0.65

FA=0.46
λ_1=1.10
λ_2 = λ_3=0.50

FA=0.82
λ_1=1.60
λ_2=λ_3=0.25

$$\mathrm{FA} = \sqrt{\frac{3}{2}} \frac{\sqrt{(\lambda_1 - \bar{\lambda})^2 + (\lambda_2 - \bar{\lambda})^2 + (\lambda_3 - \bar{\lambda})^2}}{\sqrt{\lambda_1^2 + \lambda_2^2 + \lambda_3^2}} \qquad \bar{\lambda} = \frac{\lambda_1 + \lambda_2 + \lambda_3}{3}$$

Value ranges from 0 – 1:

FA= 0 → Diffusion is spherical (i.e. isotropic)
FA= 1 → Diffusion is tubular (i.e. anisotropic)

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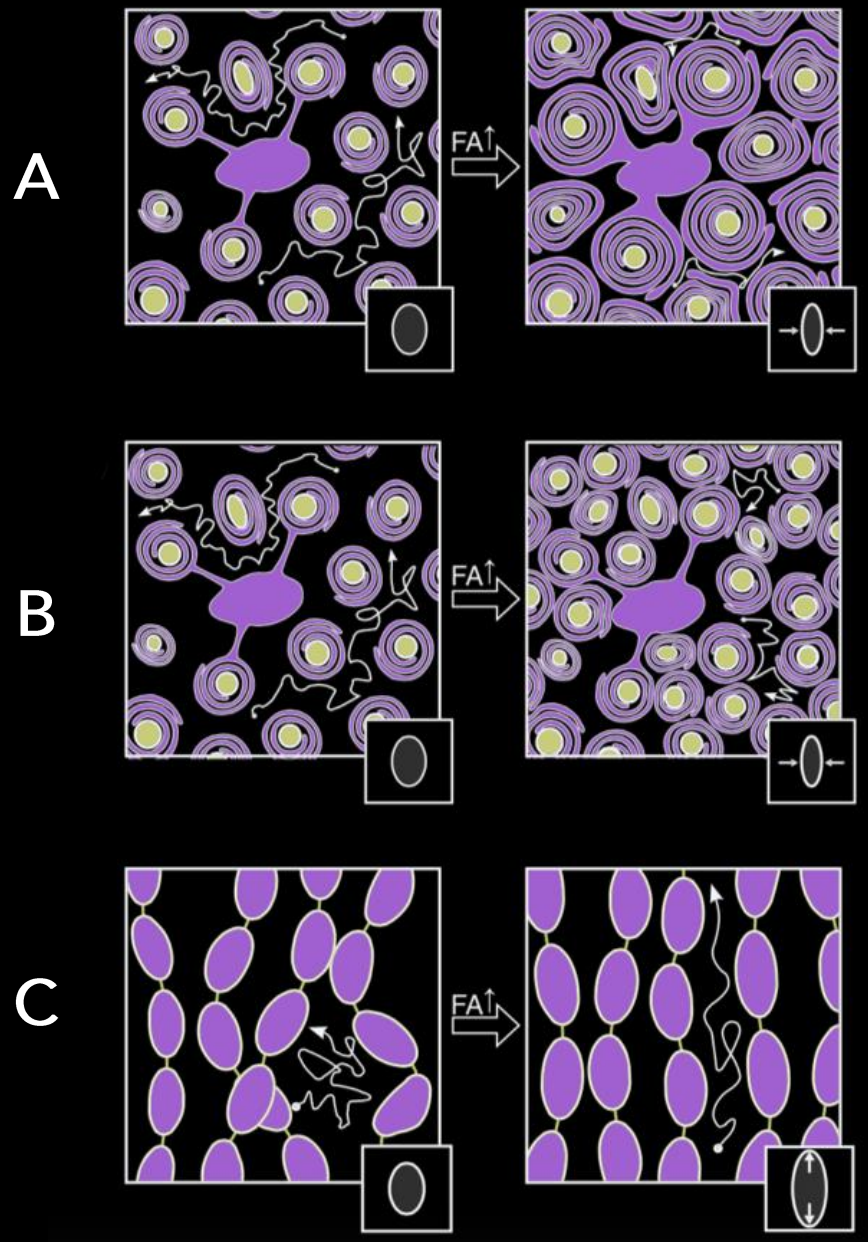
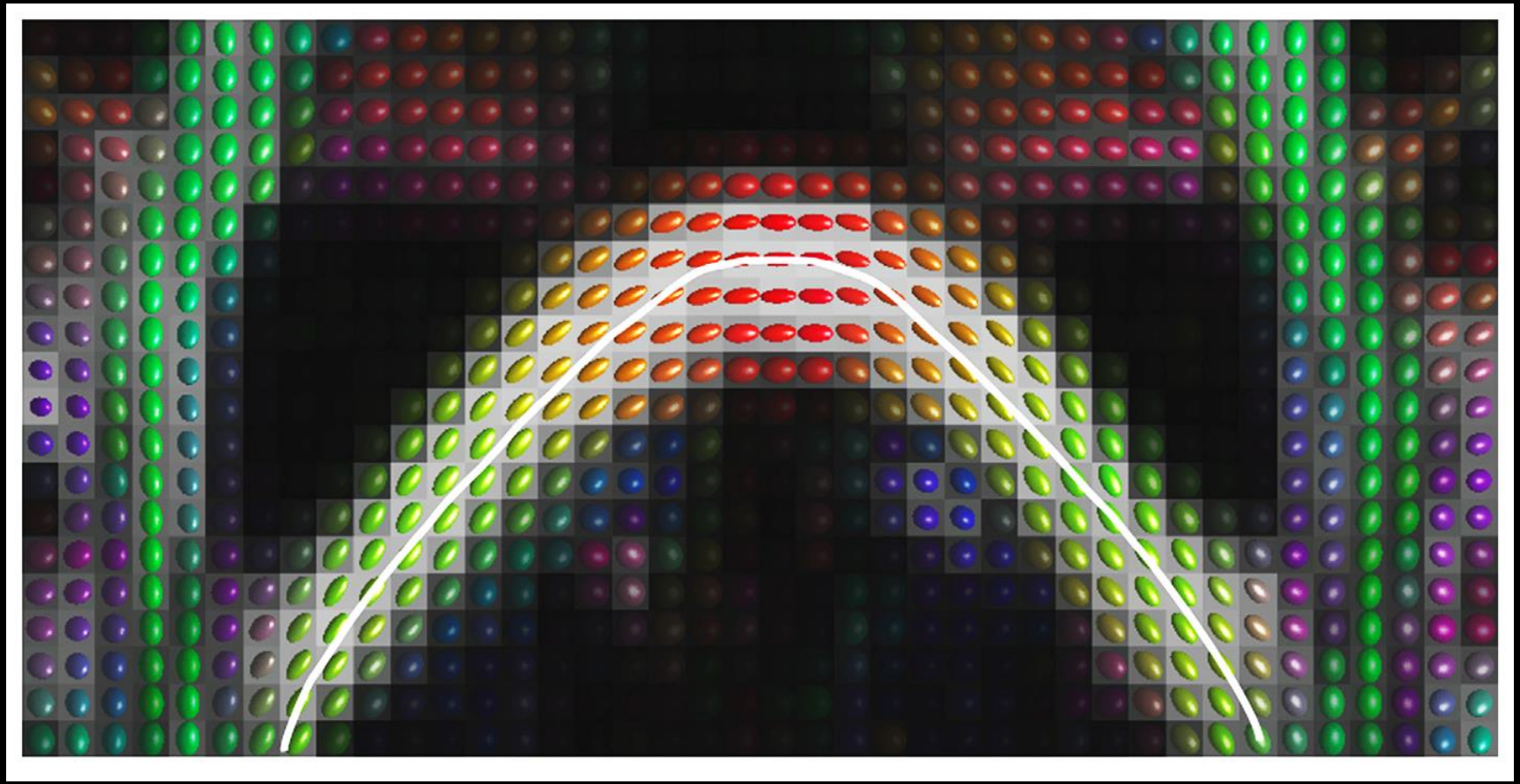
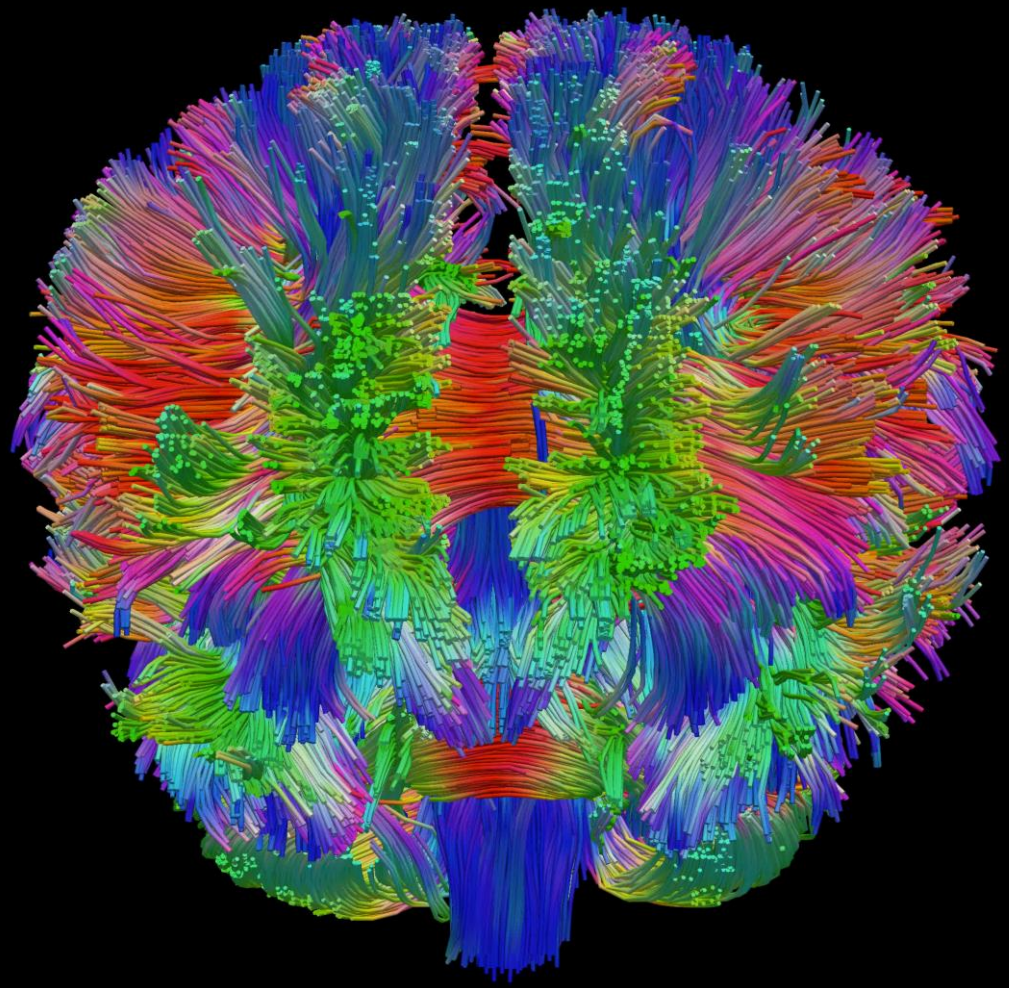
Fractional Anisotropy (FA)



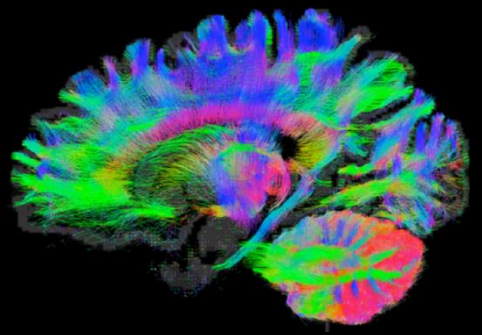
Tractography



Tractography

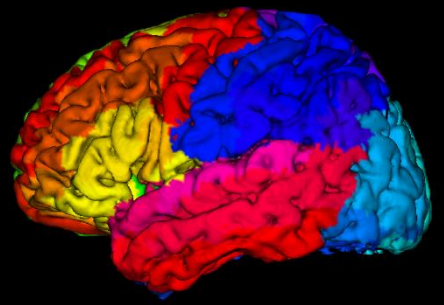


Brain Connectomics

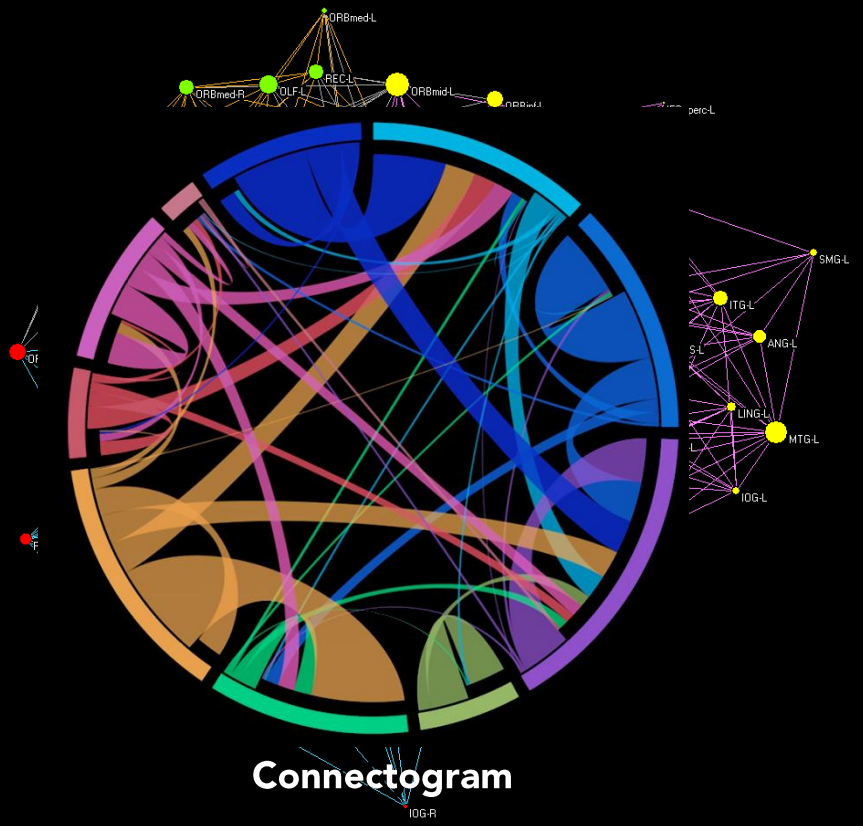


Tractography

+

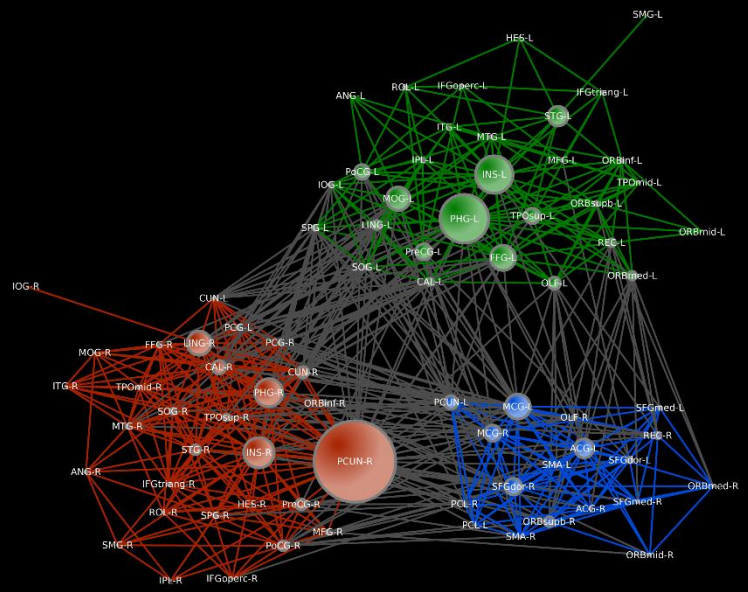


Regions of Interest (ROIs)

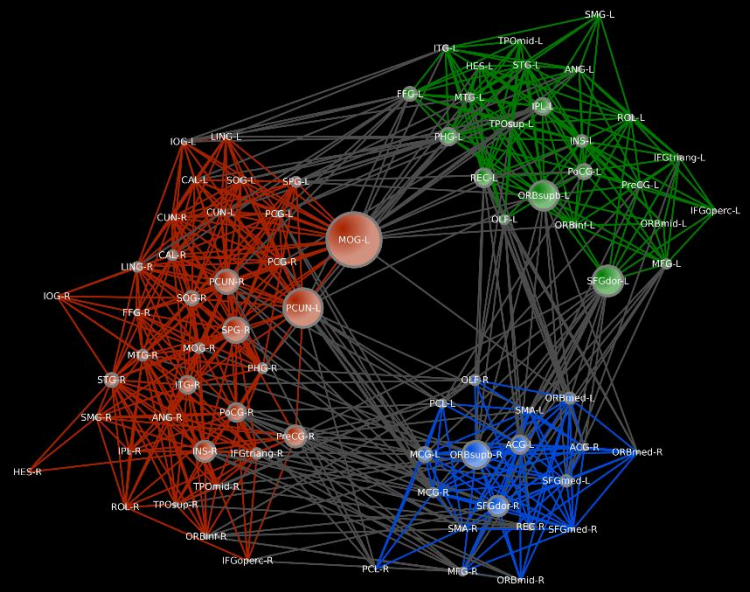


Connectogram

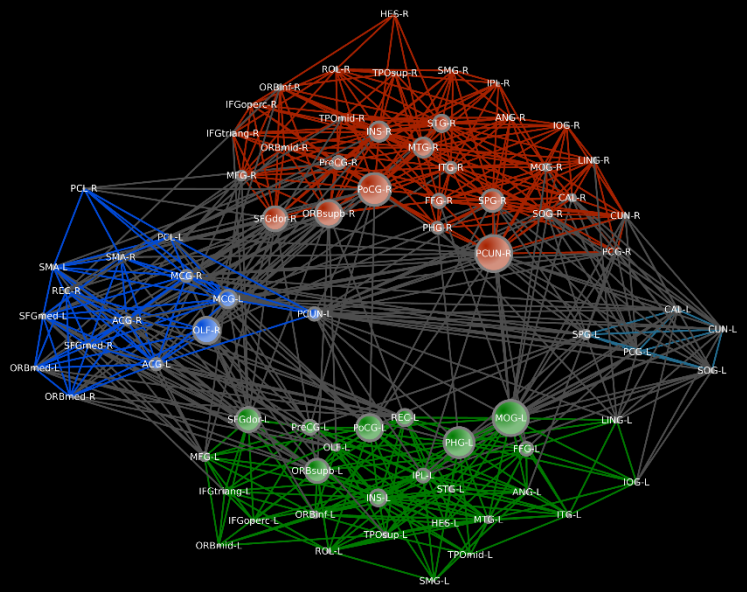
Brain Networks



2-week-olds



1-year-olds



2-year-olds

Yap et al., PLOS ONE, 2011

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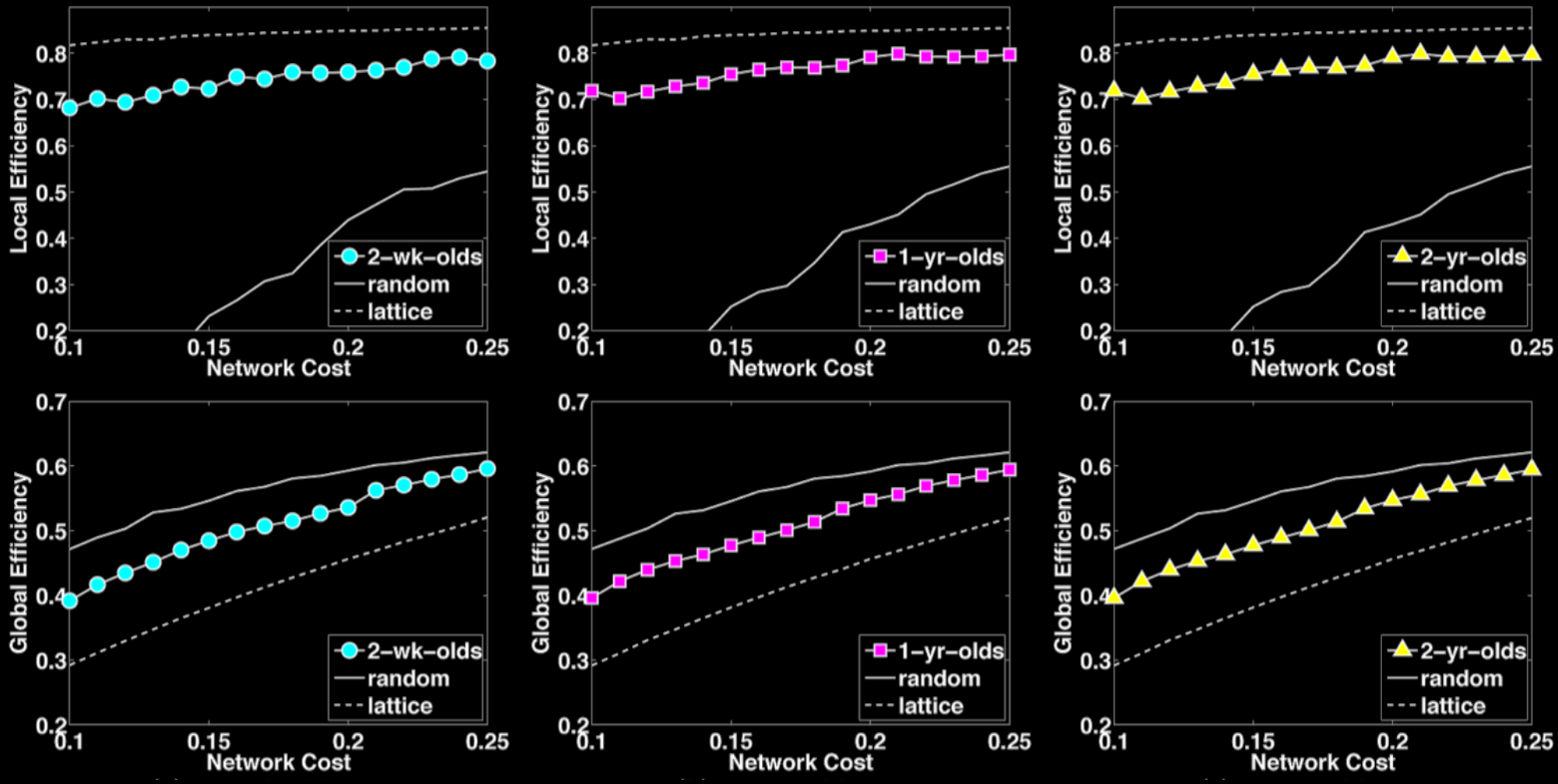
Research Lab IDEA

Small Worldness



2-week-olds 1-year-olds 2-year-olds

Yap et al., PLOS ONE, 2011

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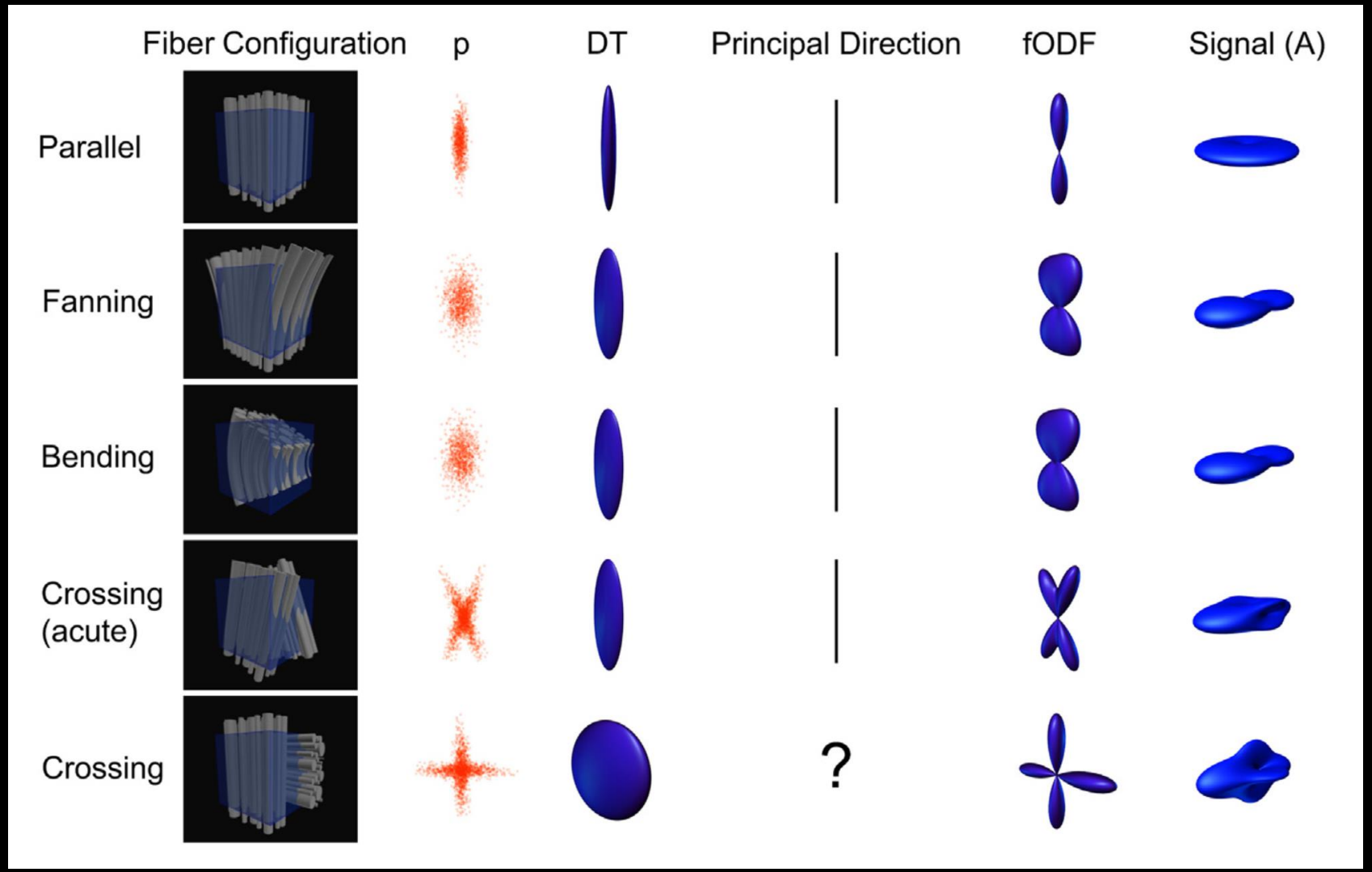
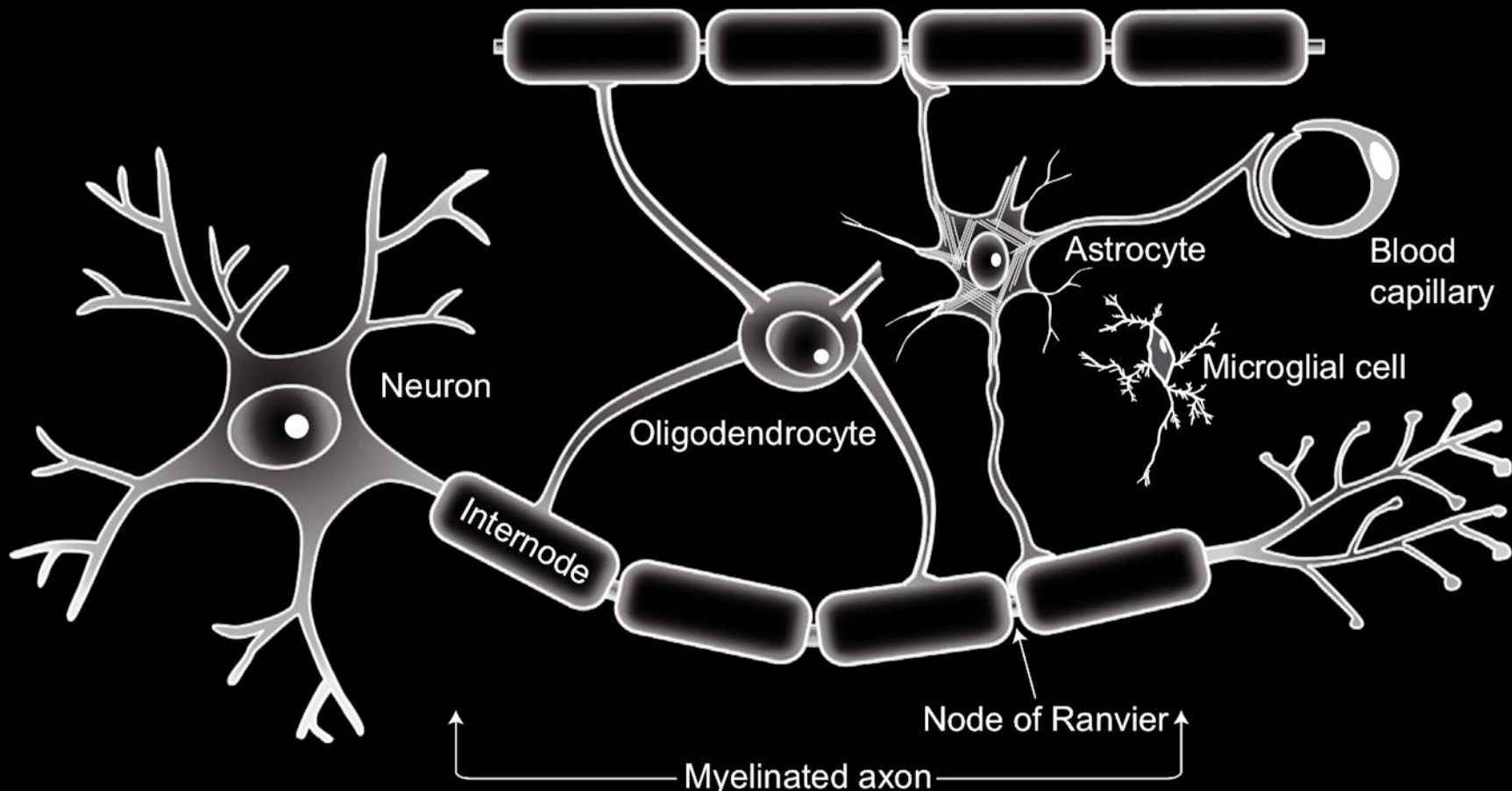
Research Lab IDEA

Microstructural Analysis

Limitations of DTI

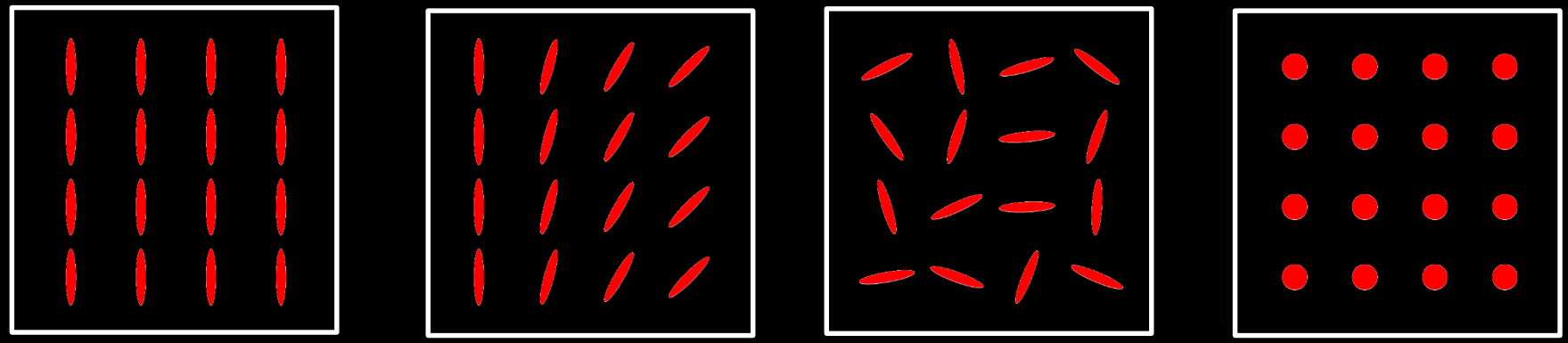


Major Cellular Elements of the White Matter

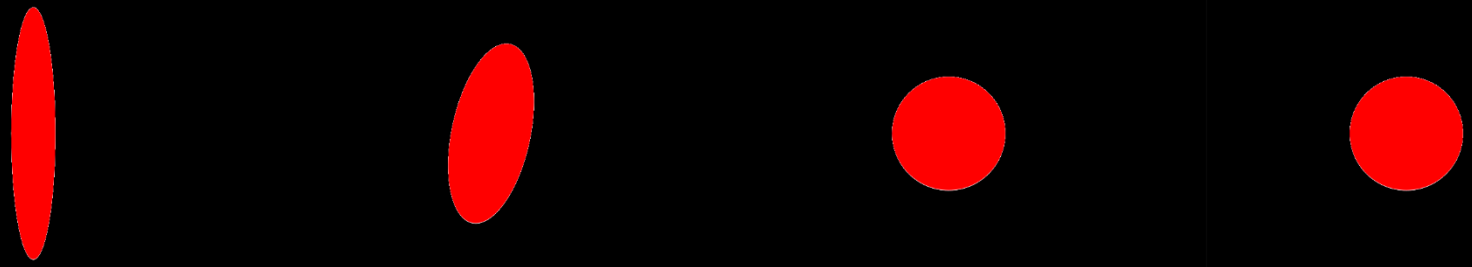


Microscopic Imaging

Microstructure



Tensor



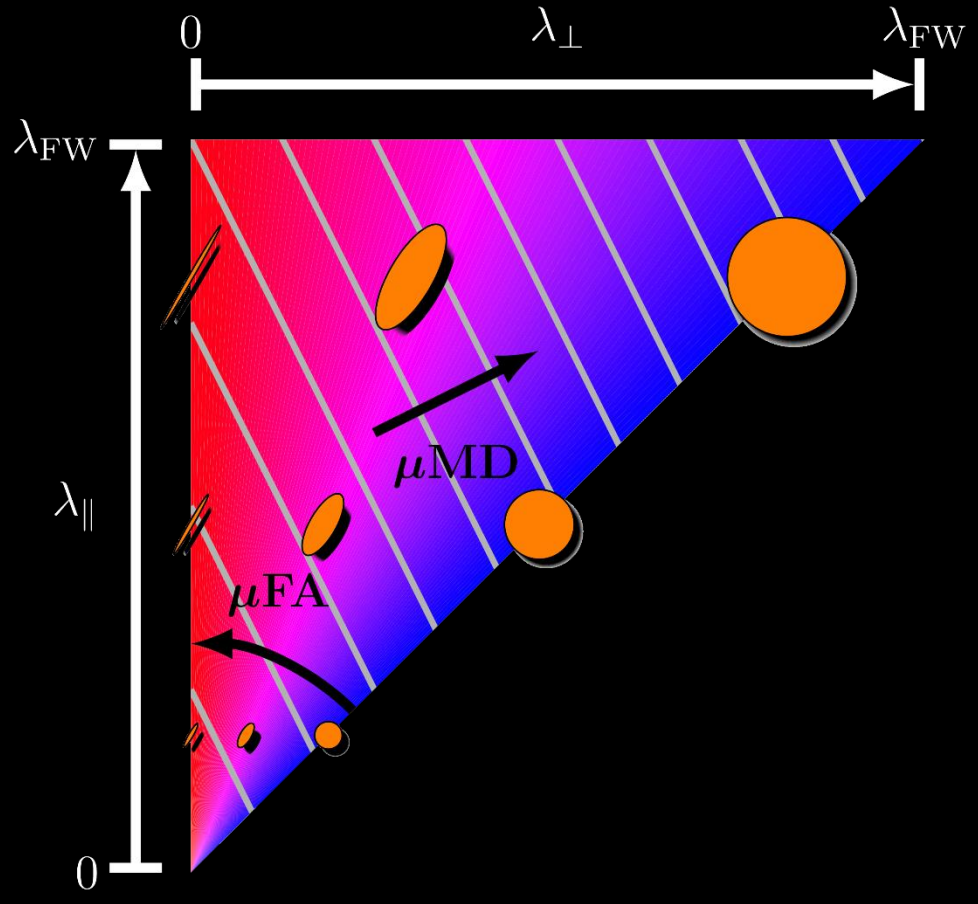
FA = 0.8, μFA = 0.8

FA < 0.8, μFA = 0.8

FA = 0, μFA = 0.8

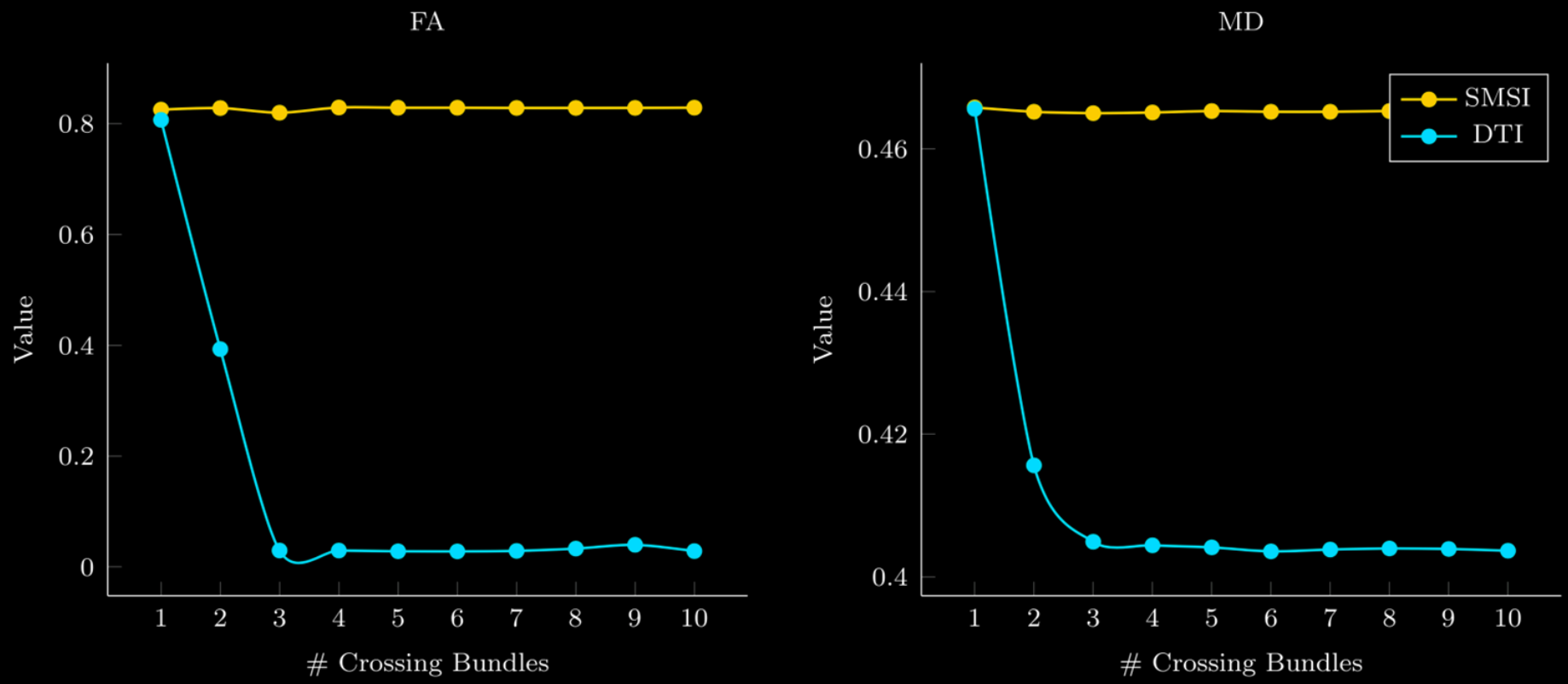
FA = 0, μFA = 0

Spherical Mean Spectrum Imaging (SMSI)



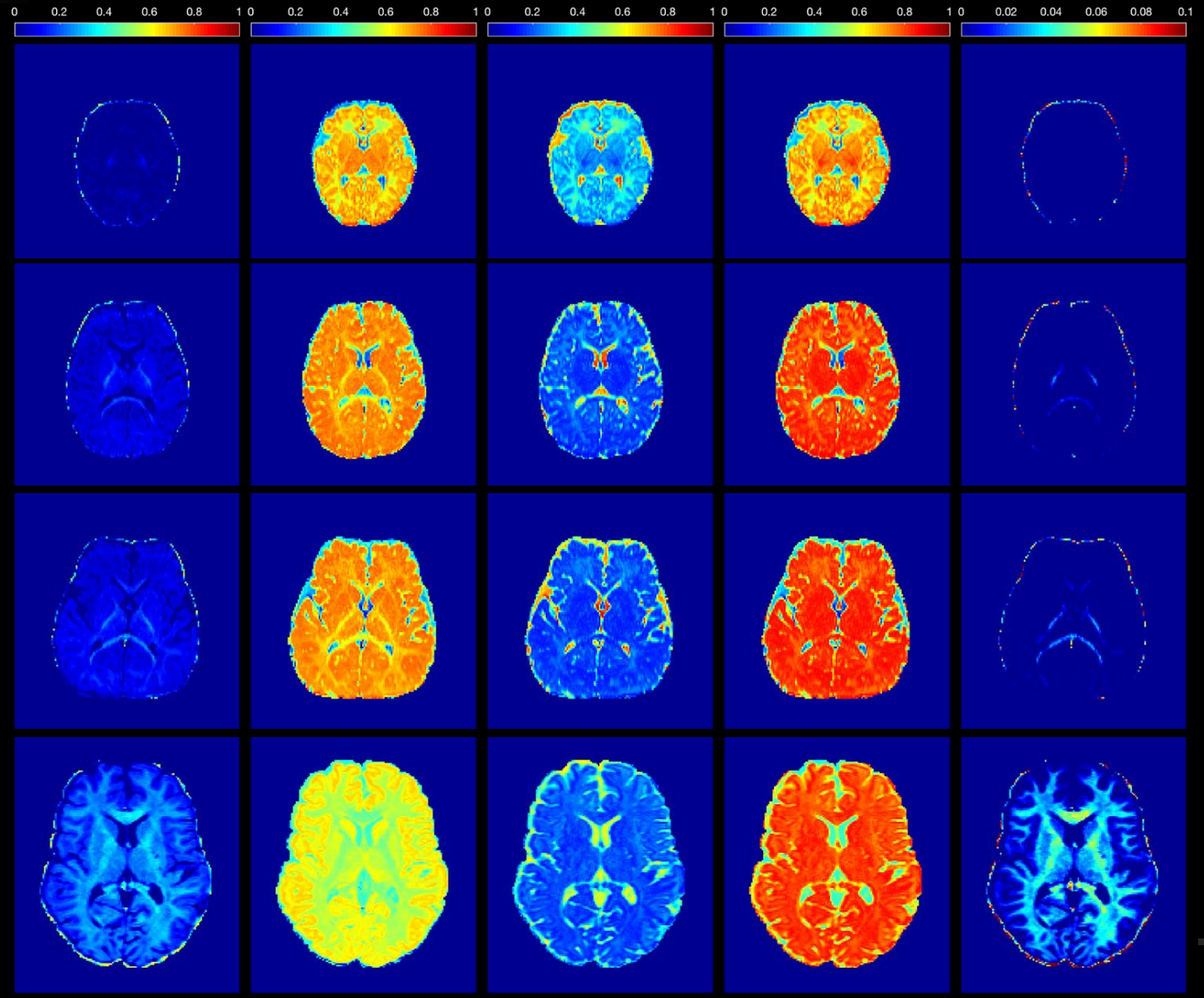
Microscopic axial diffusivity (μAD)	Microscopic anisotropic axial diffusivity (μAAD)
Microscopic radial diffusivity (μRD)	Microscopic anisotropic radial diffusivity (μARD)
Microscopic mean diffusivity (μMD)	Microscopic anisotropic mean diffusivity (μAMD)
Microscopic FA (μFA)	Microscopic isotropic mean diffusivity (μIMD)
Intra-cellular volume fraction (ICVF)	Intra-cellular axial diffusivity (ICAD)
Extra-cellular volume fraction (ECVF)	Extra-cellular axial diffusivity (ECAD)
Anisotropic volume fraction (AVF)	Extra-cellular radial diffusivity (ECRD)
Isotropic volume fraction (IVF)	Trapped-water volume fraction (TWVF)

Spherical Mean Spectrum Imaging (SMSI)



Comparison between voxel-level DTI FA and MD and microscopic-level SMSI μFA, μMD and ODI with respect to the number of crossing fiber bundles.

Spherical Mean Spectrum Imaging (SMSI)



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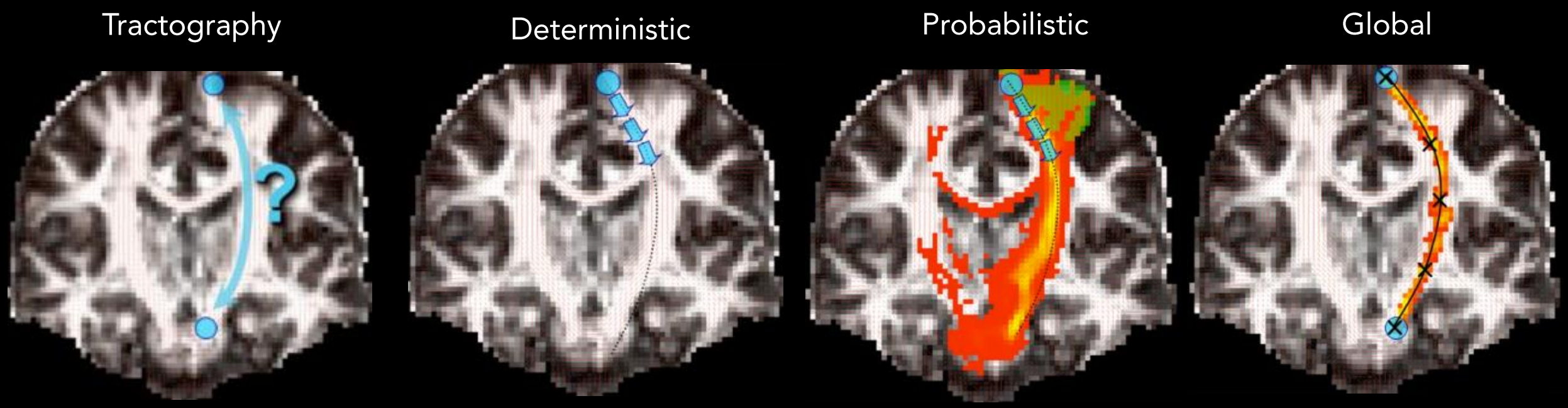
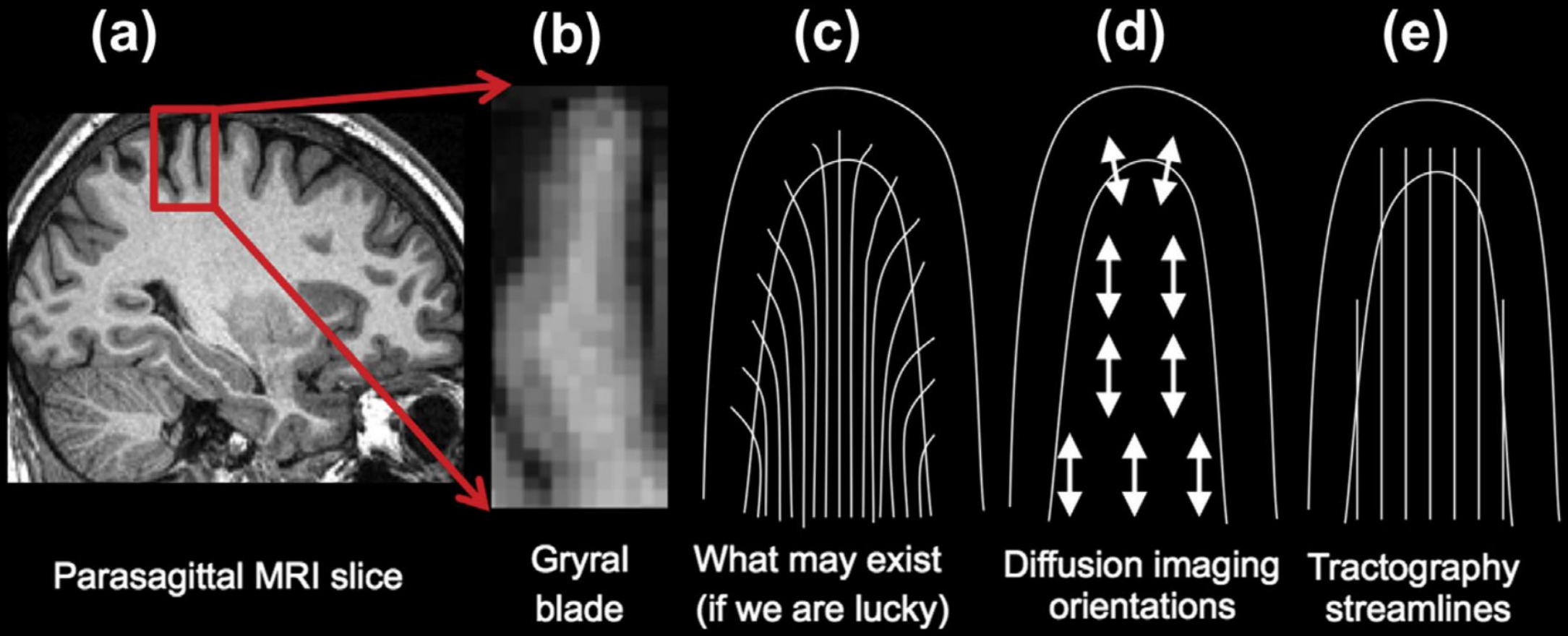
Research Lab IDEA

Tractography

Local vs Global Tractography

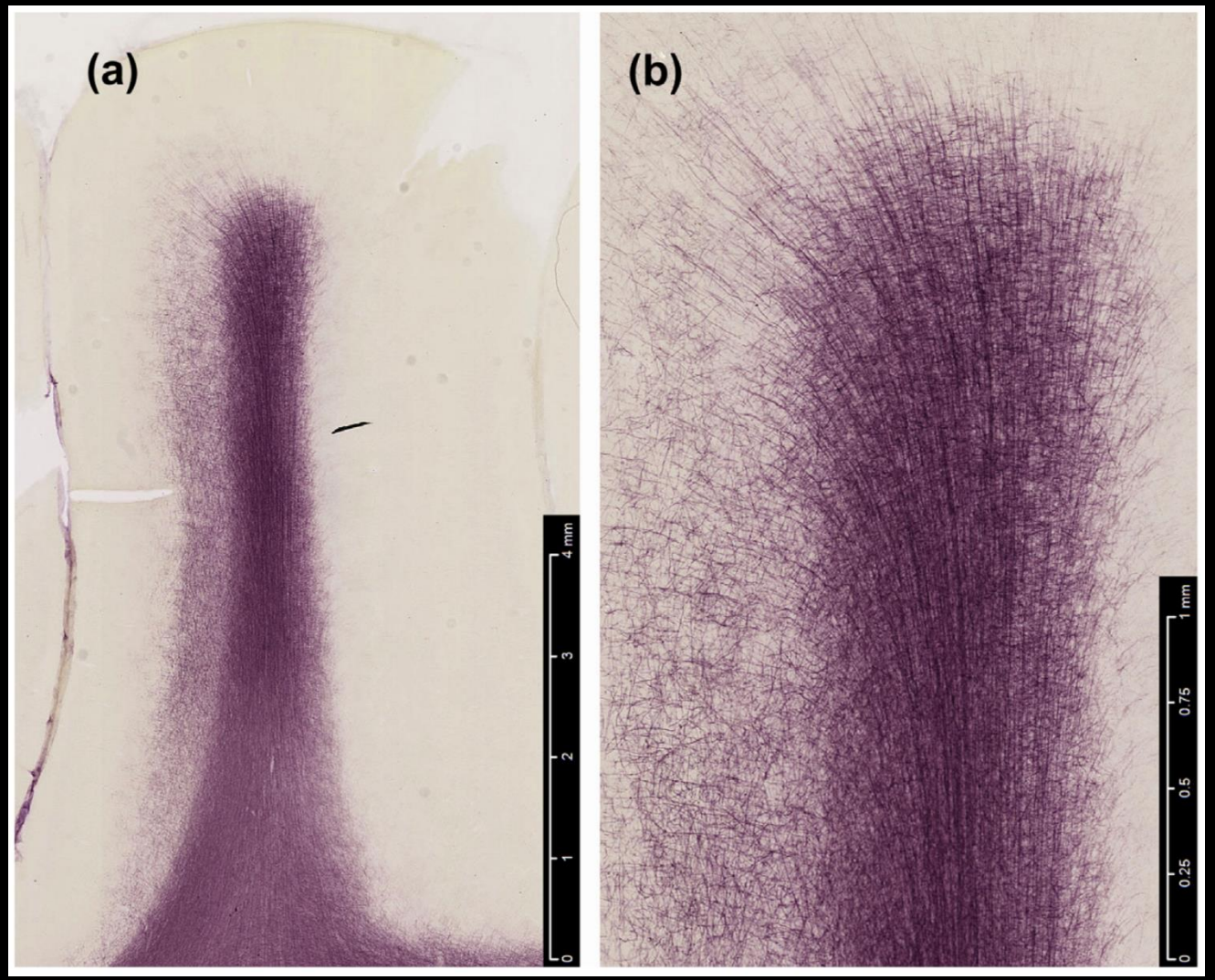


Gyral Bias

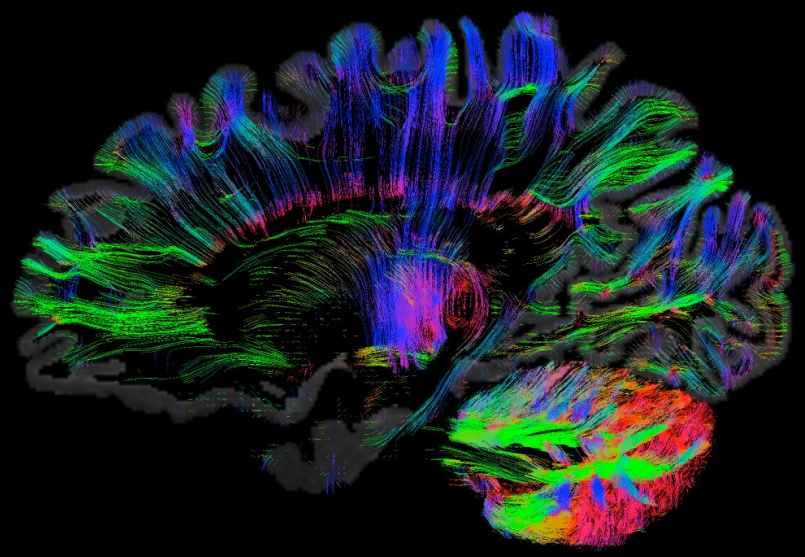


[Image] Johansen-Berg and Behrens, Diffusion MRI - From Quantitative Measurement to In-vivo Neuroanatomy, 2014

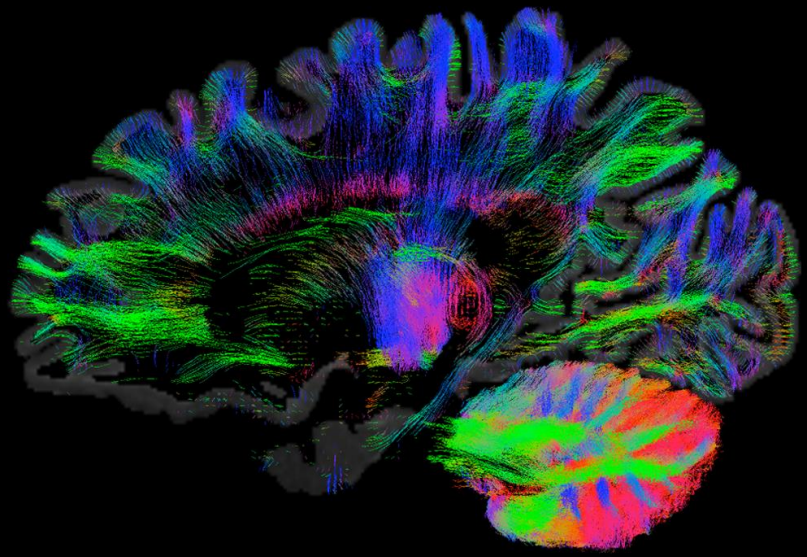
Axonal Trajectories in a Gyral Blade



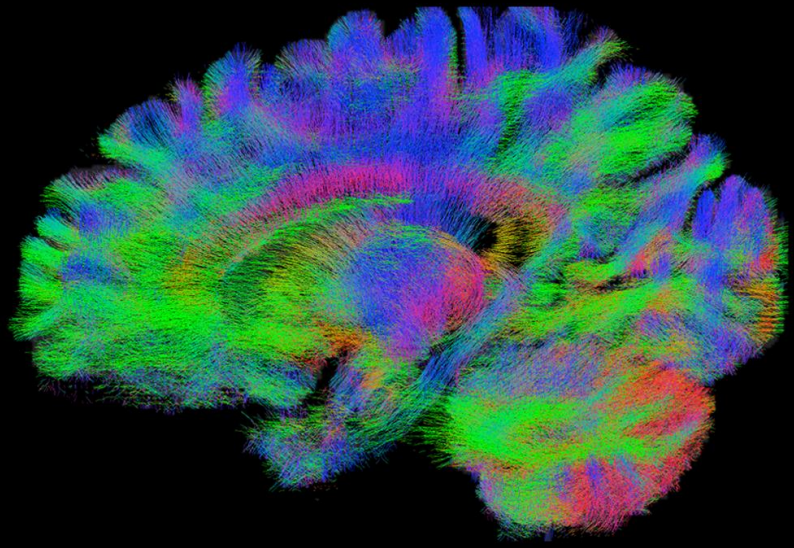
Tractography



Deterministic



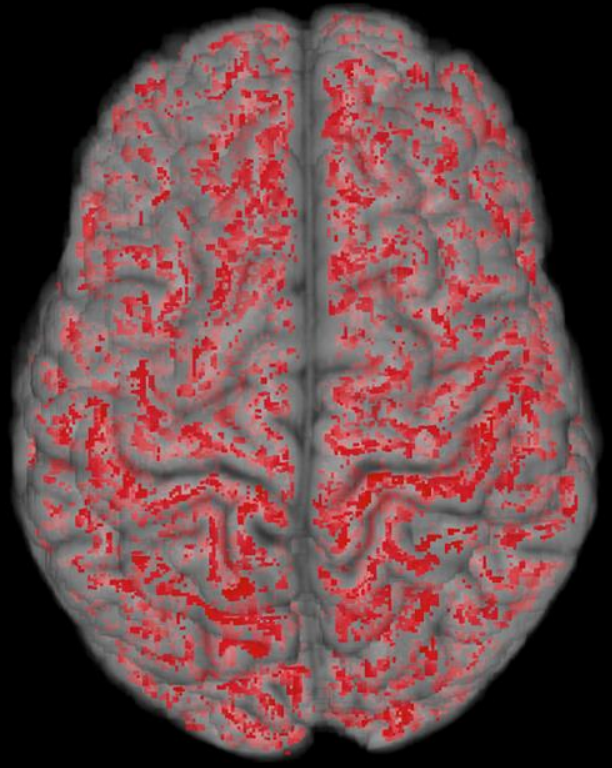
Probabilistic



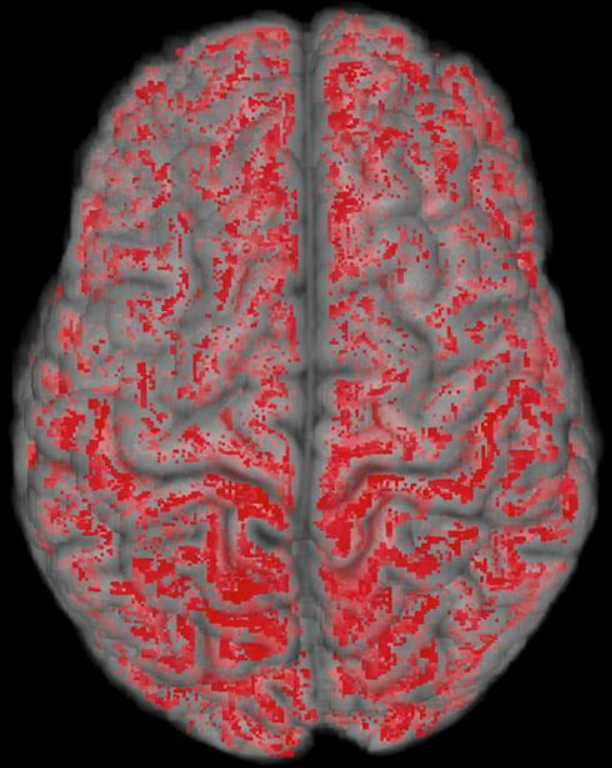
Proposed

Tractography

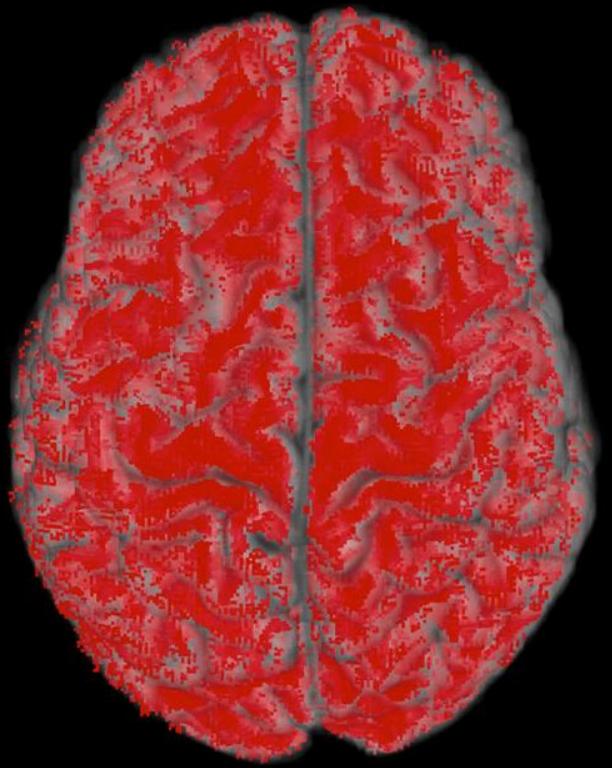
Deterministic



Probabilistic



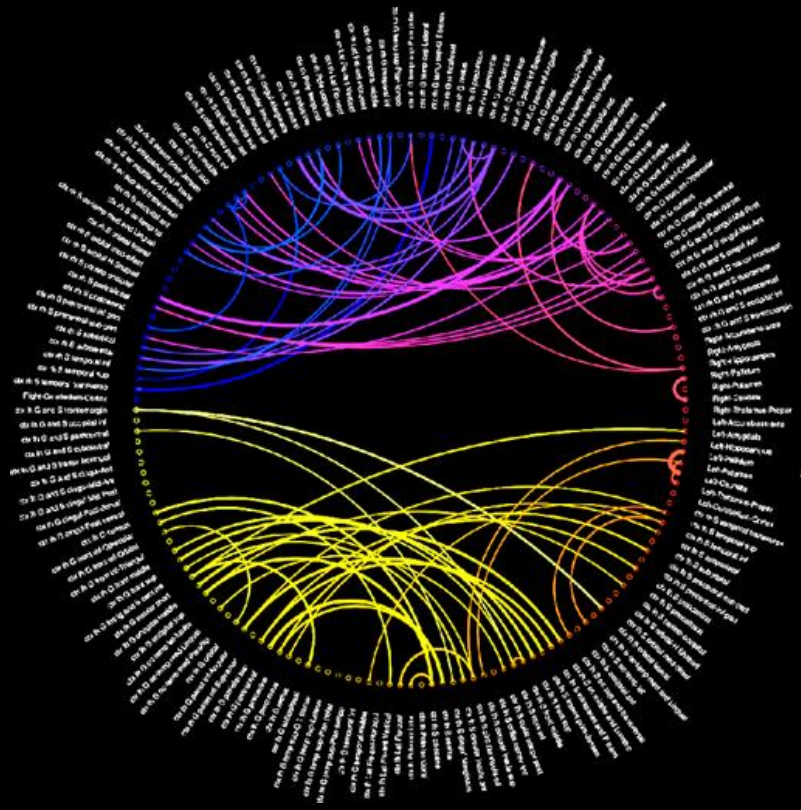
Proposed



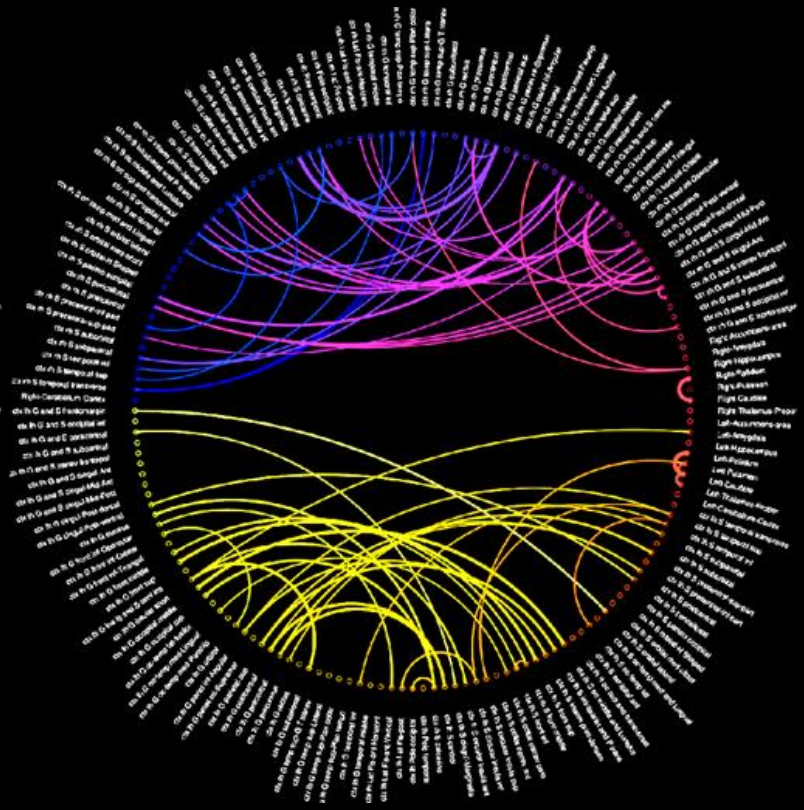
Coverage of streamline endpoints at WM-GM interface

Connectivity

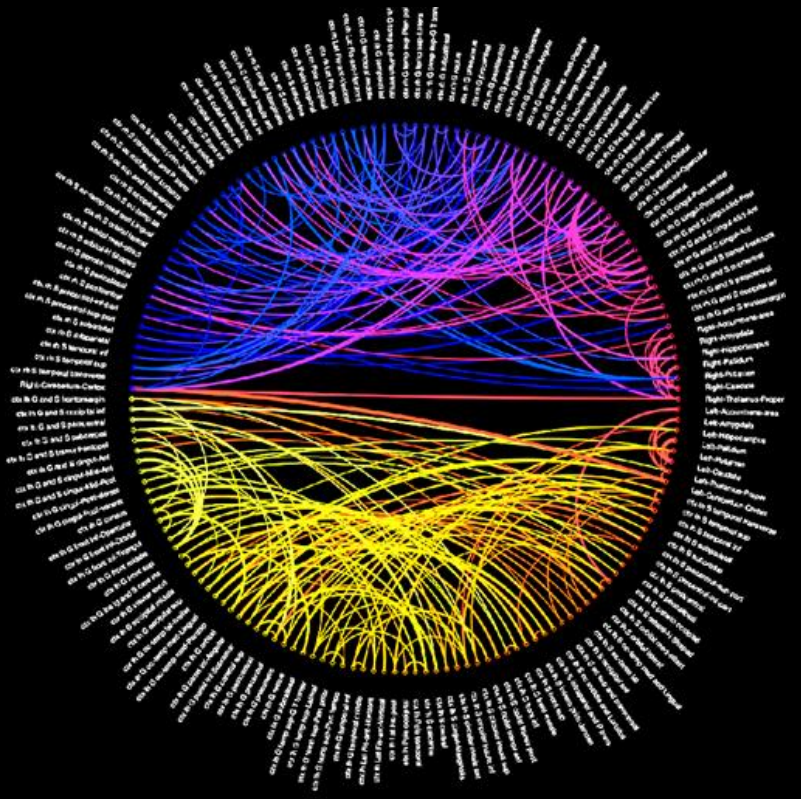
Deterministic



Probabilistic



Proposed



Connectivity between the 164 cortical regions of the Destrieux Atlas

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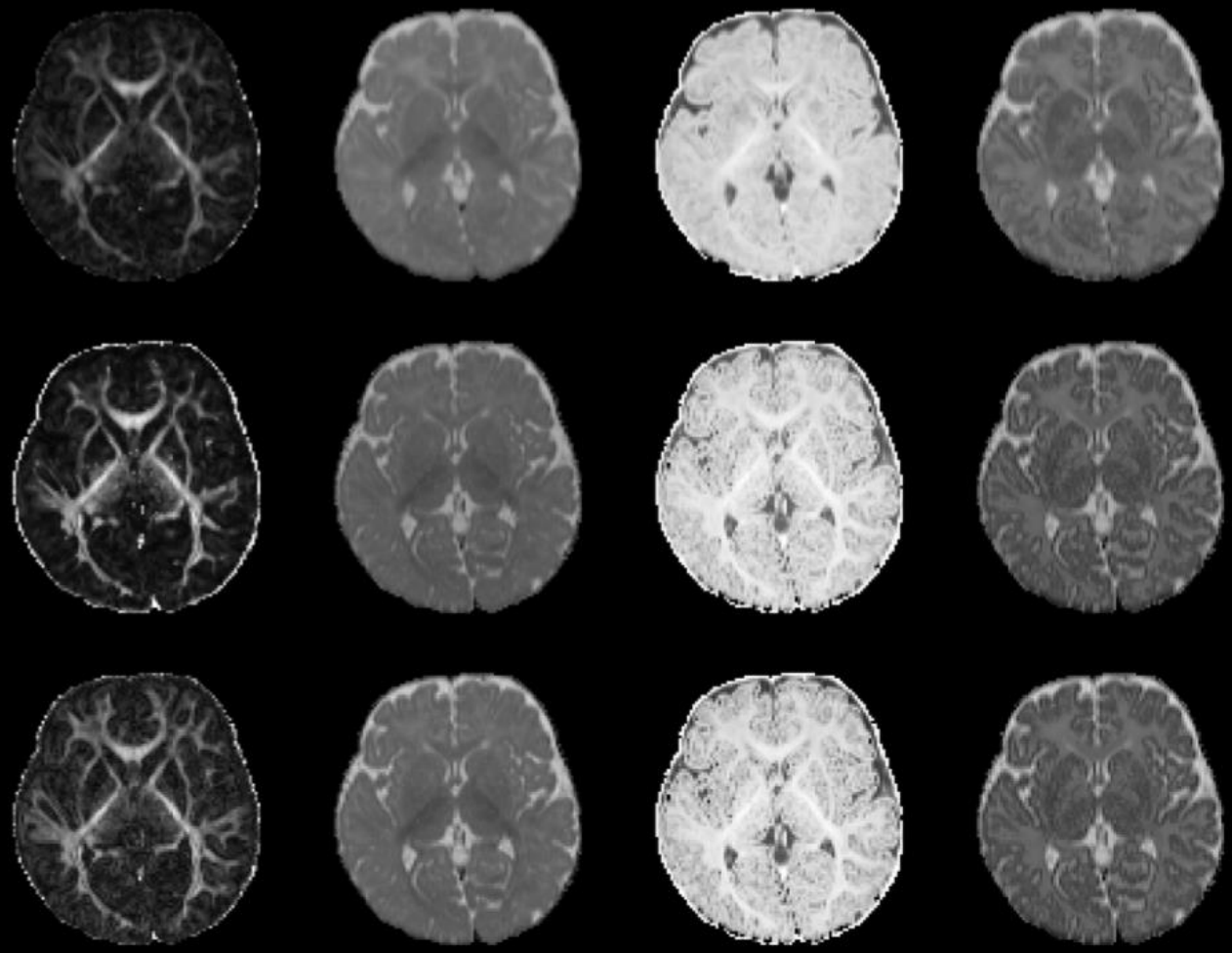
Research Lab IDEA

Longitudinal Harmonization



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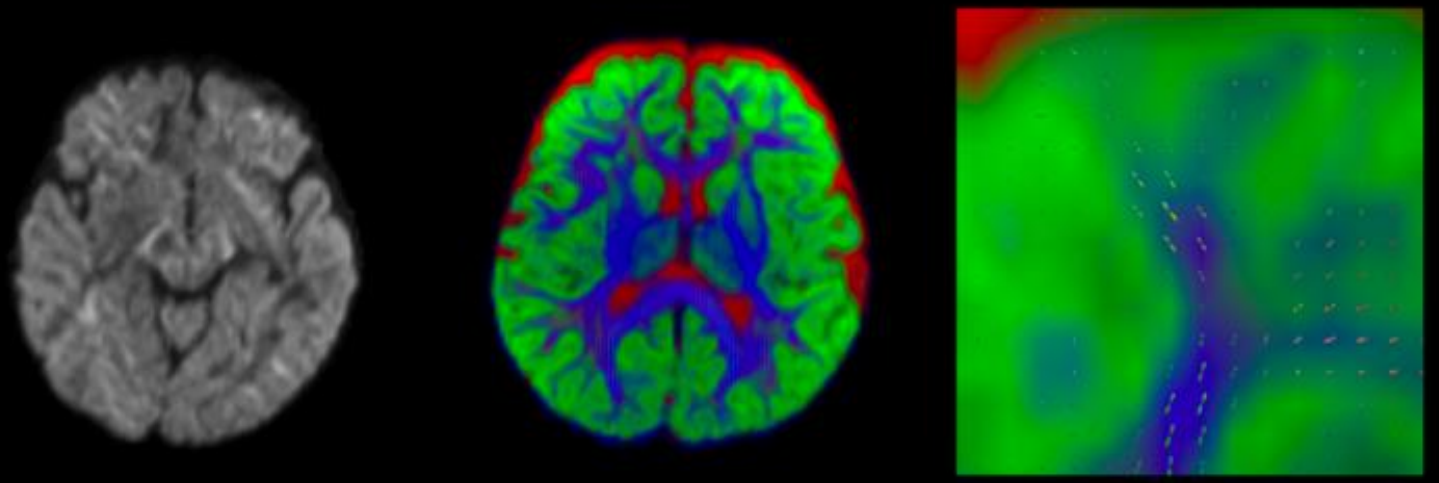
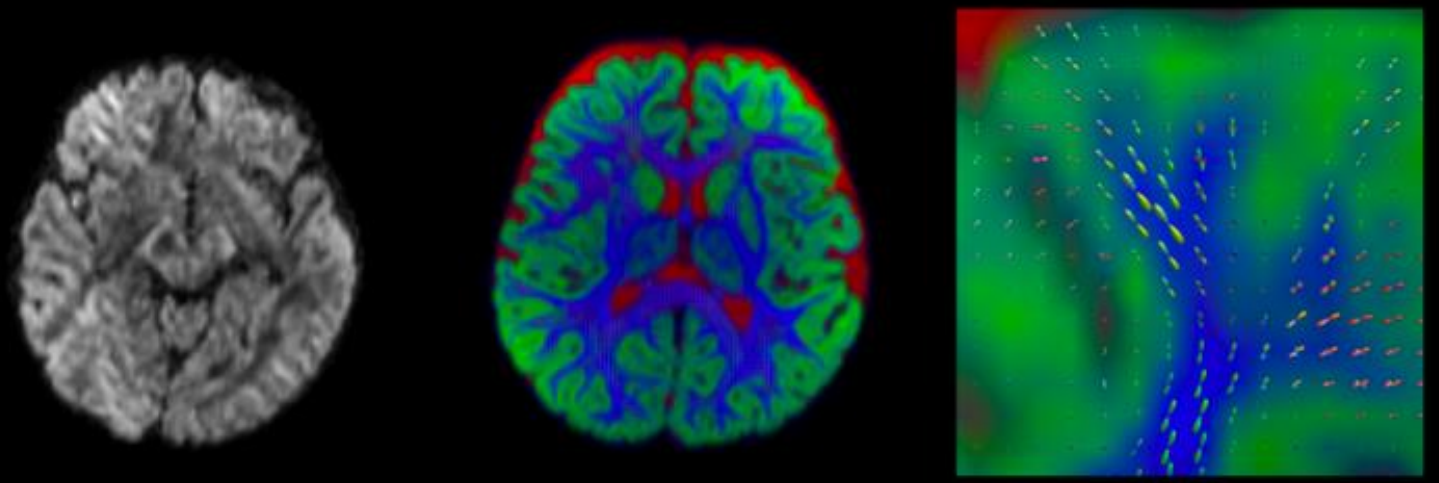
IDEA

Longitudinal Harmonization

Original
(3 Months)

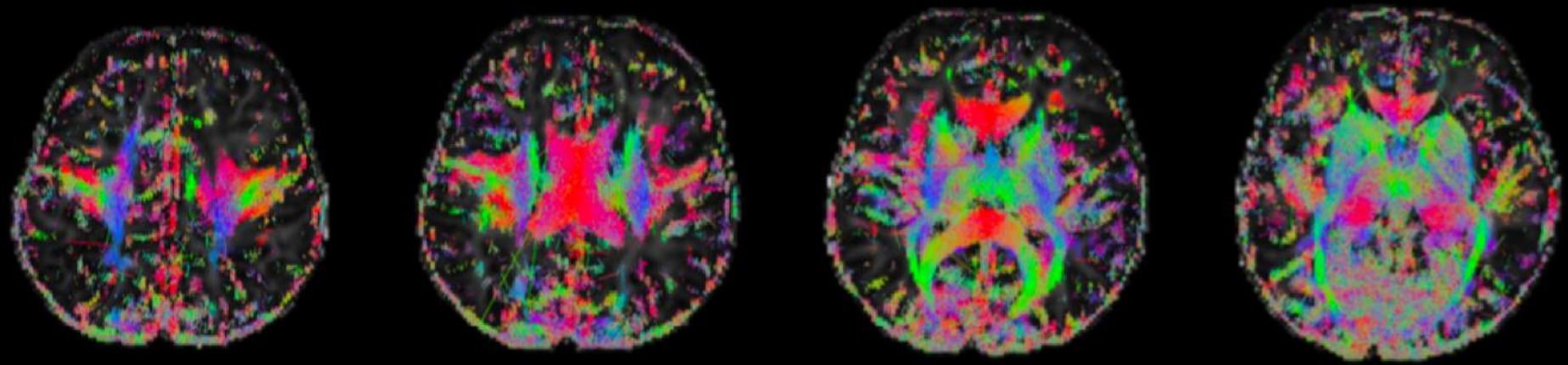


Harmonized

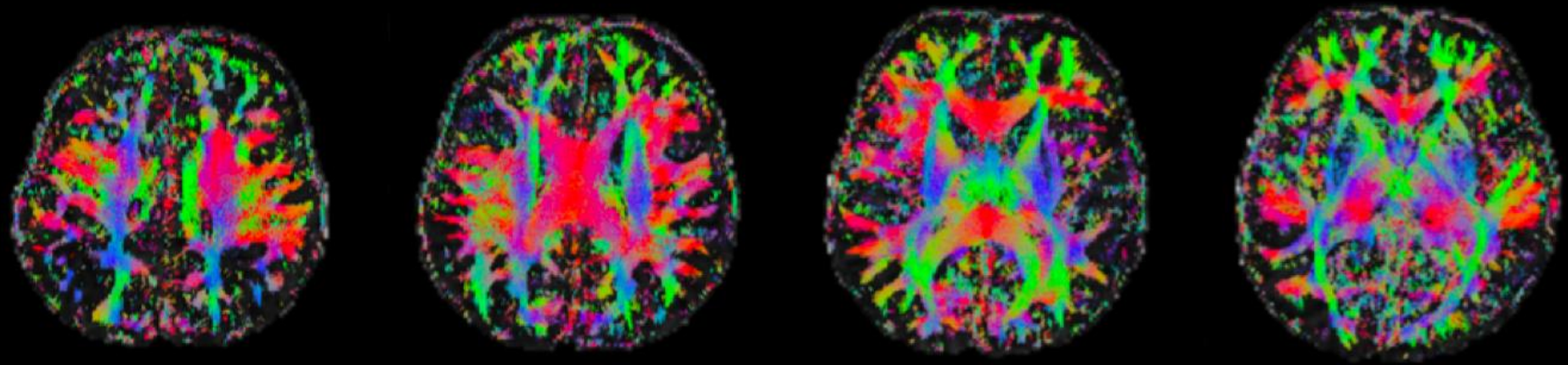


Longitudinal Harmonization

Original
(3 Months)



Harmonized



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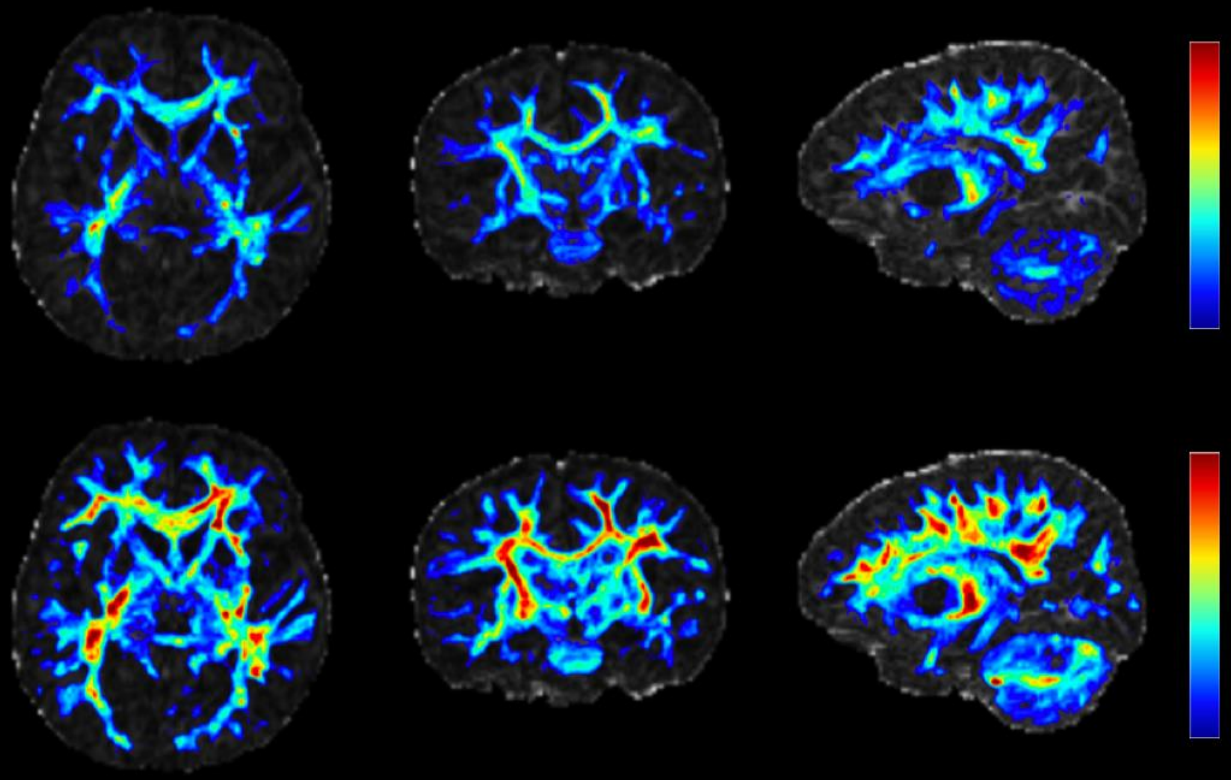
Longitudinal Harmonization

Original (3 Months)

Harmonized



Probability (%) of major tracts before and after harmonization.

Tract	Unharmonized	Harmonized
Anterior thalamic radiation (L)	10.79	20.57
Anterior thalamic radiation (R)	8.45	16.37
Corticospinal tract (L)	16.42	22.21
Corticospinal tract (R)	21.38	33.81
Cingulum (cingulate gyrus) (L)	7.63	13.96
Cingulum (cingulate gyrus) (R)	5.40	11.65
Cingulum (hippocampus) (L)	6.65	11.52
Cingulum (hippocampus) (R)	6.45	10.83
Forceps major	14.90	23.56
Forceps minor	12.86	24.03
Inferior fronto-occipital fasciculus (L)	14.24	25.37
Inferior fronto-occipital fasciculus (R)	13.78	23.77
Inferior longitudinal fasciculus (L)	9.06	16.50
Inferior longitudinal fasciculus (R)	12.27	23.57
Superior longitudinal fasciculus (L)	13.36	25.92
Superior longitudinal fasciculus (R)	13.96	25.44
Uncinate fasciculus (L)	10.48	24.25
Uncinate fasciculus (R)	9.76	16.60
Superior longitudinal fasciculus (L)	13.29	25.60
Superior longitudinal fasciculus (R)	12.70	29.82

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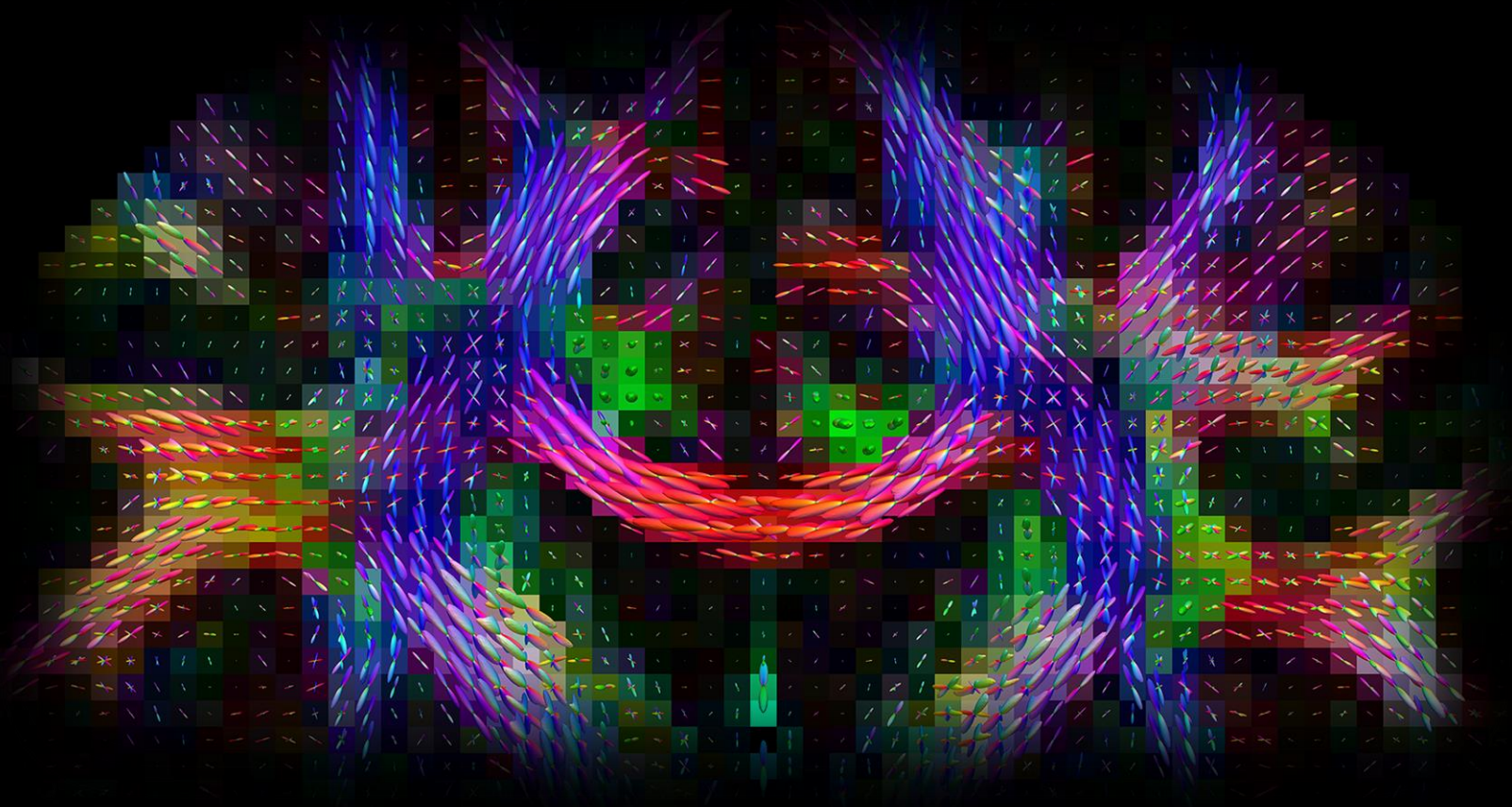

Thank You!

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